

Two Modes, Not Many Types: The Structure of Football Player Roles in Season-Aggregate Data

Zachary Chung*

Christian Chung

September 10, 2026

Abstract

Football analysts describe players with dozens of per-match statistics and then sort them into role archetypes such as the ball-playing centre back or the deep-lying playmaker. Whether such types exist has never been tested, because the evidence normally offered for them is produced just as readily by data containing no groups at all: a clustering algorithm returns groups whatever it is given, so a selection criterion, a resampling check and an interpretable average profile are satisfied by a smooth continuum. We test existence directly on 13,822 player-seasons from 5 seasons of the big five European leagues, described by 33 statistics, by calibrating cluster quality against simulated data matched to the observed sample in size, dimension and correlation but built to contain no clusters. One division survives, and it is a coarse one. Players separate into a defensive and an attacking mode of involvement, and that separation exceeds every simulation under all three clusterless references in every scope, by at least 8.01 standard deviations against the hardest. Nothing finer survives: beyond two modes the observed separation is indistinguishable from what structureless data produces. The usual stability evidence cannot make this distinction, since clusterless data resample at an adjusted Rand index of at least 0.911. The two modes are not the listed positions renamed, agreeing with them at 0.293, with defenders and forwards at opposite poles and midfielders divided almost evenly between the two. The division reproduces in every season and every league, and a club's composition along it predicts league points beyond possession out of sample. Football role space has two poles and a populated middle, and finer taxonomies are artefacts of the method used to find them.

1 Introduction

Public football data now describes every player in a major league by dozens of per-match measures, and a standard genre of analysis has grown around it. Take the measures, group the players by how similar their measures are, name the groups, and present the result as a taxonomy of roles. Published examples report four archetypes within a single position group, and in basketball nine or thirteen across the whole sport, with labels such as deep-lying progressor or advanced playmaker [1–3]. The appeal is obvious. A taxonomy turns a table of numbers into a vocabulary a coach or a recruiter can use.

The grouping step is done by a clustering algorithm, and it is worth being explicit about what one does, because the whole argument of this paper turns on it. A clustering algorithm is handed a set of points, here one point per player with one coordinate per statistic, and a number of groups to find. It then divides the points into that many groups so as to make each group as internally similar as possible. Nothing in that procedure asks whether the points were grouped to begin with. Handed

*Corresponding author: zacharytylerchung@gmail.com

a single cloud of points with no groups in it, a clustering algorithm still divides the cloud, and the division it returns will still look tidy by the measures normally used to judge one. Running the standard method on a single multivariate Gaussian, which by construction contains no groups at all, returns two clean clusters with a respectable separation score. Getting clusters out of a clustering algorithm is therefore not evidence that clusters were there to begin with.

The football literature has not tested that inference, mostly because it has been busy with a harder data problem. The most detailed work on roles uses proprietary feeds: formations and positional roles recovered from tracking coordinates, player vectors and performance ratings learned from event streams, and playing styles distinguished from match events [4–7]. Where public season aggregates are used instead, the standard output is a taxonomy, whether from k -means on a dozen indicators or from partial membership models on a season of counts [1, 8]. Two systematic reviews catalogue the genre for players and for teams, the first synthesising twenty-six styles from twelve studies and finding no shared framework behind them [9, 10]. A review of 278 sports science articles that used clustering found that 55 percent did not state the criterion by which the number of clusters was chosen, which puts the omission in context [11]. The question of existence has not been answered badly so much as not asked.

Outside sport it has been asked for forty years, and the tools already exist. Choosing the number of clusters has been studied systematically since thirty stopping rules were compared on data with known structure, and the silhouette and the gap statistic descend from that programme [12–14]. The gap statistic is the direct ancestor of what this paper does, since it compares the observed clustering against the same quantity computed on reference data drawn uniformly over the principal box. A second tradition validates a partition by its stability under resampling, and the theory reached a conclusion the applied literature has largely not absorbed: for algorithms such as k -means, stability is governed by the geometry of the objective rather than by the presence of clusters, so a partition can be perfectly stable on data with no groups in it [15, 16]. A third asks whether data are clusterable at all, finds the formal notions of it mutually inconsistent, and recommends tests built on Hartigan’s dip statistic applied to a low dimensional projection [17–19]. Underlying all three is the position that there is no true clustering independent of the purpose a clustering is meant to serve [20].

The closest methodological relative of this paper is the aggregated validity index of Akhanli and Hennig, which calibrates several criteria against their distribution over random partitions of the same data and, applied to a season of player statistics from eight European leagues, returned a coarse partition into five ordered groups [21, 22]. That coarse partition runs along the same territorial axis reported here. The difference is what the calibration is against. Comparing against random partitions of the observed points asks how good a clustering is relative to other clusterings of the same data, which is the right question for choosing among methods. Comparing against clusterless data matched to the observed sample asks whether any clustering of these points is better than a clustering of points with no structure in them, which is the question of existence. The two are complementary, and the first does not answer the second. A different tradition sidesteps the question by describing athletes as mixtures rather than as members, either as convex combinations of a few extreme profiles or as partial members of several clusters at once [8, 23]. Both are natural descriptions of a continuum and both presuppose one, since neither asks whether the data would have supported discrete groups instead.

This paper supplies the missing test. We assemble 13,822 player-seasons covering 5 seasons across the 5 major European leagues, with 33 per-90 features spanning shooting, chance creation, ball progression, territory, take-ons and defensive work. On that sample we run the standard pipeline, a selection rule fixed before any fitting followed by k -means, and then calibrate what it returns against simulated data built to contain no clusters at all. Three such references are used rather than one, ranging from a Gaussian matched on covariance to a copula that keeps every real marginal

distribution and correlation, so that a reader can see which reference the claim survives. The same simulations are used to measure what a resampling stability check is worth, which turns an assertion about the genre’s second standard piece of evidence into a number.

Our expectation, and what we find, is that the data support a real division but a far coarser one than the field reports. A single split survives calibration in every scope, separating players by defensive against attacking involvement, while every finer division is indistinguishable from what structureless data produces. That split is not the listed positions renamed, since defenders and forwards fall at opposite poles while midfielders divide almost evenly between them; it reproduces when each season and each league is fitted independently; and a club’s composition along it predicts league points beyond possession out of sample, while the finer archetypes predict nothing further. The practical implication is not that role analysis is impossible. It is that the correct output of role analysis on this kind of data is a position along an axis rather than membership of a type, and that a published taxonomy of a dozen archetypes is reporting structure its data cannot support.

2 Methods

Every procedure below is seeded from a single constant, every hyperparameter is declared in one configuration file that the analysis code reads, and nothing is tuned after seeing a result. Computation uses scikit-learn unless stated otherwise [24].

2.1 Data

The analysis uses season-aggregate player statistics for the 5 major European leagues, covering 5 seasons from 2018 to 2022, where a season is labelled by the calendar year in which it ends. The sample contains 13,822 player-seasons, of which 9,263 are outfield players above the minutes threshold of 450 and 1,035 are goalkeepers.

The tables originate from FBref, which published advanced statistics derived from a commercial event-data provider [25]. They are obtained here from a public mirror maintained by the worldfootballR project rather than from the site directly, and that choice needs stating plainly [26]. On 20 January 2026 the site’s operator removed every statistic derived from that provider, retroactively and across completed seasons, after the licence was terminated. The mirror predates the removal. It is therefore a secondary source in the strict sense: it was scraped by a third party, we did not observe the scrape, and it cannot be re-derived from the primary source because the primary source no longer holds the data. What we can do, and do, is audit it. Row counts, per-column coverage and per-season coverage are checked on load, and the checks are part of the pipeline rather than a one-off inspection. Retrieval is part of the pipeline as well, recording a SHA-256 digest per file alongside row and column counts, so that an upstream change is detectable rather than silent. One season is excluded on those grounds, because the site restructured its ball-carrying columns for the season ending 2023 and the mirror did not follow the renames, leaving every carrying column empty for that season while neighbouring columns are intact. The pipeline raises rather than imputing whenever a feature exceeds half missing within any retained season, so the exclusion is enforced by a check and not by a comment.

The outfield feature set contains 33 features, listed in full with their source table in Table S1. It spans six families: shot volume and quality including non-penalty expected goals [27, 28]; chance creation; ball progression; touches across the five pitch zones; take-ons; and defensive work together with passing accuracy by distance. Counting statistics are divided by nineties played, and quantities published as rates are used as published. Minutes and appearances are retained as context and never enter the feature set, because squad usage describes a player’s opportunity rather than the role he

performs within it. Several features are undefined for some players by construction rather than by data loss, since a player who attempted no take-ons has no take-on success rate; after the minutes filter these affect at most eight percent of players in any single feature, and are median-imputed within position group and flagged.

A second sample is collected directly from the live site for the seasons ending 2025 and 2026, after the withdrawal, with expected goals taken from Understat as an independent source [29, 30]. It retains 18 features, roughly half the primary set, and it is used only as a robustness case, because a result that holds on both a rich and an impoverished description of the same game is worth more than one that holds on either alone. That sample also carries a warning that applies beyond this study: the affected tables are still served with complete column headers and one row per player, and only the values are gone, so a feature set assembled from the published schema resolves correctly and yields an empty matrix. Feature selection against observed values instead of documentation is now a necessary step with this source. Section S8 reports the audit and the analyses that sample supports.

Season aggregates discard context, and that bounds every claim below. A tackle count does not say where on the pitch the tackles happened, and an expected goals total does not distinguish six poor chances from two good ones. The sample is recent European club football, so nothing here should be read as a claim about the game in other eras or at other levels, and the clusters are descriptive: a cluster label records how a player’s season looked, not why.

2.2 Adjusting defensive counts for possession

Defensive actions can only occur while the opponent has the ball, so a player in a side that concedes possession accumulates interceptions, tackles and fouls partly through exposure rather than through role, and since possession tracks team strength this confound would enter the feature set as a proxy for the quality of a player’s team. The correction introduced by Knutson [31], in the proportional form in common use since, multiplies a defensive rate by the ratio of a reference opponent possession to the actual one, which assumes defensive volume is directly proportional to time spent out of possession. That is an empirical claim rather than a definition. Regressing the logarithm of team defensive volume per ninety on the logarithm of opponent possession over 490 team-seasons gives the elasticity, the percentage change in defensive volume produced by a one percent change in opponent possession, for which direct proportionality corresponds to a value of one.

We adjust with the measured elasticity instead of the assumed one, raising the exposure ratio to the power of each statistic’s own fitted value. The exponent is estimated from team totals rather than player rates, because the quantity being corrected is a team-level exposure effect, and it is pooled across seasons and leagues, because it is a nuisance parameter that uses no player-level information and so cannot manufacture the cross-season and cross-league agreement reported later. Elasticities are reported with bootstrap intervals from 2,000 resamples of the team-seasons, refitted within every league and every season separately, and checked against a model with a quadratic term in log possession. Attacking statistics are deliberately left unadjusted, since they scale with a team’s own possession rather than the opponent’s; Section S2 reports the elasticities in full and tests that asymmetry rather than asserting it.

2.3 Clustering and the selection rule

Clusters are estimated with k -means as the primary method, with a Gaussian mixture model whose Bayesian information criterion is reported across the same range of component counts, and with HDBSCAN, for which we report the fraction of players assigned to noise [32]. Clustering is performed globally and within each of the three outfield position groups, which are treated separately because

correlations between features reverse sign across groups, so a single pooled geometry averages over relationships that point in opposite directions.

The number of clusters is selected by a rule fixed in advance of any fitting, which matters because four criteria applied after the fact can be made to justify almost any choice. Over k from 2 to 12 we compute the silhouette coefficient, the Davies-Bouldin index, the Calinski-Harabasz index and the gap statistic [13, 14]. The rule selects the smallest k whose silhouette lies within 0.02 of the best silhouette observed and which ranks in the top three on at least two of the silhouette, Calinski-Harabasz and gap criteria, with Davies-Bouldin breaking ties. Which criteria agreed, and which did not, is reported alongside the choice.

2.4 Testing whether clusters exist at all

Selecting a number of clusters presupposes that clusters are there to be counted, and that presupposition is the weak point of the genre. The internal criteria used to choose k are computed on the partition rather than against any alternative, so a silhouette of 0.273 sounds like evidence until one asks what silhouette structureless data of the same shape would have produced.

Let $X \in \mathbb{R}^{n \times d}$ be the standardised feature matrix for a scope, with n player-seasons and $d = 33$ features, and let $\pi_k(X)$ be the partition returned by k -means with k clusters under a fixed seed and multiple restarts. Let $s_k(X)$ be the mean silhouette of $\pi_k(X)$, computed on a fixed subsample of at most four thousand rows when n exceeds that, since the silhouette is quadratic in n . The number of clusters reported, $k^* = 2$, is not chosen by inspecting $s_k(X)$. It is returned by the selection rule declared above, and the calibration then asks a single question about it.

Three generators produce clusterless data matched to X . The Gaussian generator draws n rows from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ with the empirical mean and covariance, matching the data in size, dimension and correlation while containing no group structure by construction. The copula generator transforms each column of X to normal scores by rank, estimates the correlation $\hat{R}$ of those scores, draws n rows from $\mathcal{N}(0, \hat{R})$, and maps each simulated column back through the empirical quantile function of the corresponding observed column, so that every marginal distribution of X is reproduced exactly, skew and bounds included, and only the joint structure beyond rank correlation is destroyed. The uniform generator rotates X to its principal axes, draws n rows uniformly within the resulting bounding box, and rotates back, which is the reference distribution of Tibshirani et al. [14]. A single Gaussian is the easiest of the three to beat, because skew alone raises the silhouette a partition can reach, so the paper treats whichever is hardest in a given scope as the calibration that counts.

For each generator g and each of $B = 25$ draws $X^{(g,b)}$, the identical pipeline computes $s_{k^*}(X^{(g,b)})$, giving a null sample of size B per generator. The reported statistic is the standard score

$$z_g = \frac{s_{k^*}(X) - \bar{s}_g}{\hat{\sigma}_g},$$

where $\bar{s}_g$ and $\hat{\sigma}_g$ are the mean and standard deviation of that null sample. With B draws the null standard deviation is itself an estimate, so z_g describes the margin rather than serving as a test statistic with a known distribution. The claim the paper makes is the distribution-free one, that $s_{k^*}(X)$ exceeds every one of the B null values under every generator, which under the null of exchangeability between the observed and simulated statistics has Monte Carlo probability at most $1/(B+1)$ per generator and scope. The headline is a conjunction over three generators and four scopes rather than a disjunction, so it is conservative with respect to multiplicity: it can only be weakened by adding a generator or a scope, never strengthened.

The same generators supply two further calibrations. Hartigan’s dip statistic on the leading principal component of each $X^{(g,b)}$ gives a null sample against which the observed dip is placed in the

same way, which matters because at these sample sizes the dip test’s p-value rejects unimodality for departures too small to be of interest [19]. The bootstrap adjusted Rand index of π_{k^*} is computed on the simulated datasets by exactly the resampling procedure applied to X [33], so that the stability the genre reports for real data can be read against the stability a clusterless continuum produces. Rather than assert that a partition imposed on structureless data passes a bootstrap, the paper measures it.

One assumption should be stated because the analysis makes it silently otherwise. The rows of X are player-seasons, and the same player appears in up to 5 of them, so rows are not independent. The generators match the marginal and pairwise structure of X but not this nesting, and the bootstrap resamples rows as if exchangeable. The replication analysis below, which refits every season and every league independently, does not share this assumption, and it is the reason the paper does not rest the two-mode claim on any single pooled fit.

2.5 Stability, replication and the listed positions

We draw two hundred bootstrap resamples, refit the clustering on each, and compute the adjusted Rand index between each bootstrap partition and the reference partition on the players common to both, accumulating a consensus matrix of pairwise co-assignment alongside it. Replication is tested separately and more stringently, because a resample reuses the same players. Each season and each league is standardised inside itself and clustered from scratch, borrowing no centre, no scale and no starting point from the pooled fit. Cluster labels are arbitrary per fit, so labels are aligned to the pooled partition on the contingency table before any label-dependent quantity is computed, and an assertion fails the run if an aligned agreement drops to chance. Three comparisons are reported because they answer different questions: an independent fit scored against the pooled partition on the same rows; one season’s labels scored against another’s on the players present in both, which mixes replication with real change in those players; and one scope’s centroids imported into another scope’s players and scored against that scope’s own fit, which isolates replication. Each scope’s centroid-difference vector is also correlated with the pooled one, to check that the scopes recover the same axis rather than a similarly sized partition. Agreement with the listed positions is quantified by normalised mutual information and by the adjusted Rand index, against both the position group and the full position string.

A two-mode result on 33 features could in principle be carried by one block of them, so the features are grouped into six families, with the grouping stated in the analysis code and asserted to cover every feature exactly once. Each family is removed in turn and each is tried alone, with the selection rule and the null calibration rerun on what remains in every scope. For every ablation we record the number of clusters selected, the standard score against the null, and the adjusted Rand index between the ablated partition and the full one, so that a surviving k of two can be distinguished from the same two modes surviving.

2.6 Validation against results

The structure is tested against what happened. The archive’s team tables carry goals but not results, so wins, draws and losses are recovered by summing the goalkeeper table, which credits every match to the goalkeeper who played it, and the recovery is audited against the schedule with the exceptions listed. Each team-season is summarised by its minutes-weighted composition on the two-mode partition and on the six archetypes, and by the minutes-weighted mean position of its players along the axis between the two centroids. Prediction is out of sample under three holdouts, one season at a time, one league at a time and one club at a time, the last being the strict one since no held-out

team-season then shares a club with the training data, by ridge regression with the penalty chosen by cross-validation inside the training folds, comparing possession alone, the composition alone and both. The increment of the last over the first is the quantity of interest, because composition and dominance are confounded and only the increment says whether the modes carry information about results that dominance does not.

2.7 Profiling, supervised validation and goalkeepers

For every retained cluster we compute the centroid profile as a standardised deviation from the mean of its position group, take the eight features with the largest absolute deviation as the distinguishing features, and list the players closest to the centroid and near the boundary. Archetype names are assigned only after inspecting these profiles and the players they contain, and each name carries a justification citing the distinguishing features and the example players. The names are hypotheses about what a statistical profile represents, not observations. Outliers are identified within position group with an isolation forest alongside a Mahalanobis distance, so that a partition-based and a distance-based notion of unusualness can be compared [34].

Unsupervised structure is easy to over-interpret, so we also ask a supervised model what separates the listed positions. Logistic regression and gradient boosting are trained to predict the position group from the same features under five-fold stratified cross-validation against a majority-class baseline running through identical folds [35], and SHAP values on the boosted model identify which statistics do the separating [36]. Two properties of the confusion matrix are read as evidence about geometry rather than about accuracy: whether the two extreme classes are confused with each other, and whether the errors concentrate in the class that lies between them. An ordered space with two poles produces few errors of the first kind and many of the second, while three genuine categories would not.

Goalkeepers share almost no features with outfield players and are analysed with a separate pipeline on the 13 canonical goalkeeper features, under the same threshold, standardisation and selection rule. Saves above expectation are computed as post-shot expected goals minus goals allowed, which conditions on the difficulty of the shots faced rather than only on their number [37]. One choice needs stating in advance because the result turns on it. The goalkeeper feature block mixes quantities describing what a goalkeeper does with quantities describing what his defence permits him to face, and a partition fitted on both will separate teams rather than goalkeepers. We therefore fit twice, once on the full block and once on technique alone with the exposure variables excluded, and report for each partition how much variance membership explains in situational quantities against technical ones.

2.8 Reproducibility

The analysis runs from a single command against pinned dependencies and a fixed seed. Every figure and every number in this paper is generated by code in the accompanying repository and read into the document from generated definitions at build time, so no value in the text was transcribed by hand. The code, the retrieval step that fetches the archive from its stated upstream with a recorded digest per file, the cluster assignments for both samples, and the sources of this document are at <https://github.com/ZacharyTChung/pl-player-profiling>.

3 Results

3.1 Is there any structure at all?

The first question is not how many clusters there are but whether there are any. Figure 1 answers it against the three clusterless references of Section 2.4.

Real structure exists, and it is confined to a single division. At two clusters the observed silhouette is 0.273 for the outfield pool against a Gaussian null mean of 0.207, and it exceeds every one of the 25 simulations; the same holds in each position group. A Gaussian is the softest reference available and those are the numbers a sceptic should discount, so the two harder nulls are put through the same procedure. The copula null keeps every feature’s real marginal distribution, skew and bounds included, along with the real rank correlations, which removes the objection that skew alone produced the separation. The uniform reference keeps only the data’s principal bounding box. The copula is the hardest of the three in every scope, and the observed partition clears it everywhere, by 13.87 standard deviations on the pool and by 8.60, 10.13 and 8.01 within the three position groups. The observed silhouette at two clusters exceeds every simulation under every null in every scope, and the smallest margin anywhere over all four scopes and all three nulls is 8.01 standard deviations, for forwards against the copula.

Beyond two clusters the curves converge, which is the central negative result of this paper stated precisely. By around six clusters the observed silhouette lies inside the simulated band in every position group, so the third, fourth and subsequent clusters that a taxonomy would name are not distinguishable from what a partitioning algorithm extracts from structureless data.

The second standard piece of evidence for a published taxonomy is a bootstrap, and the same simulations show what it is worth. Computed on 5 clusterless datasets from each null in each scope, by exactly the procedure applied to the real data, the bootstrap adjusted Rand index of the two-cluster partition runs from 0.930 to 0.985, and no single clusterless dataset anywhere falls below 0.911. The observed data run from 0.962 to 0.990. The gap between real data and data built to contain no groups is as small as 0.003 and never larger than 0.033, and a stability screen at 0.8, demanding by the genre’s standards, is passed by every clusterless dataset in every scope. A resampling check does not distinguish real structure from a partitioned continuum. It measures whether an algorithm is deterministic on similar inputs, which k -means is, and it is reported here only because readers will expect it.

Three further instruments agree, with one exception reported as such. Hartigan’s dip test rejects unimodality on the leading principal component at $p < 0.0001$ for the pool, but at these sample sizes a p-value rejects for departures too small to matter, so the dip statistic is the quantity to read and it is read against the nulls. The observed dip on the pool is 0.0176, which sits 13.67 null standard deviations above the hardest of the three references; defenders clear it by 8.84 and midfielders by 5.66, so the dominant axis is two-moded rather than merely wide. Forwards are the exception in both readings: the test returns $p = 0.9936$, and the observed dip of 0.0053 sits -2.69 standard deviations against the uniform reference, which is to say below it. The forward split is real by the silhouette calibration, clearing the hardest null by 8.01, and its leading axis is not bimodal by the dip. Both halves are reported because together they say what the forward split is: a cut through a continuum of involvement that the null cannot produce, rather than two kinds of forward. Density-based clustering assigns 0.865 of the pool to noise and every position group entirely to noise, which is what a density method does when handed a continuum with a density gradient instead of separated groups. The Gaussian mixture criterion declines monotonically to the top of the candidate range in every scope, the signature of a model absorbing extra components into a smooth density rather than identifying real ones.

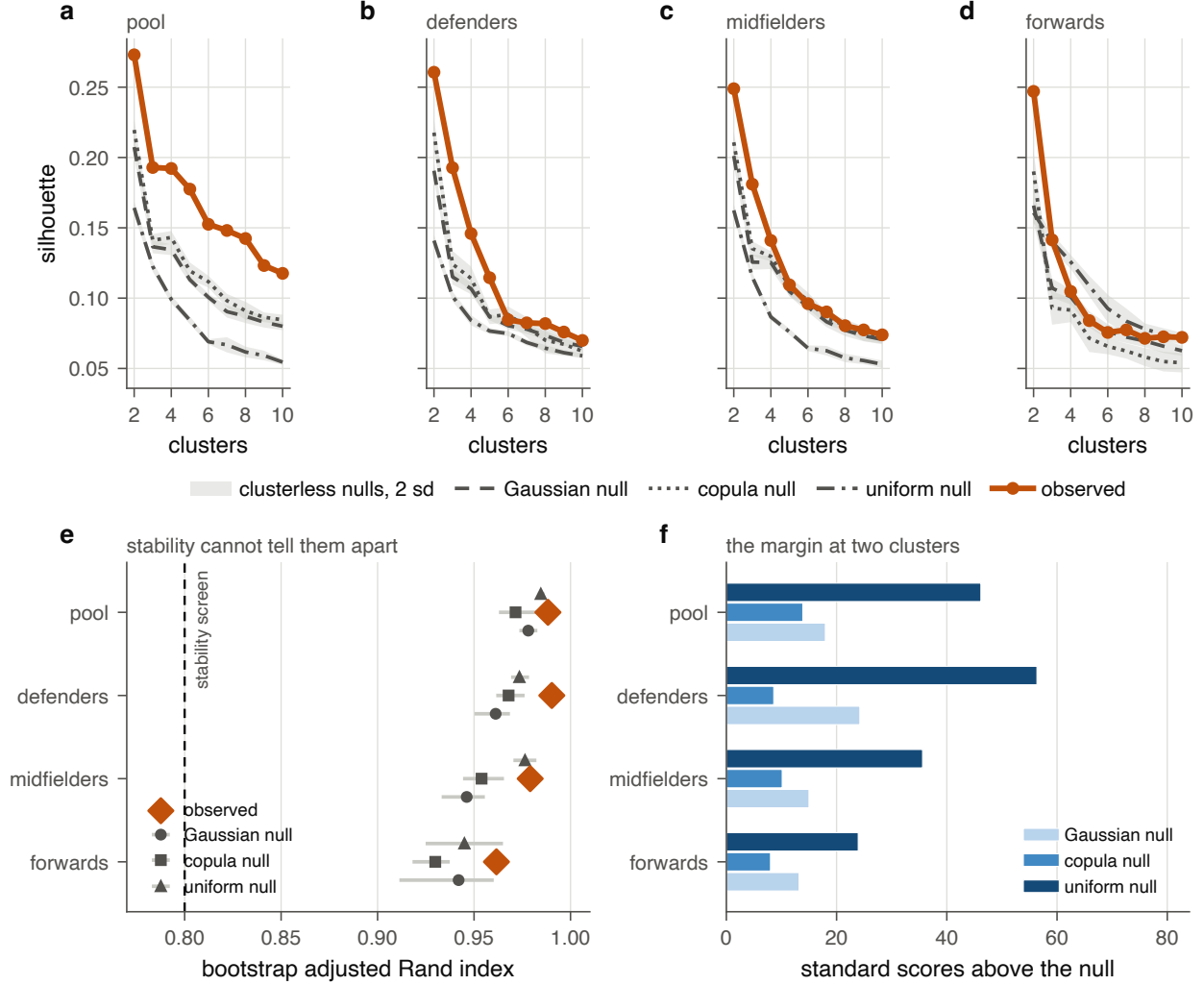


Figure 1: Two modes are real and nothing finer is. **a** to **d**, observed silhouette against three kinds of data built to contain no clusters, one panel per scope. Each null is simulated 25 times and drawn as its mean with a band of two standard deviations: a Gaussian matched on covariance, a copula that keeps every real marginal and correlation, and the uniform reference of the gap statistic. The observed curve separates from all three at two clusters and rejoins the bands as the number of clusters grows. **e**, bootstrap adjusted Rand index of the two-cluster partition on the real data against the same quantity computed on the clusterless simulations, with the range over simulated datasets: the stability check the taxonomy genre relies on cannot tell them apart. **f**, the margin at two clusters, as standard scores of the observed silhouette above each null.

Taken together the instruments support one claim and refuse a stronger one. Football role data contains a real division into two modes. It does not contain a taxonomy.

3.2 The shape of the space, and what the modes are not

A two-mode description invites two objections. If the data reduce to two groups, perhaps they reduce to one number. If they do not, perhaps the two groups are the listed positions under another name. Figure 2 answers both, and the answer to each is no.

The leading principal component carries 39.9 percent of the variance, far more than any other single direction, which confirms that one axis dominates. It is nowhere near the whole space. The first three components together reach only 64.2 percent, and 14 are required to reach ninety percent of the variance, so the role space has one dominant axis and a long tail of smaller ones, which is a different object from a line. The content of the leading components matches the football. The first separates players by where on the pitch they touch the ball, loading positively on progressive receptions and touches in the attacking third and the opposition box, and negatively on touches in the defensive third and in the defending team’s own box. The second is a progression axis close to orthogonal to the first. A deep midfielder who moves the ball forward and a striker who receives it in the box sit at opposite ends of the second component while sharing territory on the first. That pair explains how two modes and a multidimensional space coexist: the first component is bimodal and drives the partition, while the components behind it carry real and interpretable variation that is continuous rather than clustered.

Projecting every player onto the axis between the two centroids shows the distribution the paper is describing. Density is concentrated at two ends with a populated middle between them, and the listed positions are laid across that shape rather than aligned with it. Agreement with the listed position group is well above chance and well below recovery, at an adjusted Rand index of 0.293 and a normalised mutual information of 0.338. A partition that simply relabelled position would return values near one and a partition unrelated to position would return values near zero.

Where the line falls is the informative part. 92.9 percent of defenders fall on one side and 99.8 percent of forwards on the other, while midfielders divide 1,920 to 1,846, which puts 51.0 percent of them on the attacking side and is as close to an even split as the sample permits. It is that even split that drags the agreement indices down. The partition is not a relabelling of position but a cut through it: defenders and forwards sit at the ends of the same axis and are assigned almost deterministically, while midfielders are spread along its middle and assigned by which end they lean toward.

A supervised model recovers the same geometry from machinery that shares nothing with the clustering. Gradient boosting predicts the listed position group at 93.4 percent accuracy and a macro F1 of 0.921 out of fold, against a majority-class baseline of 40.9 percent, and multinomial logistic regression is barely behind at 93.1 percent. The near equality of a linear model and a boosted tree ensemble is itself informative, since it means almost all the recoverable signal is linearly separable and there is little interaction structure left for the trees to find, where a taxonomy of many discrete types would be expected to leave more. The errors are where the argument is. Of 9,263 player-seasons the model assigns 0 defenders to the forward class and 8 forwards to the defender class, while every other confusion runs through midfield, which is involved in 603 errors against the 8 that link the two poles directly. Only 1.31 percent of misclassifications skip the middle. That is the confusion structure of an ordered line whose ends are far apart and whose middle overlaps both, not of three distinct categories, which would scatter errors more evenly across the off-diagonal. The feature attributions agree: the three strongest features overall are Touches def third, Clearances (adj) and Ball recoveries, all territorial rather than technical. Position, as the data record it, is mostly a statement about where a player stands.

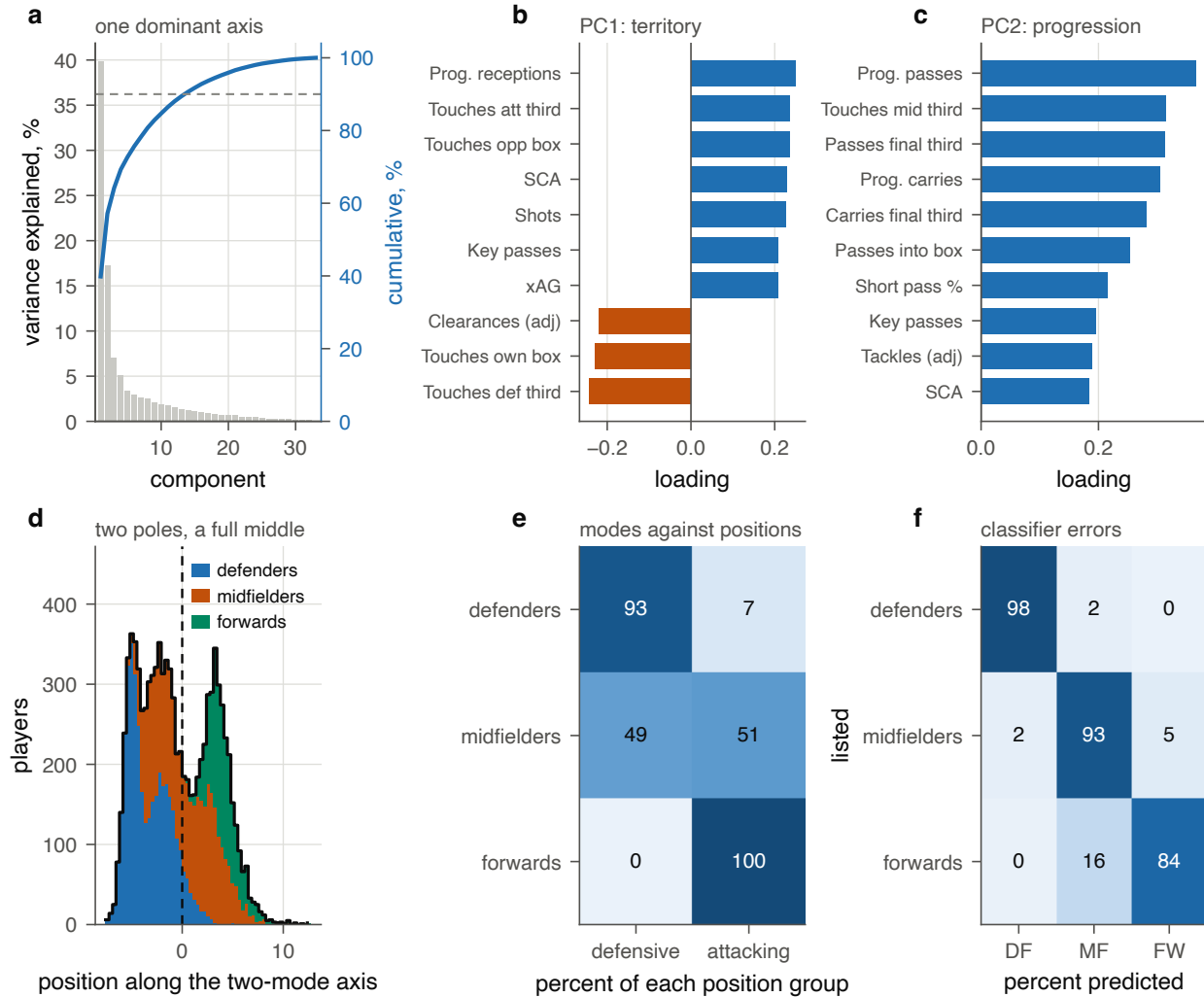


Figure 2: The space has one dominant axis and many smaller ones, and that axis is not the listed positions. **a**, variance explained per principal component with the cumulative curve, over 9,263 eligible player-seasons on 33 features: one dominant component and a long tail rather than a sharp elbow. **b** and **c**, the ten strongest loadings on the first two components, which order players by pitch territory and by ball progression respectively. **d**, every player projected onto the axis between the two centroids, stacked by listed position group with the total drawn as an outline; the dashed line is the cut the partition makes. **e**, the share of each listed position group falling in each mode. **f**, out-of-fold confusion for the gradient boosted position classifier, in percent of each listed class. The corners linking defenders and forwards are close to empty in both, and midfield is where the disagreement lives.

3.3 What the two modes are

Two modes within each of three position groups give six descriptions, and with 33 features they can be characterised sharply. Names were assigned only after inspecting centroid profiles and member players, and the justifications are generated from the computed centroids rather than composed for the paper. Figure 3 shows the features that separate each pair.

Among defenders the division is between 2,094 Deep central defenders and 1,699 Wide attacking

defenders. The first take three times as many touches in their own defensive third as the second, more than twice as many in their own box, twice the possession-adjusted clearances, and win considerably more aerial duels. The second take five times as many touches in the attacking third, attempt ten times as many crosses, receive far more progressive passes and attempt four times as many take-ons. This is the centre back against wide defender distinction, and the notable point is that the source's own position label does not encode it, since the great majority of both groups are listed simply as defenders.

Among midfielders the division separates 1,939 Deep ball winning midfielders from 1,827 Advanced attacking midfielders. The first recover the ball more often, tackle more and take far more touches in their own third. The instructive detail is that they also make more passes into the final third and more progressive passes than the second group. They are not less involved in progressing the ball, they progress it differently, by passing forward from deep rather than by receiving high. A description that treated attacking involvement as one axis would miss that, and it is the clearest case in the paper of the richer feature set buying something a poorer one could not.

Among forwards the division separates 1,241 Finishing centre forwards from 463 Creating and carrying forwards, and it is not a division by scoring. Non-penalty expected goals are slightly higher in the first group. What separates them is carrying and creation, with the second group recording nearly three times the progressive carries, almost three times the passes into the box, twice the take-ons and substantially more shot creating actions while shooting slightly less. Forward is not one role, but the axis that splits it is involvement in build-up rather than goal threat.

One structural property constrains how far the six can be read as independent. Features are standardised within position group before a two-way split, so the two centroids in a group are exactly anti-parallel and the larger cluster necessarily carries the smaller deviations. Each pair is therefore the two ends of one axis, and the six archetypes describe three axes rather than six independently shaped profiles. The three are also not equally well supported. The defender split is the strongest on every measure applied to it: the largest centroid separations, the lowest churn, the highest agreement across leagues, and a dip test that rejects unimodality. The midfield split is well supported with a more populated boundary. The forward split is the weakest and should be read with the most caution, since its dip test does not reject unimodality, the division is uneven, and one league's refit falls well below the others.

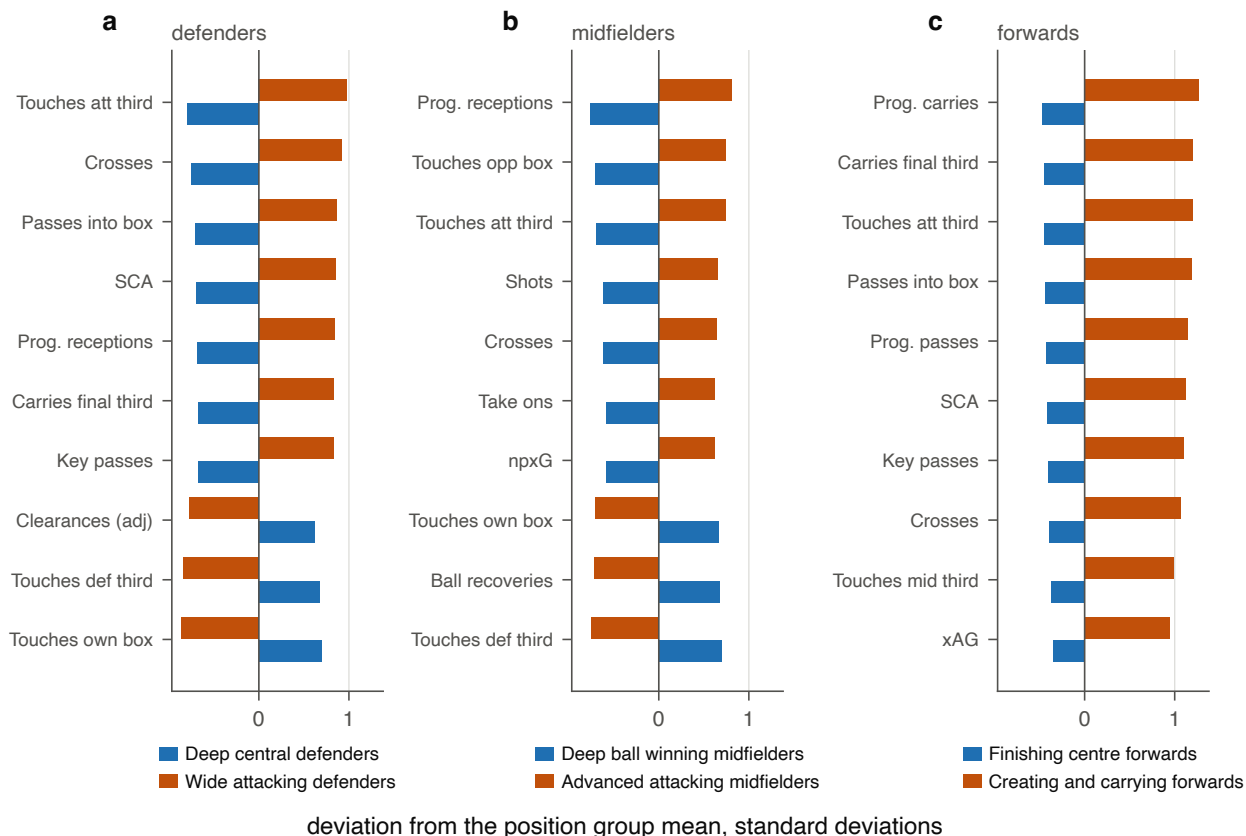


Figure 3: What separates the two modes inside each position group, as the standardised deviation of each archetype centroid from its group mean on the ten features with the largest separation. **a**, defenders. **b**, midfielders. **c**, forwards. The two profiles within a group are mirror images by construction, since features are standardised within group before the split, so each panel describes one axis and its two ends rather than two independent profiles. Radar views of the same centroids are in Figure S7.

3.4 The same division in every season, league and feature family

A partition that appears once is a description of one dataset. A partition that appears independently in five seasons and five competitions is a description of the sport. Each season and each league is therefore standardised inside itself and clustered from scratch, borrowing no centre, no scale and no starting point from the pooled fit.

Every scope recovers the pooled partition. The weakest season agrees with the pool at an adjusted Rand index of 0.916 and the weakest league at 0.932, on 9,263 player-seasons in total. The separating axis is the same object each time rather than merely a similarly sized one, since the correlation between each scope’s own discriminant direction and the pooled direction never falls below 0.9983. Importing one scope’s centroids into another scope’s players separates replication of the solution from change in the players, and it is the cleaner of the two tests: across the twenty ordered league pairs the transferred centroids reproduce the receiving league’s own fit at a mean adjusted Rand index of 0.944, never falling below 0.893. A model of role fitted on Ligue 1 assigns Bundesliga players almost exactly as a model fitted on Bundesliga does.

The season-against-season comparison on players present in both is lower, at a mean of 0.693

with a range from 0.570 to 0.753, and the pattern explains why. Agreement decays monotonically with the gap between seasons while the transfer index over the same pairs stays flat, so the decay is players changing role over four years rather than a weakness of the partition. Across the same pairs 91.6 percent of players present in both seasons are assigned to the same mode in both, falling no lower than 87.8 percent on the pair furthest apart in time. Roughly one player in twelve moves between modes over a four-year span, which is a plausible rate of genuine role change and not the churn an unstable partition would produce. Figure 4 shows all of this, the season and league refits above and the feature-family ablation below.

A result on 33 features could also be carried by one block of them, and a reviewer who suspected the modes were the expected goals block wearing a costume would be right to ask. The features fall into 6 families, and each was removed in turn and each was tried alone. Removing any one family removes nothing that matters. The rule returns two clusters in every leave-one-out in every scope, and of the 52 feature sets tested across four scopes, 51 exceed every one of the 25 clusterless simulations; the weakest case is MF without passing, where the observed silhouette still sits 10.15 null standard deviations above the clusterless mean. The survivors are the same two modes rather than two different ones, since the adjusted Rand index between each ablated partition and the full one runs from 0.703 to 0.990.

The converse is more instructive than the headline. Judged by whether a single family produces a two-cluster solution that beats the null, most families carry structure alone, and creation, progression, territory, defending do so in every scope. That criterion is generous, and the agreement with the full partition shows why. Shooting alone in defenders separates from the null at 19.79 standard deviations while agreeing with the paper’s partition at only 0.000: it finds two groups, and they are the finishers against everyone else, not the deep against the wide. Only territory reproduces the paper’s own modes by itself in every scope, at an adjusted Rand index from 0.732 to 0.856. The two modes are where a player touches the ball, and every other family either sharpens that division or, alone, cuts a different one. The two failures are forwards, where shooting alone clears the null by only 2.92 standard deviations and passing accuracy alone reaches 1.42 without exceeding every simulation, which is a further reason to read the forward split as a cut through a continuum of involvement.

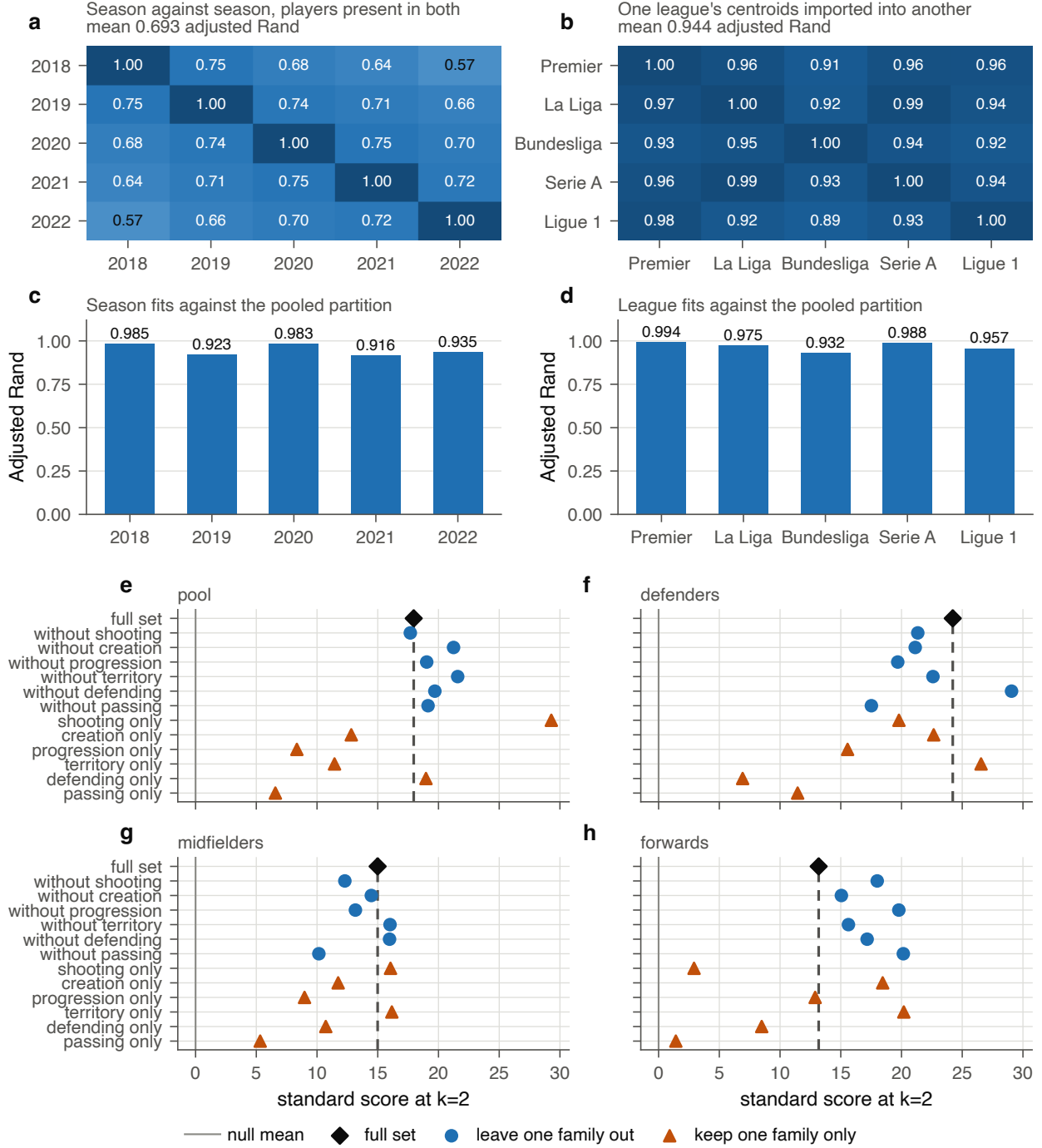


Figure 4: The division reproduces across every partition of the data and every part of the feature set. **a**, agreement between seasons on the players common to both, which declines as the gap widens because players change. **b**, one league's centroids imported into another league's players, which does not decline. **c** and **d**, each season and each league fitted from scratch and scored against the pooled partition. **e** to **h**, standard score of the observed two-cluster silhouette against the clusterless null under every feature-family ablation, one panel per scope: circles remove one family, triangles keep one family alone, and the vertical line is the full feature set. The adjusted Rand index against the full partition, which distinguishes the same two modes from a different pair, is in Table 1.

Table 1: Standard score of the observed two-cluster silhouette against the clusterless Gaussian null, by scope, under every feature-family ablation. A dagger marks a cell where the pre-declared rule chose a number of clusters other than two.

Feature set	All	DF	MF	FW
<i>Full feature set</i>				
all 33 features	18.0	24.2	15.0	13.2
<i>Leave one family out</i>				
without shooting	17.7	21.3	12.3	18.0
without creation	21.2	21.1	14.5	15.1
without progression	19.0	19.7	13.2	19.8
without territory	21.6	22.6	16.0	15.6
without defending	19.7	29.1	16.0	17.2
without passing	19.1	17.5	10.2	20.1
<i>One family alone</i>				
shooting only	29.3	19.8	16.1	2.9
creation only	12.8	22.6	11.7	18.4
progression only	8.3	15.6	9.0	12.9
territory only	11.4	26.6	16.2	20.2
defending only	19.0	6.9	10.7	8.5
passing only	6.6	11.4	5.3	1.4

3.5 Do the modes predict results?

Everything above is a claim about the structure of the data. The archive also carries what happened, so the two modes can be tested against real results and not only against listed positions. Each of 490 team-seasons is summarised by the minutes-weighted share of its outfield minutes played by the attacking mode, by the minutes-weighted mean position of its players along the axis between the two centroids, and by its share in each of the six archetypes, with eligible players covering a median 96.9 percent of each club’s outfield minutes and never less than 89.5.

The share alone correlates with points per match at a Spearman rank correlation of 0.430 (0.348 to 0.502) and with goal difference per match at 0.442. The continuous axis position is stronger, at 0.655 (0.594 to 0.708) against points, which is what a continuum should show: where a club’s players sit along the axis carries more than which side of the cut they fall on. Within leagues the share correlation runs from 0.19 in La Liga to 0.61 in Premier League, and every season lies inside that range. Table 2 lists the ten team-seasons at the extremes so a reader can check them against memory, and they are the sides one would name.

A correlation between attacking composition and points is partly circular, because good teams have good attackers, so the test that matters is whether the modes carry anything beyond team possession, the simplest summary of dominance. Ridge regression of points per match on possession alone, on the composition alone and on both gives out-of-sample coefficients of determination of 0.566, 0.545 and 0.664 when one season is held out at a time, and 0.557, 0.534 and 0.656 when one league is. The strictest holdout leaves one club out at a time, so that no held-out team-season shares a club with the training data and a model cannot score by remembering which clubs are good; there the three figures are 0.558, 0.546 and 0.661. The modes add 0.099 (0.067 to 0.131) to what possession explains under the first holdout, 0.098 under the second and 0.102 under the club holdout. The composition on its own predicts slightly worse than possession on its own, which should be said plainly: the modes are not a substitute for dominance, they are information about how it is

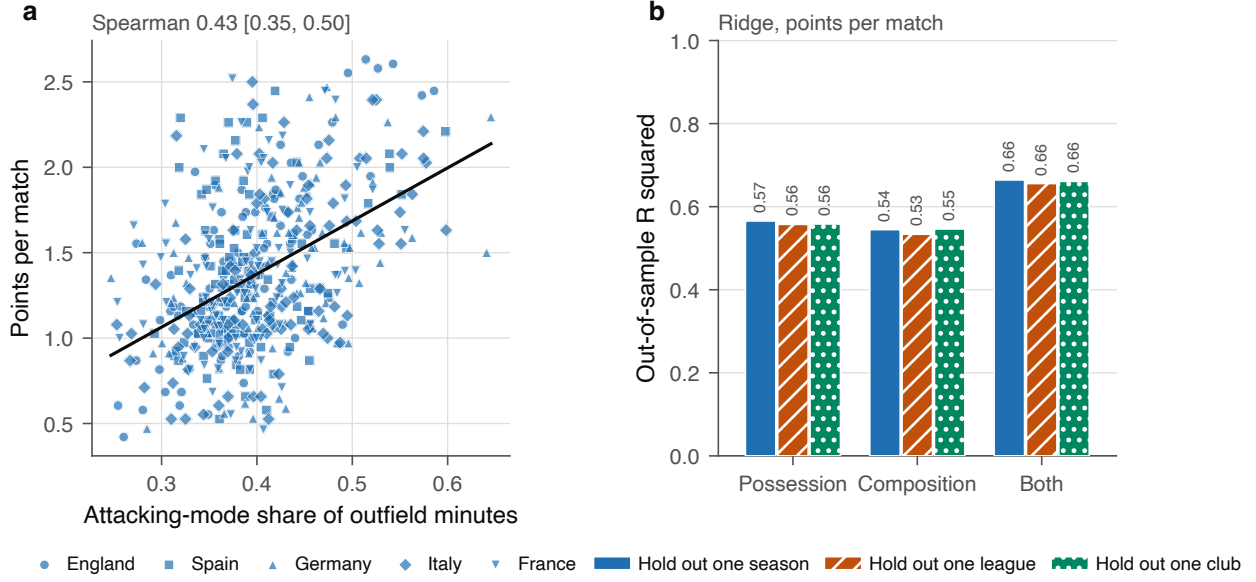


Figure 5: The two modes against real results over 490 team-seasons. **a**, the minutes-weighted share of outfield minutes in the attacking mode against points per match, one marker shape per league. **b**, out-of-sample coefficient of determination for points per match from possession alone, the composition alone and both, under leave-one-season-out, leave-one-league-out and leave-one-club-out holdouts.

Table 2: The five team-seasons with the highest and the five with the lowest minutes-weighted share of outfield minutes in the attacking mode, with their real points per match and final league position.

Club	League	Season	Share	Points per match	Position
<i>Highest attacking-mode share</i>					
Bayern Munich	Bundesliga	2019	0.645	2.29	1
Hoffenheim	Bundesliga	2019	0.641	1.50	9
Fiorentina	Serie A	2022	0.598	1.63	7
Real Madrid	La Liga	2021	0.598	2.21	2
Manchester City	Premier League	2022	0.586	2.45	1
<i>Lowest attacking-mode share</i>					
Mainz 05	Bundesliga	2022	0.247	1.35	8
Parma	Serie A	2019	0.253	1.08	16
Reims	Ligue 1	2022	0.254	1.21	12
Sheffield Utd	Premier League	2021	0.254	0.60	20
Troyes	Ligue 1	2022	0.256	1.00	15

produced that dominance does not contain.

Where the information lives is the sharpest result in this section. Possession plus the two two-mode measures alone reach 0.660, an increment of 0.094 over possession that is the whole of the gain. Adding the six archetype shares on top adds 0.005 (-0.006 to 0.015), an interval that contains zero, and 0.002 under the club holdout, while the archetypes without the two-mode measures reach only 0.579. Real results say what the null calibration said: two modes carry information and a finer taxonomy carries none that the two modes did not already carry. Figure 5 shows the share against points and the out of sample comparison.

3.6 Goalkeepers

Goalkeepers are described by a different vocabulary from outfield players, and the primary sample supplies all of it, with 1,039 keeper-seasons carrying post-shot expected goals, sweeper actions and their average distance, cross stopping, launch rate and average pass length, at a worst per-column coverage of 0.962.

The first attempt at a partition walks into a trap, and the trap is itself a finding. Clustering on all available features returns 2 clusters at a silhouette of 0.207, and the partition looks interpretable until it is asked what it is made of. Membership explains a mean of 0.386 of the variance in situational quantities, the volume and quality of shooting a defence permits, against 0.109 in quantities describing what the goalkeeper does. The partition is largely a ranking of the teams in front of the goalkeepers. Named as archetypes it would be a league table wearing a tactical vocabulary. Excluding the exposure variables and clustering on technique alone reverses that completely. The same rule again returns 2 clusters, at a silhouette of 0.189 and a bootstrap adjusted Rand index of 0.570, and membership now explains 0.135 of the variance in technique against 0.019 in situation, a ratio of about seven.

What that partition describes is not the expected taxonomy. The familiar trio of shot stopper, sweeper keeper and distributor is not what the data produces. The division is between Line keepers, 411 keeper-seasons who stay on their line, and Sweeper claimers, 282 who come off it, separating on cross stopping and sweeping together. Distribution style barely separates the clusters at all, and neither does shot stopping ability. Both are real and measurable, and both are continua lying across the off-line division rather than dimensions of it. A goalkeeper who sweeps is not thereby a better or worse shot stopper, and how far he kicks the ball is close to independent of both.

The results test discriminates between the two partitions in the direction the effect sizes predict. The share of a club's goalkeeper minutes in the full-feature partition's second cluster correlates with points per match at 0.713 (0.667 to 0.753), which is what a partition sorting keepers by the team in front of them must do. The technique partition's share correlates at 0.087 (0.002 to 0.174), an interval that barely clears zero, so the line keeper against sweeper claimer division is unrelated to how good the team is, which is the property a description of goalkeeping rather than of goalkeepers' teams should have. Figure 6 shows the technique partition and the two profiles. That is a substantive finding rather than a taxonomy, and it echoes the outfield result: the data support one binary distinction plus continuous variation, not a set of types.

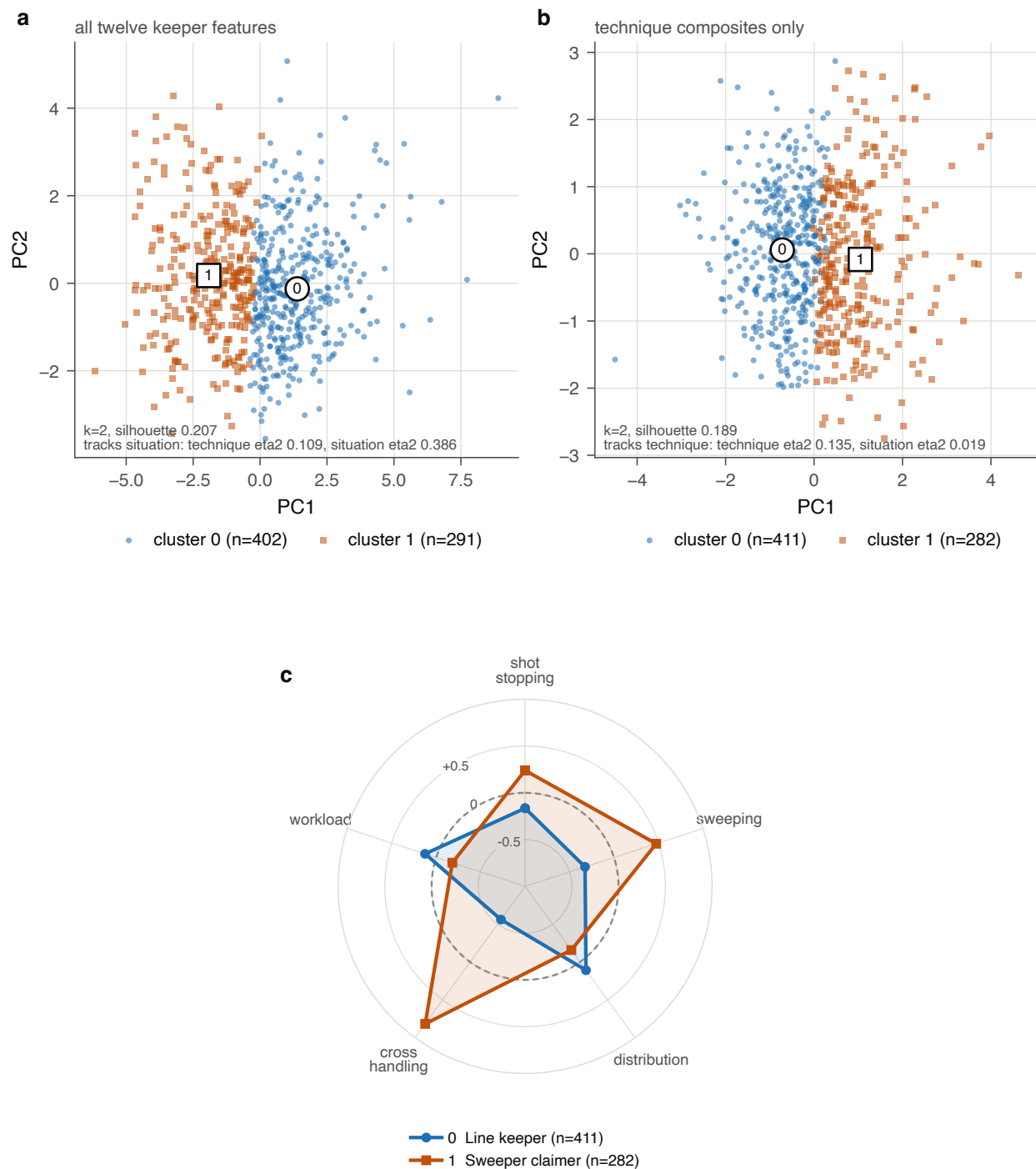


Figure 6: Goalkeeper structure on the primary sample. **a** and **b**, the technique partition, which separates goalkeepers who stay on their line from those who come off it. **c**, the two cluster centroids on the five composite axes, in standard deviations from the eligible pool mean, with the dashed ring marking that mean and the distribution axis signed so that a higher value means shorter passing. The clusters separate on cross handling and sweeping together, while distribution and shot stopping vary continuously across both rather than distinguishing them, which is why the familiar trio of shot stopper, sweeper keeper and distributor is not what the data returns.

3.7 Robustness

Four checks bound what the result depends on, and each is reported in full in the supplementary material.

The result does not depend on the richness of the description. The reduced sample of Section 2.1 provides an unusually clean test, because the withdrawal removed a specific and large part of the feature set without touching the population. Repeating the analysis there returns the same answer: the declared rule selects two clusters globally and within every position group, agreement with listed positions remains weak at 0.226, and density-based clustering again assigns every position group entirely to noise. Removing expected goals, all passing and possession volume, shot and goal creating actions and the entire advanced goalkeeping table changes what the two modes are made of without changing that there are two of them, which bounds what richer data buys: the modern feature block sharpens the description considerably but reveals no additional modes that the poorer description was hiding.

The result does not depend on the eligibility threshold either. Repeating the entire selection and stability procedure at 270, 450 and 900 minutes, treating each threshold as an independent pool, returns two clusters at every threshold with the silhouette moving only from 0.286 to 0.302, and the partition itself agreeing with the one reported here at 0.967 and 0.986 on the common players (Section S3).

The possession adjustment behind the defensive features is measured rather than assumed, and the measurement matters. Four of the five defensive elasticities have bootstrap intervals excluding one, so the conventional proportional correction overcorrects them; for clearances the interval reaches one and the paper does not claim otherwise. Applying the conventional exponent reverses the sign of the residual correlation with team possession rather than removing it. The correction is also one sided, and that asymmetry is tested rather than asserted: attacking output responds to a team’s own possession more than proportionately, so dividing it out would remove signal along with any bias. Section S2 gives the estimates, the intervals, the league refits and the counterfactual in which both sides are corrected.

Finally, the partitions are not artefacts of sampling noise in per-90 rates computed over partial seasons. Empirical Bayes shrinkage toward the position group mean, with the degree of shrinkage set by minutes played, leaves the global partition essentially unchanged, while the within-group splits survive unequally; Section S8 reports which, and reports the within-season role drift analysis that bounds the assumption that a player’s role is constant across a season.

4 Discussion and conclusion

The result is easy to overstate in either direction. It does not say that football has two kinds of player, and it does not say that role is a single number. It says that the distribution of players over 33 measured statistics has two regions of concentrated density joined by a populated middle, that the axis running between them is the dominant direction in the space, and that no division finer than that one is distinguishable from what a partitioning algorithm produces on data containing no divisions. That is a description a mixture representation expresses well and a taxonomy does not, which is why the archetypal and partial membership approaches to athlete description are better matched to this data than the genre that dominates it [8, 23].

The axis itself is territorial rather than technical. Its strongest loadings are touches by pitch third and progressive receptions, not passing accuracy or duel success, and the same ordering appears in the supervised attributions, where Touches def third carries more weight than any technical statistic for all three position groups. Football’s own vocabulary for role mixes territory with technique freely,

describing a player as a ball-playing centre back or a pressing forward, but the season aggregate record is far better at recording where a player was than what he did there. Any role structure recovered from it will be mostly a statement about territory, and reading it as a statement about ability is the most common way to misuse this kind of analysis. That the discovered division cuts across the listed positions rather than reproducing them is consistent with what tracking and event data show about modern roles, where positional assignment is fluid within a match and formations are recovered as distributions rather than as slots [4, 5].

4.1 Why finer taxonomies keep being published

The methodological point generalises past football. A published role taxonomy is typically supported by three things: a selection criterion that picks the reported number of clusters, a stability check that shows the partition survives resampling, and centroid profiles that read sensibly against domain knowledge. Every one of those is satisfied here by partitions this paper shows to be artefacts.

None of the three tests existence. A selection criterion compares candidate partitions against each other and is never offered the option of returning none, which is the limitation the stopping-rule literature has documented since it began [12, 13]. A bootstrap measures whether an algorithm applied twice to similar data behaves similarly, which a deterministic procedure on a smooth unimodal distribution does very reliably, exactly as the theory of stability-based validation predicts [15, 16]. Interpretability is the weakest of the three, because a cut through any continuous distribution produces two centroids that differ along the direction of the cut, and a domain expert can narrate that difference whether or not it corresponds to a real boundary. The evidence here makes the concern measured rather than hypothetical: on data built to contain no groups, the same pipeline returns a silhouette maximum at two clusters, two nameable centroids, and a bootstrap adjusted Rand index of at least 0.911 on every clusterless dataset in every scope, against 0.962 on the real data, a difference as small as 0.003. A stability check that passes a continuum at 0.911 is not a test of existence, and a taxonomy that rests on one rests on evidence a continuum satisfies equally well. Given that 55 percent of sports science clustering studies do not state how the number of clusters was chosen at all, the gap between the applied genre and the validity literature is wide [11].

Calibration against matched clusterless data closes it at low cost. It requires no assumption the analysis does not already make, since the simulated data are drawn using the observed mean and covariance; it costs a few dozen refits; and it is the single step that separates the two-mode result reported here from the six-way taxonomy the same pipeline would have produced with the same apparent confidence. The gap statistic already does a version of this against a uniform reference [14], and what this paper adds is not a new statistic but harder references and a reporting convention, that the observed value be placed against all of them so a reader can see which the claim survives. Hartigan’s dip test and a density-based method permitted to answer that nothing is clustered are useful supplements [18, 19, 32], and the agreement of all three is what makes the negative half of the result credible. The position is the one Hennig argues for, that a partition should be judged by whether it has the properties the analyst needs rather than by whether it is correct [20]; the property asked for here is that a partition be distinguishable from one imposed on a continuum.

The comparison with the closest prior work is instructive. Akhanli and Hennig’s aggregated validity index, applied to a season of player statistics from eight European leagues, returned a coarse partition into five ordered groups running along the same territorial axis found here [21, 22]. Two independent calibrations, one against random partitions of the observed data and one against clusterless data matched to it, agree that the recoverable structure in season aggregates is coarse and ordered. The taxonomies of a dozen named types that the applied genre reports are not what either calibration supports [1–3, 9].

4.2 What the modes mean in practice

The two modes are not the listed positions renamed, and where they disagree is the interesting part. The division leaves the two extreme groups intact and cuts the group between them close to in half, and the players it reassigns are recognisable: holding midfielders sit with the defenders and attacking midfielders with the forwards. That the supervised model makes its errors in the same place is corroboration from machinery sharing nothing with the clustering. The practical reading is that listed position is a reasonable but coarse summary at the extremes and close to arbitrary in the middle, which is what anyone who has watched a modern midfield would expect and which the data now support quantitatively.

The test against real outcomes closes the gap between describing the statistics and describing the sport, and it closes it in two directions at once. The two-mode composition of a club carries information about its points that possession does not, out of sample under three different holdouts, so the axis the partition cuts is one along which football is actually decided. The six archetypes carry no information about points that the two modes did not already carry, so the finer taxonomy fails against results exactly as it failed against the clusterless null. Those are independent tests, one internal and one external, and they agree. Two cautions belong with that. The composition alone predicts slightly worse than possession alone, so the increment is the right thing to cite. The within-league correlation also varies, from 0.19 in La Liga to 0.61 in Premier League, a reminder that the same composition buys different amounts in different competitions.

For recruitment the honest use of this work is narrow and real. The involvement axis is stable enough to support questions about squad composition, such as whether a club’s minutes are concentrated at one end of the distribution. What the work does not support is a scouting taxonomy, since a shortlist built by matching a target’s archetype label matches a label the data do not establish, and a similarity search over the continuous profile is the better instrument for the same question because it does not force a boundary that is not there. For anyone doing quantitative football analysis the two directly transferable items are the possession elasticities, which apply to any defensive count comparison and correct a convention that has stood since it was proposed without being measured [31], and the null calibration, which costs little and should be run before any archetype claim is made.

The goalkeeper analysis carries its own caution. The division that emerges runs along what a goalkeeper is asked to do rather than how well he does it, and a partition of goalkeepers built from the full public feature block is mostly a partition of the teams in front of them, with a situational effect size of 0.386 against 0.109 for technique. Anyone building goalkeeper archetypes from public data will produce labels that describe a role assignment made by a coach rather than a property of the goalkeeper, and it is easy to present the former as the latter. Conditioning on shot difficulty through post-shot expected goals rather than on shot volume is the minimum requirement for saying anything about shot stopping at all [37].

4.3 Limitations and future work

Season aggregates discard context. A tackle count does not say where the tackles happened, an expected goals total does not distinguish six poor chances from two good ones, and a touch map summarised into three thirds is a coarse instrument for a question that turns out to be territorial. Everything here is a claim about what season aggregate data can support, not about what football contains, and the two should not be conflated: it remains entirely possible that finer roles exist and that this description cannot see them. Event and tracking data would be the way to find out [6, 7].

The confound that possession adjustment does not remove is team strength. Correcting the

defensive counts removes the exposure effect from those five features, but attacking statistics still scale with a team’s own share of the ball, so the leading component remains related to team quality. The unit of analysis is the player-season, and player-seasons are nested, with the same player appearing in up to 5 rows and clubs nested within leagues. The pooled fits and the bootstrap treat rows as exchangeable, which they are not, so pooled intervals are narrower than a fully nested analysis would give. The defence against this is design rather than adjustment: every central claim is also made on independent refits by season and by league, and the test against results holds out whole seasons, whole leagues and whole clubs. The archive is a mirror rather than the source, and although its coverage was audited it cannot be re-derived from the live site, and one season was excluded outright because a whole feature block was absent from it. Season aggregation also assumes a player’s role is constant across a season, which the drift analysis of Section S8 shows holds for most players and fails for a minority.

Three directions follow. Event and tracking data would restore the spatial detail that season aggregates discard and would test the two-mode claim against a description rich enough to reveal finer structure if it exists. The most interesting version of that test is the one this paper cannot run, whether the modes are a property of football or a property of the measurement, which requires a dataset in which a finer taxonomy is known to be present. Extending beyond the big five, to competitions with different possession profiles, would test whether the elasticities and the two-mode result hold where the league is less concentrated.

The question this paper set out to answer is whether football player roles, as season aggregate statistics describe them, form types or a continuum. They form neither cleanly. The useful description sits between the two: an ordered space with one dominant axis, density concentrated at two ends, and a populated middle that no partition divides well. Two modes are real, they reproduce in every season and every competition, they predict what happened on the pitch, and nothing finer than them survives a test that structureless data would fail. What follows for practice is a caution about method rather than about football. A silhouette maximum, a high bootstrap adjusted Rand index and an interpretable centroid are all obtainable from data containing no groups whatsoever, because the algorithm supplies the groups.

Author contributions

C.C. conceived the research question. Z.C. designed the study, carried out the analysis, and wrote the paper. Both authors approved the final version.

Data and code availability

The analysis code, the retrieval step that fetches the archive from its stated upstream and records a digest per file, the cluster assignments for both samples and the sources of this document are available at <https://github.com/ZacharyTChung/pl-player-profiling>. The primary sample is the public worldfootballR mirror of FBref described in Section 2.1, and the reduced sample was retrieved from FBref and Understat as described there.

Acknowledgements

The statistics analysed here originate with FBref and were retrieved from the public mirror maintained by the worldfootballR project [26]. The analysis code and the manuscript were developed with

the assistance of an AI coding assistant, Claude, under the direction of the authors, who take full responsibility for the content.

References

- [1] Soumaya El Foukahi, Kaoutar El Handri, Mohamed El Ghmary, and Hassan Echoukairi. Unveiling midfielder roles in elite football: A data-driven clustering approach. In *Artificial Intelligence and Mathematics*, Springer Proceedings in Mathematics & Statistics, pages 465–477. Springer, 2026. doi: 10.1007/978-3-032-18042-1_37.
- [2] Samuel Kalman and Jonathan Bosch. NBA lineup analysis on clustered player tendencies: A new approach to the positions of basketball and modeling lineup efficiency of soft lineup aggregates. In *MIT Sloan Sports Analytics Conference*, 2020. URL <https://www.sloansportsconference.com/research-papers/nba-lineup-analysis-on-clustered-player-tendencies-a-new-approach-to-the-positions-of-basketball-modeling-lineup-efficiency>.
- [3] Muthu Alagappan. From 5 to 13: Redefining the positions in basketball. In *MIT Sloan Sports Analytics Conference*, 2012. URL <https://www.sloansportsconference.com/research-papers/from-5-to-13-redefining-the-positions-in-basketball>.
- [4] Alina Bialkowski, Patrick Lucey, Peter Carr, Yisong Yue, Sridha Sridharan, and Iain Matthews. Large-scale analysis of soccer matches using spatiotemporal tracking data. In *Proceedings of the 2014 IEEE International Conference on Data Mining (ICDM)*, pages 725–730, 2014. doi: 10.1109/ICDM.2014.133. URL <https://doi.org/10.1109/ICDM.2014.133>.
- [5] Tom Decroos and Jesse Davis. Player vectors: Characterizing soccer players’ playing style from match event streams. In *Machine Learning and Knowledge Discovery in Databases: ECML PKDD 2019*, Lecture Notes in Computer Science, pages 569–584. Springer, 2020. doi: 10.1007/978-3-030-46133-1_34. URL https://doi.org/10.1007/978-3-030-46133-1_34.
- [6] Luca Pappalardo, Paolo Cintia, Paolo Ferragina, Emanuele Massucco, Dino Pedreschi, and Fosca Giannotti. PlayeRank: Data-driven performance evaluation and player ranking in soccer via a machine learning approach. *ACM Transactions on Intelligent Systems and Technology*, 10(5), 2019. doi: 10.1145/3343172. URL <https://doi.org/10.1145/3343172>. arXiv:1802.04987.
- [7] Bart Aalbers and Jan Van Haaren. Distinguishing between roles of football players in play-by-play match event data, 2018. URL <https://arxiv.org/abs/1809.05173>.
- [8] Emiliano Seri, Roberto Rocci, and Thomas Brendan Murphy. Partial membership models for soft clustering of multivariate football player performance data, 2024. URL <https://arxiv.org/abs/2409.01874>.
- [9] Spyridon Plakias, Serafeim Moustakidis, Christos Kokkotis, Marina Papalexi, Themistoklis Tsatalas, Giannis Giakas, and Dimitrios Tsaopoulos. Identifying soccer players’ playing styles: A systematic review. *Journal of Functional Morphology and Kinesiology*, 8(3):104, 2023. doi: 10.3390/jfmk8030104.
- [10] Linda Gu, Wenlong Wang, Spyridon Plakias, and Shaoliang Zhang. Playing style identification in team sports: A systematic review from statistical dimensionality reduction to unsupervised machine learning. *International Journal of Sports Science & Coaching*, 20(6):2742–2761, 2025. doi: 10.1177/17479541251372586.

- [11] Daniel Fernández, Martí Casals, Martí Oliver, Montse Plensa, and Marica Manisera. Reporting of clustering techniques in sports sciences: a scoping review. *Electronic Journal of Applied Statistical Analysis*, 17(3):653–675, 2024. doi: 10.1285/i20705948v17n3p653.
- [12] Glenn W. Milligan and Martha C. Cooper. An examination of procedures for determining the number of clusters in a data set. *Psychometrika*, 50(2):159–179, 1985. doi: 10.1007/BF02294245.
- [13] Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7. URL [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7).
- [14] Robert Tibshirani, Guenther Walther, and Trevor Hastie. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(2):411–423, 2001. doi: 10.1111/1467-9868.00293. URL <https://doi.org/10.1111/1467-9868.00293>.
- [15] Asa Ben-Hur, André Elisseeff, and Isabelle Guyon. A stability based method for discovering structure in clustered data. In *Pacific Symposium on Biocomputing*, volume 7, pages 6–17, 2002. URL <https://psb.stanford.edu/psb-online/proceedings/psb02/benhur.pdf>.
- [16] Ulrike von Luxburg. Clustering stability: An overview. *Foundations and Trends in Machine Learning*, 2(3):235–274, 2010. doi: 10.1561/22000000008.
- [17] Margareta Ackerman and Shai Ben-David. Clusterability: A theoretical study. In *Proceedings of the Twelfth International Conference on Artificial Intelligence and Statistics (AISTATS 2009)*, volume 5 of *Proceedings of Machine Learning Research*, pages 1–8, 2009. URL <https://proceedings.mlr.press/v5/ackerman09a.html>.
- [18] Andreas Adolffson, Margareta Ackerman, and Naomi C. Brownstein. To cluster, or not to cluster: An analysis of clusterability methods. *Pattern Recognition*, 88:13–26, 2019. doi: 10.1016/j.patcog.2018.10.026.
- [19] J. A. Hartigan and P. M. Hartigan. The dip test of unimodality. *The Annals of Statistics*, 13(1):70–84, 1985. doi: 10.1214/aos/1176346577.
- [20] Christian Hennig. What are the true clusters? *Pattern Recognition Letters*, 64:53–62, 2015. doi: 10.1016/j.patrec.2015.04.009.
- [21] Serhat Emre Akhanli and Christian Hennig. Comparing clusterings and numbers of clusters by aggregation of calibrated clustering validity indexes. *Statistics and Computing*, 30(5):1523–1544, 2020. doi: 10.1007/s11222-020-09958-2.
- [22] Serhat Emre Akhanli and Christian Hennig. Clustering of football players based on performance data and aggregated clustering validity indexes. *Journal of Quantitative Analysis in Sports*, 19(2):103–123, 2023. doi: 10.1515/jqas-2022-0037.
- [23] Manuel J. A. Eugster. Performance profiles based on archetypal athletes. *International Journal of Performance Analysis in Sport*, 12(1):166–187, 2012. doi: 10.1080/24748668.2012.11868592.
- [24] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard

- Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12: 2825–2830, 2011. URL <https://jmlr.org/papers/volume12/pedregosa11a/pedregosa11a.pdf>.
- [25] Sports Reference LLC. FBref.com: Football statistics and history. <https://fbref.com/en/>, 2026. Advanced statistics were licensed from a commercial event-data provider, StatsBomb until 2022 and Opta thereafter; the provider-derived columns were withdrawn on 20 January 2026.
 - [26] Jason Zivkovic. worldfootballR: Extract and clean world football (soccer) data. R package and public data mirror, 2026. URL https://github.com/JaseZiv/worldfootballR_data. Archived FBref advanced season tables retrieved 7 September 2026.
 - [27] Harm Eggels, Ruud van Elk, and Mykola Pechenizkiy. Explaining soccer match outcomes with goal scoring opportunities predictive analytics. In *Proceedings of the 3rd Workshop on Machine Learning and Data Mining for Sports Analytics (MLSA 2016), co-located with ECML/PKDD 2016*, volume 1842 of *CEUR Workshop Proceedings*, 2016. URL https://ceur-ws.org/Vol-1842/paper_07.pdf.
 - [28] Alex Rathke. An examination of expected goals and shot efficiency in soccer. *Journal of Human Sport and Exercise*, 12(Proc2), 2017. doi: 10.14198/jhse.2017.12.proc2.05. URL <https://doi.org/10.14198/jhse.2017.12.proc2.05>.
 - [29] Pieter Robberechts. soccerdata: A collection of scrapers for gathering soccer data. Python package, version 1.9.1, 2026. URL <https://github.com/probberechts/soccerdata>.
 - [30] Understat. Understat: Expected goals statistics. <https://understat.com>, 2026. Player and team expected goals for the seasons ending 2025 and 2026, retrieved 2026.
 - [31] Ted Knutson. Introducing possession-adjusted player stats. StatsBomb, June 2014. URL <https://blogarchive.statsbomb.com/articles/soccer/introducing-possession-adjusted-player-stats/>. Published 23 June 2014.
 - [32] Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*, volume 7819 of *Lecture Notes in Computer Science*, pages 160–172. Springer, 2013. doi: 10.1007/978-3-642-37456-2_14. URL https://doi.org/10.1007/978-3-642-37456-2_14.
 - [33] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. doi: 10.1007/BF01908075. URL <https://doi.org/10.1007/BF01908075>.
 - [34] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining (ICDM)*, pages 413–422. IEEE, 2008. doi: 10.1109/ICDM.2008.17. URL <https://doi.org/10.1109/ICDM.2008.17>.
 - [35] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
 - [36] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html. arXiv:1705.07874.

- [37] Dinesh Vatvani. Upgrading Expected Goals. StatsBomb Blog, 2022. URL <https://blogarchive.statsbomb.com/articles/soccer/upgrading-expected-goals/>.

Supplementary Information

S1 Feature dictionaries and data availability

The two samples are described by different feature sets drawn from different pulls, so each has its own dictionary. Table S1 lists the 33 features of the primary sample with the FBref table each column was read from, and Table S2 lists the 18 features of the reduced sample, which combines what survived the withdrawal with an independent expected goals source. In both, counting statistics are expressed per ninety minutes played and rate features are used as published rather than divided a second time.

Every feature in the reduced set was confirmed to hold non-null values in both of its seasons before being admitted, which is the check that distinguishes a live column from one whose values were withdrawn. Table S3 reports that audit. The primary set is audited differently, by per-column and per-season coverage on load, because its failure mode is a block missing from one season rather than a column emptied across all of them.

Table S1: The primary feature set, with the FBref table each column was read from. Counting statistics are per ninety minutes played; the seven rate features are used as published. Five defensive counts are possession adjusted, as described in Section 2.2.

Feature	FBref table	Definition
np xg p90	standard	Non-penalty expected goals (per 90)
shots p90	shooting	Shots taken, excluding penalties (per 90)
goals non penalty p90	standard	Goals excluding penalties (per 90)
xag p90	standard	Expected assisted goals, the expected goals of shots a pass created (per 90)
key passes p90	passing	Passes leading directly to a shot (per 90)
sca p90	gca	Shot creating actions, the two offensive actions before a shot (per 90)
gca p90	gca	Goal creating actions, the two offensive actions before a goal (per 90)
passes final third p90	passing	Completed passes into the final third (per 90)
passes penalty area p90	passing	Completed passes into the penalty area (per 90)
crosses p90	misc	Crosses attempted (per 90)
progressive passes p90	passing	Completed passes moving the ball substantially toward goal (per 90)
progressive carries p90	possession	Carries moving the ball substantially toward goal (per 90)
progressive receptions p90	possession	Passes received in a progressive position (per 90)
carries final third p90	possession	Carries into the final third (per 90)
touches def pen p90	possession	Touches in the defending team's own penalty area (per 90)
touches def third p90	possession	Touches in the defensive third (per 90)
touches mid third p90	possession	Touches in the middle third (per 90)
touches att third p90	possession	Touches in the attacking third (per 90)
touches att pen p90	possession	Touches in the opposition penalty area (per 90)
take ons p90	possession	Take-ons attempted (per 90)
tackles padj p90	defense	Tackles, possession adjusted (per 90)
interceptions padj p90	misc	Interceptions, possession adjusted (per 90)
blocks padj p90	defense	Blocks of a pass or shot, possession adjusted (per 90)
clearances padj p90	defense	Clearances, possession adjusted (per 90)
ball recoveries p90	misc	Loose balls recovered (per 90)
fouls committed padj p90	misc	Fouls committed, possession adjusted (per 90)
shot accuracy pct	shooting	Shots on target as a share of shots
take on success pct	possession	Take-ons completed as a share of those attempted
tackle win pct	defense	Duels won as a share of those contested
aerials won pct	misc	Aerial duels won as a share of those contested
pass cmp short pct	passing	Short passes completed as a share of those attempted
pass cmp medium pct	passing	Medium passes completed as a share of those attempted
pass cmp long pct	passing	Long passes completed as a share of those attempted

Table S2: The outfield feature set. Counting statistics are per ninety minutes. The two rate features are used as published.

Feature	Source	Definition
np_xg_p90	Understat	Non-penalty expected goals
xa_p90	Understat	Expected assists
key_passes_p90	Understat	Passes leading directly to a shot
xg_chain_p90	Understat	Expected goals of possessions the player was involved in
xg_buildup_p90	Understat	As xGChain, excluding shots and key passes
shots_p90	FBref	Shots taken, excluding penalties
shots_on_target_p90	FBref	Shots on target
goals_non_penalty_p90	FBref	Goals excluding penalties
assists_p90	FBref	Assists
crosses_p90	FBref	Crosses attempted
interceptions_padj_p90	FBref	Interceptions, possession adjusted
tackles_won_padj_p90	FBref	Tackles won, possession adjusted
fouls_committed_padj_p90	FBref	Fouls committed, possession adjusted
fouls_drawn_p90	FBref	Fouls drawn
offsides_p90	FBref	Times caught offside
cards_yellow_p90	FBref	Yellow cards
shot_accuracy_pct	FBref	Shots on target as a share of shots, a rate
goals_per_shot	FBref	Goals per shot, a rate

Table S3: Populated columns in each FBref player table. The tables are still served with complete headers, so the empty counts are the number of columns that resolve correctly and contain no values.

Table	2024-25		2025-26	
	Columns	Empty	Columns	Empty
standard	17	0	17	0
shooting	11	0	11	0
passing	21	19	21	19
passing types	16	14	16	14
goal shot creation	17	16	17	16
defense	17	14	17	14
possession	21	20	21	20
playing time	17	0	17	0
misc	13	2	13	2
keeper	19	0	19	0
keeper adv	26	23	26	23

S2 Possession elasticities

Before any structure is estimated, the defensive counts have to be separated from the opportunity to record them. Team defensive volume does rise as opponent possession rises, so the confound is real, and the conventional correction, in the proportional form that has succeeded the original of Knutson [31], assumes the relationship is proportional: a side with twice the ball is taken to concede half the defensive events.

It is much weaker than that for four of the five statistics, and it is not distinguishable from proportional for the fifth. Fitted on the primary sample over 490 team-seasons, with 2,000 bootstrap resamples giving each estimate an interval, the elasticities are 0.172 for tackles (0.109 to 0.235), 0.353 for interceptions (0.171 to 0.540), 0.439 for blocks (0.379 to 0.499) and 0.443 for fouls committed (0.359 to 0.526). All four intervals exclude one, and no bootstrap resample of any of them reached it. Clearances are different: the estimate is 0.958, the interval runs from 0.823 to 1.091, 28.2 percent of resamples land at or above one, and the bootstrap probability that the true elasticity is at least one is 0.2975. For clearances the conventional exponent cannot be rejected, and the paper does not claim otherwise.

That pattern fits the football. A clearance is the most purely reactive action in the set, made when the ball arrives rather than sought, so it is the statistic for which the exposure story should hold best, and it is the one where proportionality survives. A tackle is the most volitional, and it is furthest from proportional. The spread from 0.172 to 0.958 is itself the point: a single exponent applied to every defensive statistic is wrong for four of them.

The reduced sample supplies the check the primary sample cannot, because it retains the team-level possession series needed to measure what is left after correcting. Across its 40 team-seasons, team possession ranges from 40.6 to 61.3 percent, and its fitted elasticities agree with the primary sample in falling well below one, at 0.525 for interceptions, 0.432 for tackles won and 0.283 for fouls committed. Table S4 shows what each choice of exponent leaves behind. The three raw correlations with team possession are -0.122, -0.111 and -0.102, all negative, since a player whose team keeps the ball records fewer defensive actions. Applying the conventional exponent overshoots every one of them, giving 0.070, 0.105 and 0.129, reversing sign in all three cases and, for fouls committed, ending further from zero than where it started. The correction passes through the confound and out the other side. Adjusting with the measured elasticity instead leaves -0.022, -0.018 and -0.037, three to four times closer to zero than the conventional exponent achieves.

Two qualifications belong with this. The residual-dependence check runs on one league, whose possession shares are relatively compressed, so the exact residuals should not be transported to a competition with a wider spread without refitting. Refitting the elasticities within each league tests transport directly, and the answer is mixed. For tackles, interceptions, fouls and clearances every league's interval overlaps the pooled one. Blocks do not transport: the league estimates run from 0.263 in La Liga to 0.615 in Premier League, and the intervals for both of those leagues miss the pooled interval entirely. Clearances are also the statistic where leagues disagree most, with 3 of the five league point estimates at or above one, which is a second reason not to claim sub-proportionality for them.

The constant elasticity form is itself an assumption, and for 3 of the five statistics it fails. Adding a quadratic term in log opponent possession improves the fit for tackles, blocks and fouls, with a coefficient whose interval excludes zero in each case, and in each the response flattens toward zero for the sides that see least of the ball. Fouls committed bend most: the implied elasticity is 0.90 at 37.6 percent opponent possession and -0.07 at 58.9 percent. The exponents are therefore averages over the possession range rather than constants. Figure S1 gathers the pooled estimates, their intervals and the league refits, and places both families of statistic either side of the proportional line.

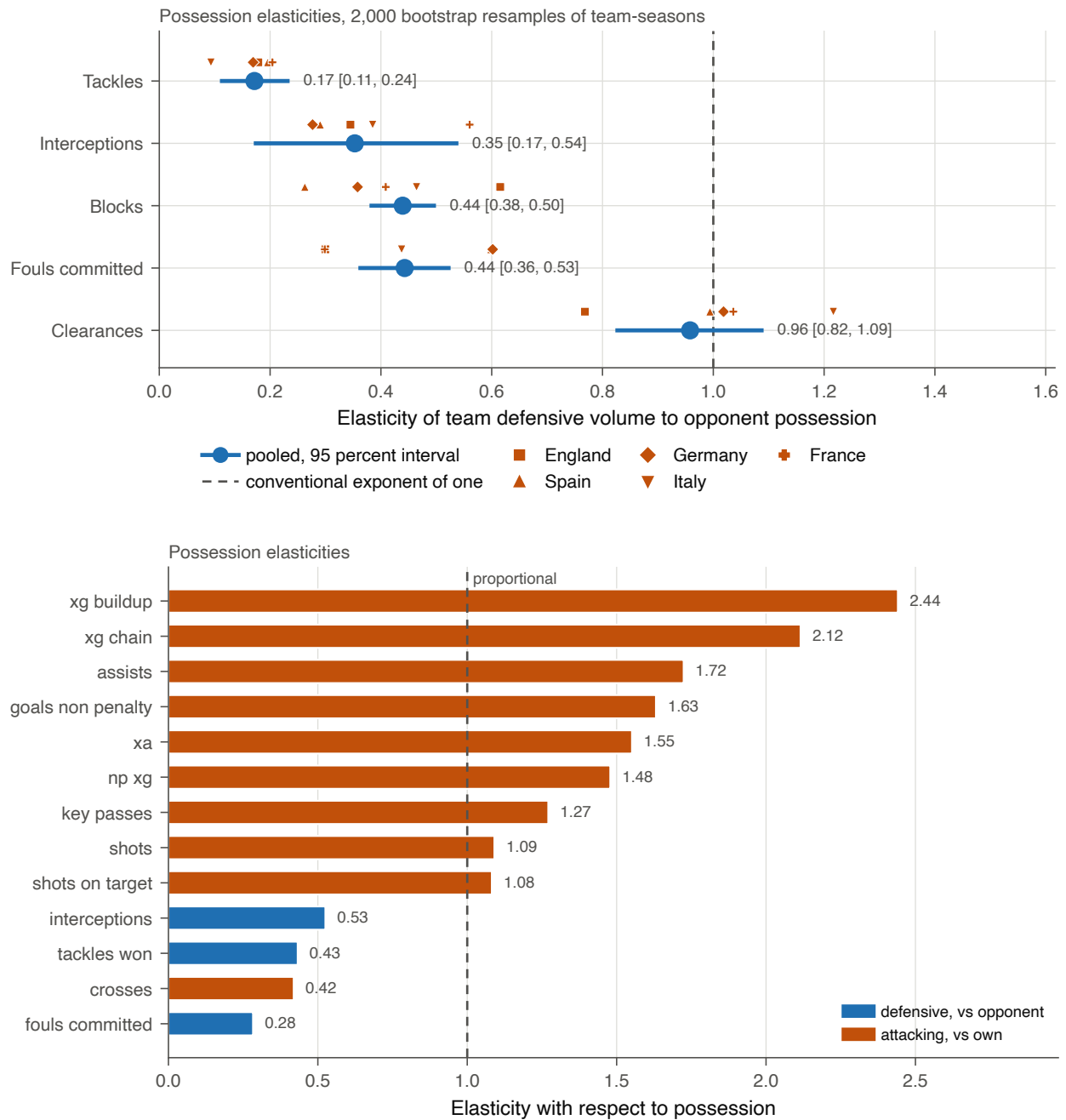


Figure S1: Possession elasticities. Top, the five defensive elasticities with bootstrap intervals from 2,000 resamples of 490 team-seasons, against the conventional value of one, with each league's own refit shown as a smaller marker. Bottom, both families of statistic on the reduced sample: defensive volume responds to opponent possession less than proportionately, while attacking output responds to a team's own possession more than proportionately.

Table S4: Possession adjustment. The elasticity is fitted from team totals. The final three columns give the correlation between each per-90 rate and team possession: unadjusted, adjusted with the fitted elasticity, and adjusted with the conventional exponent of one. A value near zero is the goal.

Statistic	Elasticity	Correlation with team possession		
		Unadjusted	Adjusted	Exponent one
Interceptions	0.525	-0.122	-0.022	0.070
Tackles won	0.432	-0.111	-0.018	0.105
Fouls committed	0.283	-0.102	-0.037	0.129

S2.1 Why the correction is one sided

Correcting only the defensive statistics invites an obvious objection. Attacking output also depends on possession, so the feature set is still team dependent and the correction looks like half a correction. The objection deserves a test rather than an assertion. The test runs on the reduced sample, which is the only one of the two carrying an independent expected-goals source with involvement measures, and its conclusion concerns the shape of the two relationships rather than their exact magnitudes.

The two sides of the ball do not behave alike. Fitted the same way, against a team’s own possession rather than the opponent’s, the attacking elasticities are 1.09 for shots, 1.48 for non-penalty expected goals, 1.63 for non-penalty goals, 1.72 for assists, 2.12 for xGChain and 2.44 for xGBuildup. Every one except crossing, at 0.42, exceeds one, while four of the five defensive elasticities fall well below it and the fifth cannot be distinguished from it.

That asymmetry is the whole argument. For defending, opponent possession is an exposure denominator in the plain sense that a tackle cannot occur while your own side holds the ball, and the sub-proportional response says exposure is what it measures. For attacking, an elasticity above one says the opposite: ten percent more of the ball buys appreciably more than ten percent more output. Returns to possession are increasing, which is a statement about football rather than about measurement, so dividing it out would remove signal along with any bias. The two involvement measures make the point most starkly, since both exceed two.

The counterfactual was nonetheless computed. Applying the fitted exponents to the attacking block as well moves the correlation between club centroids and final league position from -0.861 to -0.610, so a symmetric correction does remove a real part of the team relationship rather than nothing at all. It does not remove all of it, the partition moves at an adjusted Rand index of 0.872 against the one reported here, and the silhouette drifts from 0.285 to 0.281. The choice is a judgement rather than a fact, and it is recorded as one: we treat possession driven productivity as something a role profile should reflect, and a reader who takes the opposite view has the numbers to see what that would cost.

S3 Minutes threshold sensitivity

Both samples exclude outfield players below 450 minutes. That threshold balances two errors, since a low threshold admits players whose per-90 rates are dominated by sampling noise while a high threshold discards genuine squad roles such as the impact substitute. Table S5 repeats the entire selection and stability procedure at 270, 450 and 900 minutes, treating each threshold as an independent pool in which imputation and standardisation are recomputed and the declared rule is reapplied from scratch.

The threshold matters very little. The rule selects two clusters at every threshold, the silhouette

Table S5: Sensitivity of the analysis to the minutes threshold in the earlier of the two reduced-sample seasons. Agreement is the adjusted Rand index against the partition at the threshold actually used, computed on the players common to both pools.

Minutes	Eligible	k	Silhouette	Bootstrap ARI	Agreement
270	403	2	0.286	0.930	0.967
450	366	2	0.286	0.924	baseline
900	296	2	0.302	0.947	0.986

moves only from 0.286 to 0.302, and bootstrap stability is between 0.924 and 0.947 throughout. The partition itself barely moves: measured on the players common to both pools, the adjusted Rand index against the partition at the threshold actually used is 0.967 at 270 minutes and 0.986 at 900. Figure S2 plots the same quantities.

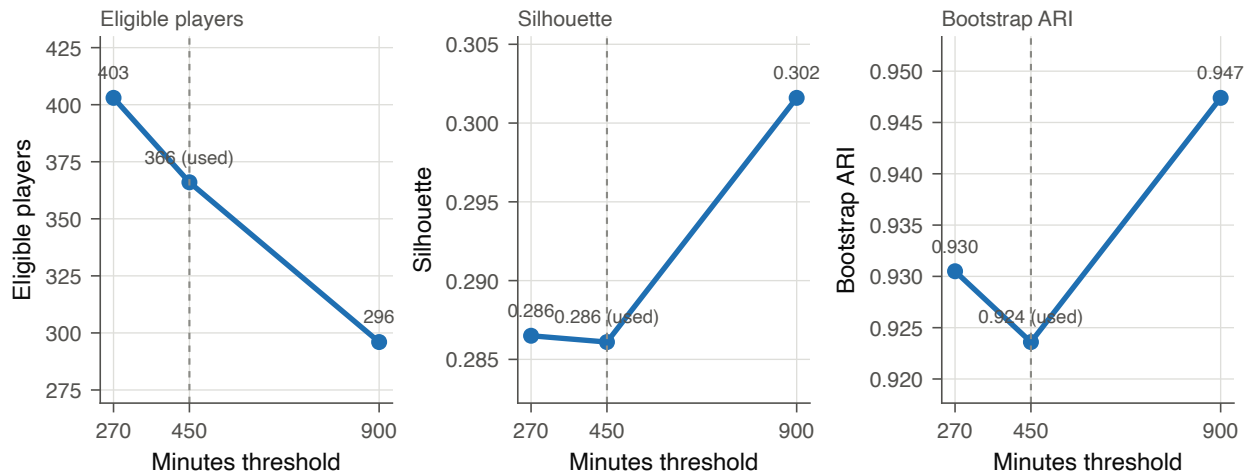


Figure S2: Eligible sample, silhouette and bootstrap stability across the candidate minutes thresholds. The dashed line marks the threshold used in the main analysis.

S4 Selection criteria and embedding choices

Table S6 gives the silhouette coefficient, Davies-Bouldin index, Calinski-Harabasz index and gap statistic for every candidate number of clusters on the reduced sample, for the global analysis and within each position group, with the row the declared rule selects marked in each scope. The corresponding evidence for the primary sample is Figure 1, which carries more information than a criteria table would, because the question the primary sample answers is not which candidate scores best but whether any of them beats what a continuum produces.

Table S7 reports trustworthiness for all twelve configurations of the neighbour count and minimum distance, on the reduced sample, which is the only one for which a manifold embedding is reported. Selection was by trustworthiness alone, decided before any embedding was inspected visually, which matters because a manifold method will produce a visually appealing picture at almost any setting.

Table S6: Cluster selection criteria for every candidate number of clusters, by scope. The consensus rule is applied to these values.

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
global*	2	0.287	1.451	145.6	1.216
global	3	0.215	1.695	101.9	1.283
global	4	0.154	1.954	86.1	1.339
global	5	0.145	1.861	75.4	1.381
global	6	0.144	1.797	68.3	1.415
global	7	0.135	1.905	61.3	1.438
global	8	0.118	1.881	56.1	1.458
global	9	0.123	1.840	53.5	1.477
global	10	0.123	1.805	49.8	1.484
global	11	0.128	1.657	47.6	1.506
global	12	0.121	1.911	44.6	1.517
DF*	2	0.204	2.037	25.8	0.943
DF	3	0.151	1.919	23.5	1.011
DF	4	0.142	1.983	19.7	1.036
DF	5	0.120	1.961	18.0	1.059
DF	6	0.117	1.795	16.9	1.083
DF	7	0.106	1.898	15.6	1.109
DF	8	0.120	1.792	14.7	1.136
DF	9	0.120	1.754	14.1	1.154
DF	10	0.112	1.636	13.5	1.159
DF	11	0.098	1.636	12.8	1.162
DF	12	0.105	1.627	12.3	1.196
MF*	2	0.243	1.543	69.4	1.035
MF	3	0.147	1.969	49.0	1.091
MF	4	0.145	1.895	40.4	1.130
MF	5	0.145	1.873	34.7	1.153
MF	6	0.132	1.939	30.3	1.167
MF	7	0.122	2.054	27.4	1.199
MF	8	0.107	2.030	24.9	1.202
MF	9	0.103	1.923	23.8	1.212
MF	10	0.106	1.983	21.9	1.220
MF	11	0.094	1.944	20.4	1.227
MF	12	0.097	1.835	19.5	1.226
FW*	2	0.187	1.803	12.4	0.764
FW	3	0.175	1.576	10.8	0.837
FW	4	0.176	1.599	10.2	0.908
FW	5	0.196	1.181	9.3	0.909
FW	6	0.124	1.490	8.5	0.944
FW	7	0.132	1.350	7.8	1.017
FW	8	0.101	1.389	7.7	1.021
FW	9	0.127	1.229	7.7	1.070
FW	10	0.125	1.350	7.8	1.051

Table S7: UMAP hyperparameter grid, ranked by trustworthiness. Selection was by trustworthiness alone and was decided before any embedding was inspected.

Neighbours	Minimum distance	Trustworthiness
10	0	0.9322
10	0.10	0.9316
15	0	0.9306
15	0.10	0.9303
30	0.10	0.9276
30	0	0.9257
10	0.50	0.9186
15	0.50	0.9175
50	0.10	0.9169
50	0	0.9156
30	0.50	0.9083
50	0.50	0.9045

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
FW	11	0.128	1.199	7.1	1.064
FW	12	0.116	1.126	6.8	1.096

S5 Fuller views of the main-text figures

Figure S3 gives the three panels of Figure 2 at full size and with the detail a six-panel composite cannot carry: the principal component loadings across all five leading components rather than the first two, the mode against position contingency normalised by row as well as by column, and the confusion matrices for both classifiers rather than only the boosted one. Figure S6 gives the centroid deviation of each archetype from its position group mean across the entire feature set, rather than only the ten features with the largest separation shown in Figure 3. Figure S7 gives the same centroids as radars on fixed axes within each position group. Within a group the two profiles are mirror images by construction, since the features are standardised within group before clustering, so the useful reading is of shape rather than of extent.

Figure S9 gives the full feature attribution for the position classifier. The direction of each attribution adds a detail that magnitude alone hides. Ranking the correlation between a feature’s value and the attribution it makes to a class separates a class identified by what its members do from one identified by what they do not, and the three classes differ on exactly that. 4 of the five strongest features for defenders are positively signed, so defenders are recognised by the presence of what they do. Forwards and midfielders are recognised as much by the absence of defensive volume as by the presence of attacking output, with only 2 and 2 of their five positive respectively. A class defined largely by negations is a class occupying one end of a scale rather than a category with a signature of its own. Figure S10 splits the same attribution by class.

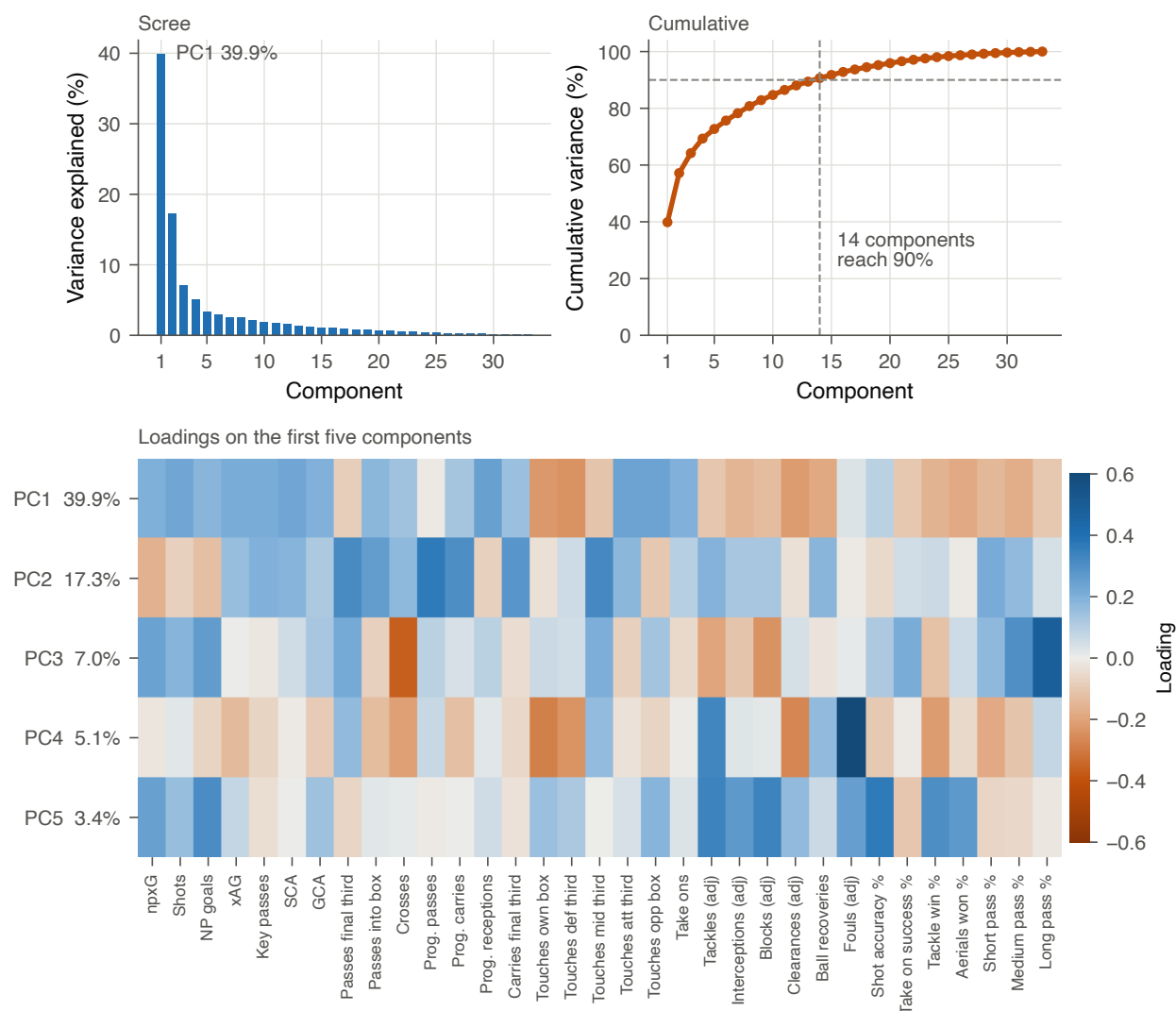


Figure S3: Principal components of the 33 archive features over 9,263 player-seasons, with the scree curve and the loadings of the leading components at full size. Figure S4 gives the contingency and Figure S5 the confusion matrices.

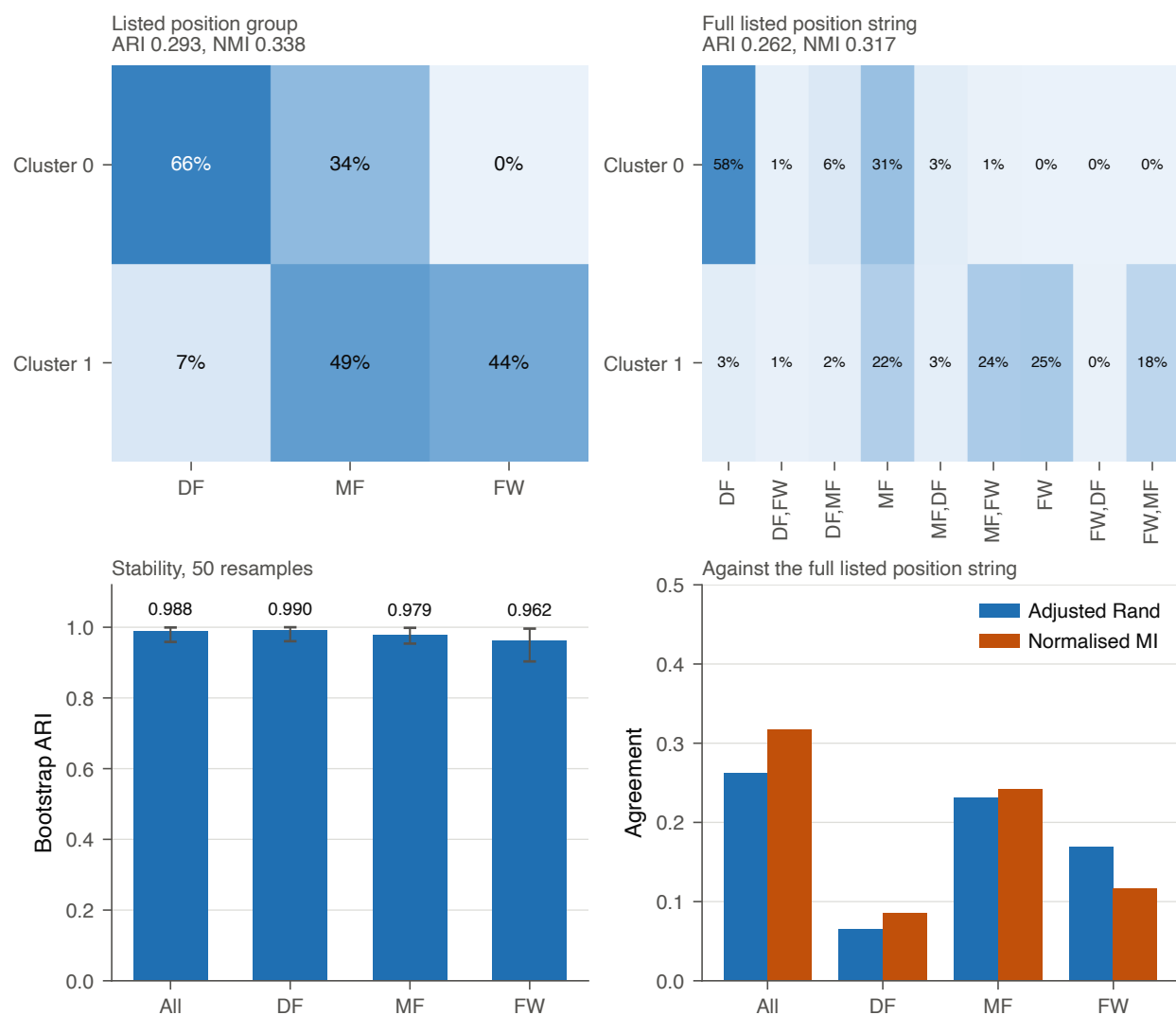


Figure S4: The pooled two-mode partition against listed position, with the heat maps normalised by row so that each gives the position composition of one mode. The main text reads the same table by column, which is where the 92.9 and 99.8 percent figures come from.

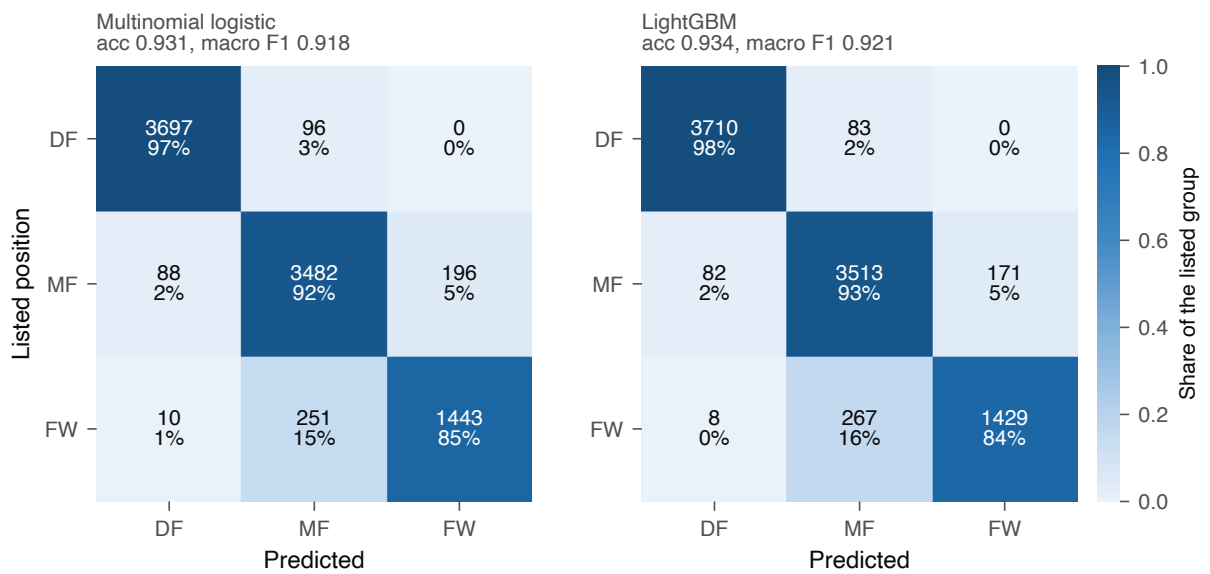


Figure S5: Out-of-fold confusion for both position classifiers on 9,263 player-seasons. The near equality of the linear and the boosted model is itself evidence that little interaction structure remains for the trees to find.

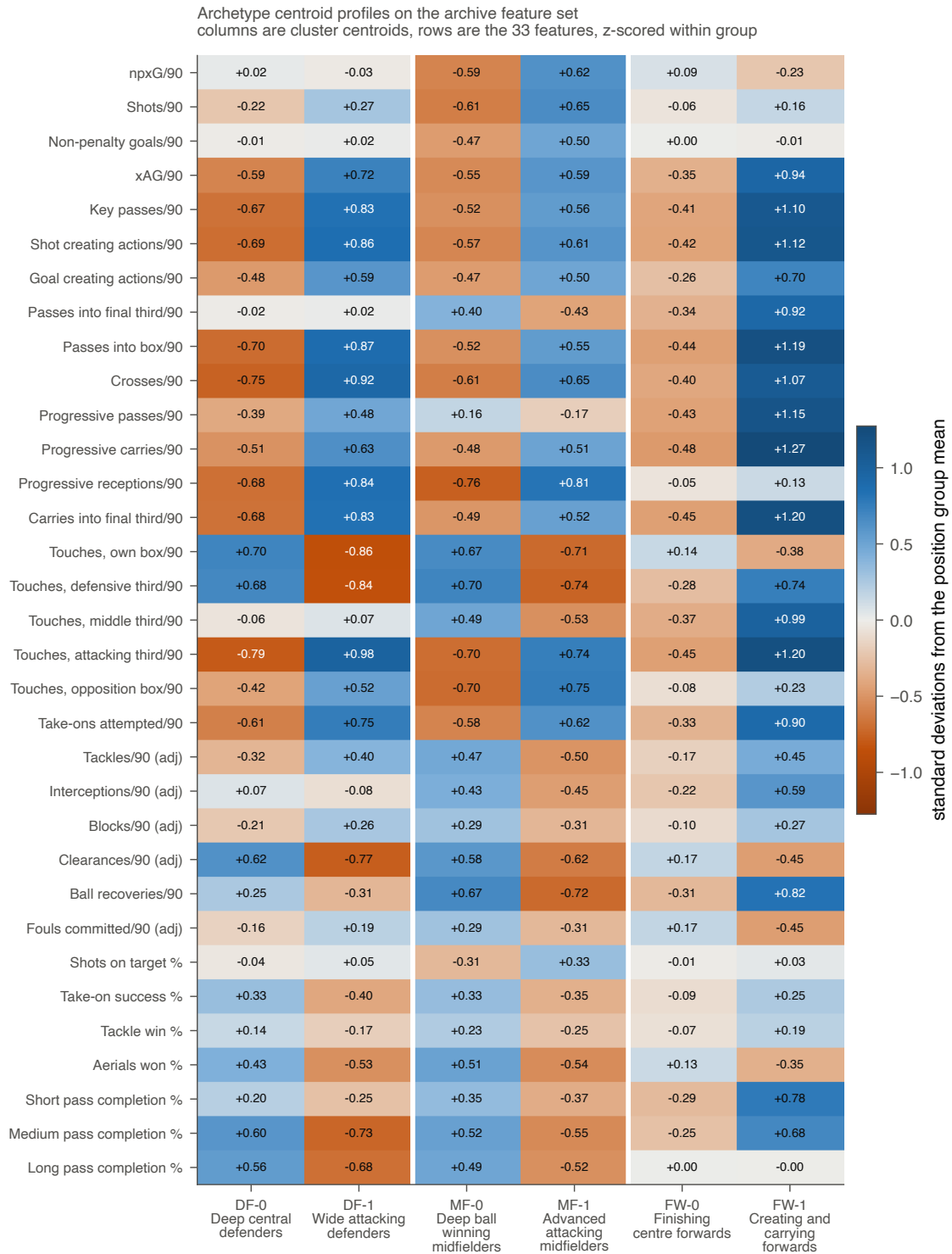


Figure S6: Centroid deviation of each archetype from its position group mean across the full 33 feature set, primary sample. Within a group the two profiles are mirror images by construction.

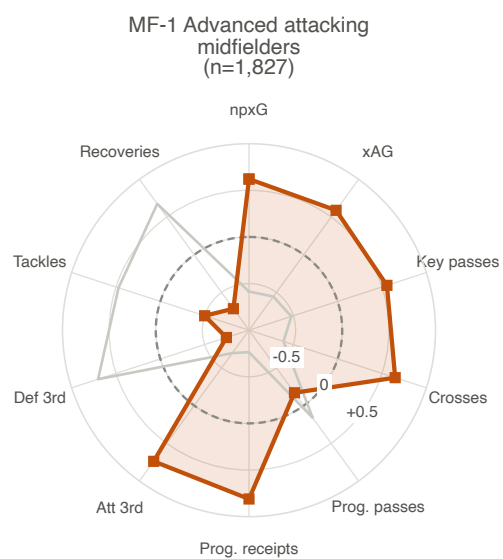
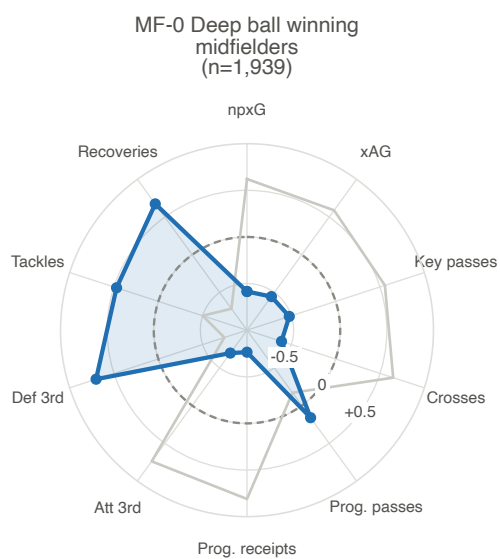
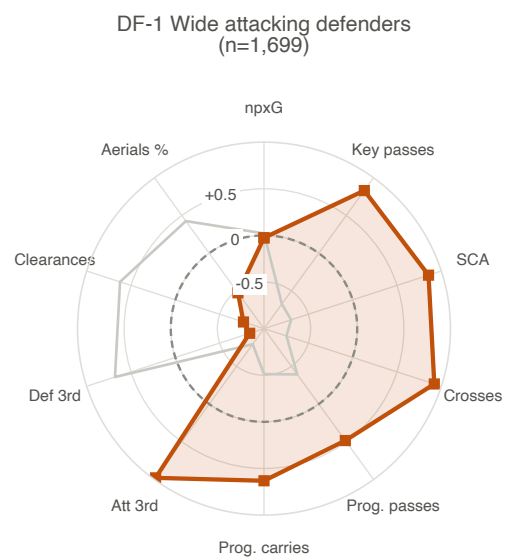
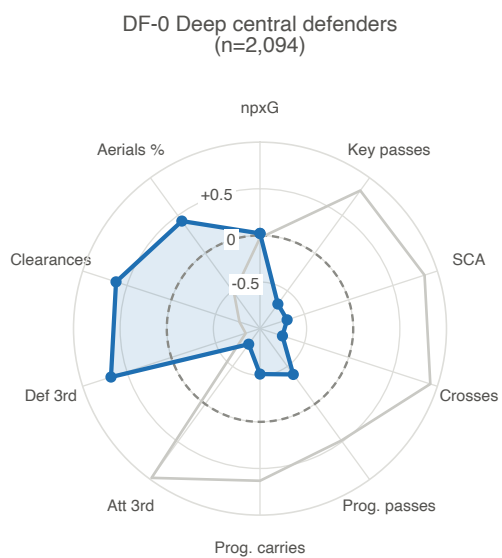


Figure S7: Archetype radars on fixed axes, primary sample. Top, defenders. Bottom, midfielders. Forwards are in Figure S8.

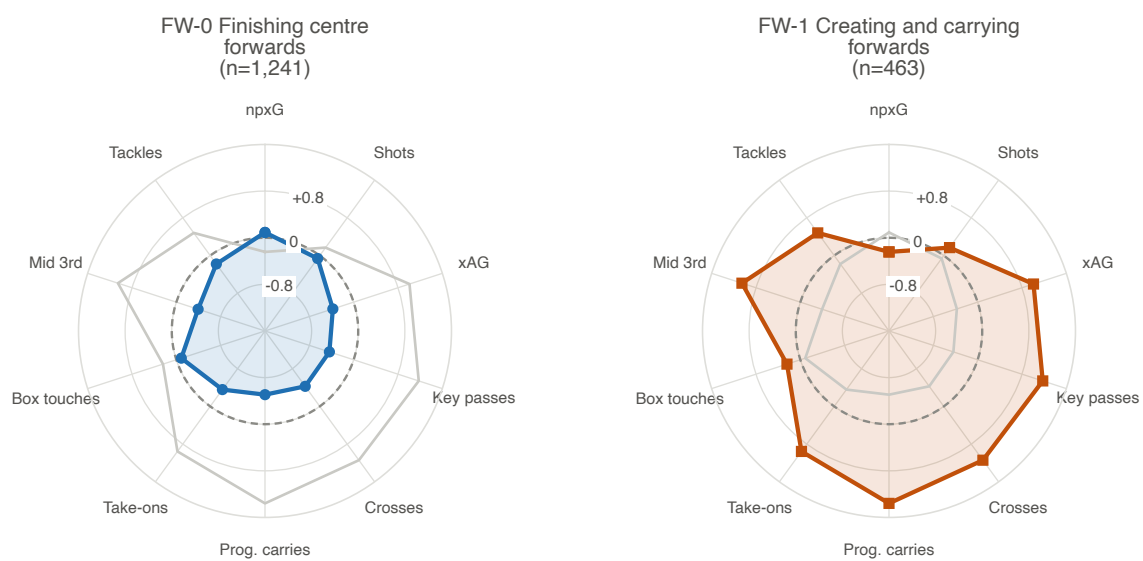


Figure S8: Archetype radars for forwards on the primary sample, on the same fixed axes as Figure S7.

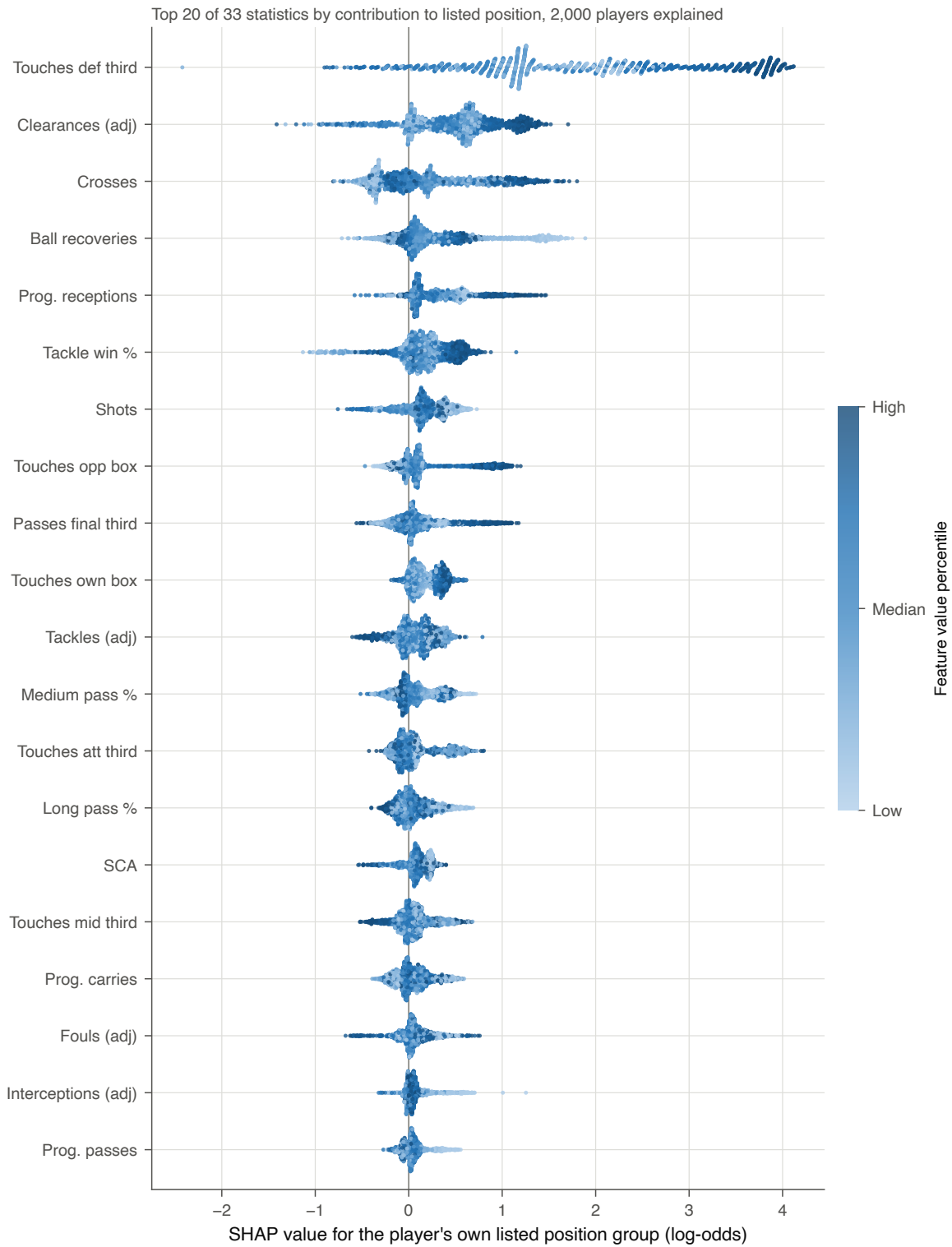


Figure S9: SHAP values for the gradient boosted position classifier on the primary sample, showing for each player the contribution of each feature toward the position group the player is actually listed at. Territorial features dominate technical ones throughout.

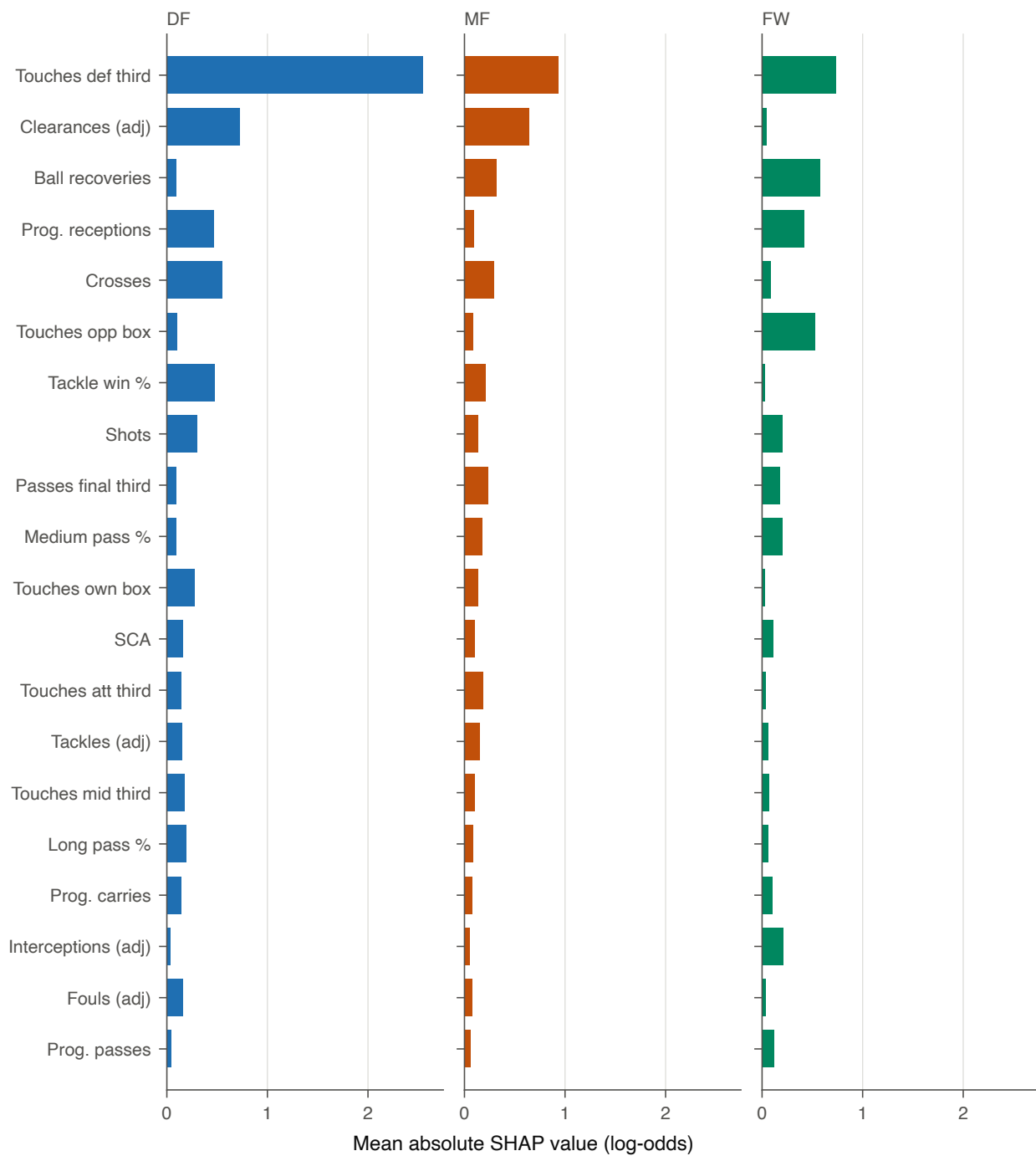


Figure S10: Mean absolute attribution per feature, split by position group, primary sample. Touches in the defensive third dominate for all three, and the profile of what matters differs sharply between them: defenders are separated largely by defensive volume and crossing, forwards by where they receive the ball. The panel shows magnitude only, and the direction each feature pushes is reported in the text.

S6 Players the description fits worst

A two-mode description of a continuum will fit some players badly, and which players those are is a test of whether the description is measuring football or noise. Isolation forests and Mahalanobis distance under a shrinkage covariance were fitted separately within each position group, and the two detectors agree on 80.5 percent of the players they flag.

The 15 strongest outliers are 12 distinct players, headed by Neymar. Two features of the list stand out. 11 of the 15 carry a compound listed position such as midfielder and forward together, against a much smaller share in the pool, so the players the partition handles worst are the ones the listing system also declines to place. The detector is also not simply finding stars: it finds attacking full backs whose output profile resembles a winger's, which is the same boundary the defender split describes and a genuine tactical category rather than an artefact.

One caveat belongs with the list rather than after it. 9 of the 15 played under nine hundred minutes in the season concerned, so their per-90 rates carry more sampling noise than a full season's would, and extreme rates are easier to reach on a small denominator. The list should be read as evidence about the kind of player the two-mode description struggles with, not as a ranking in which the fifteenth entry means what the first does. Figure S11 shows both detectors against each other.

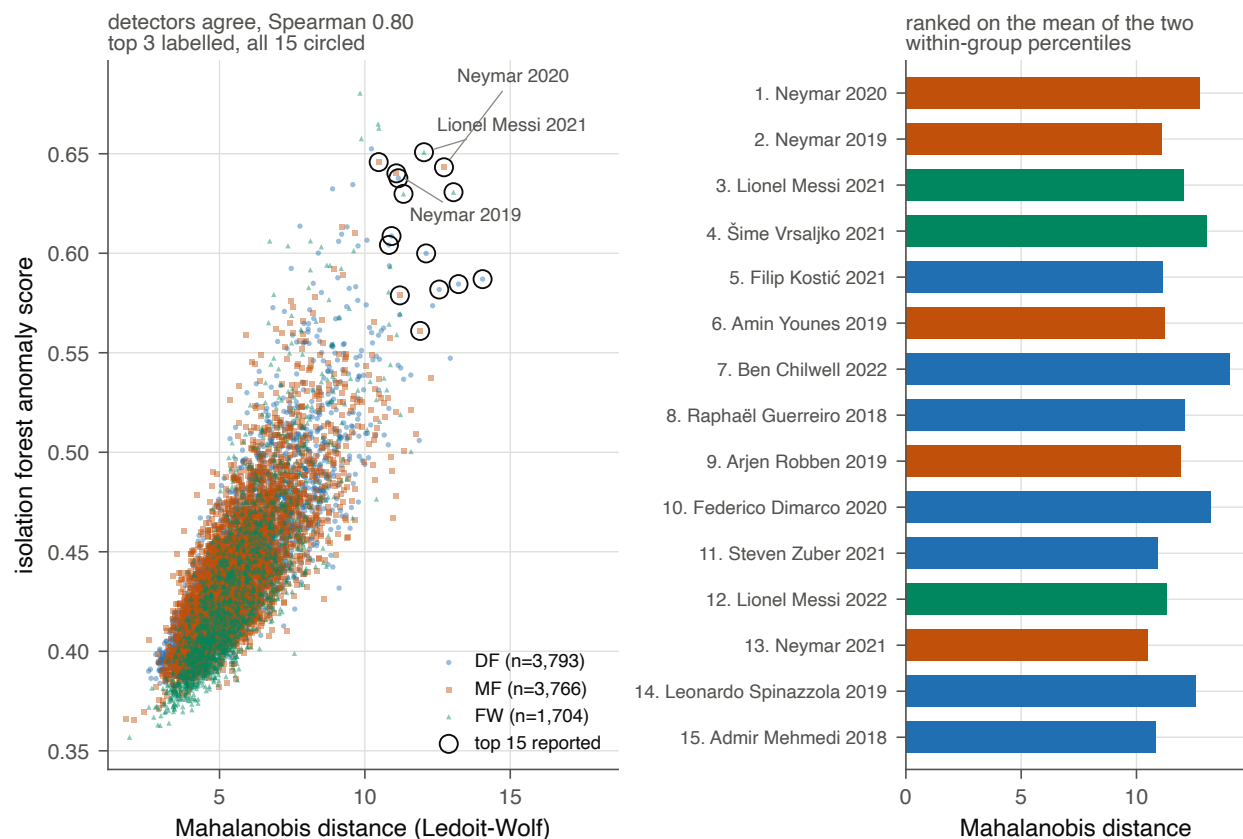


Figure S11: Isolation forest score against Mahalanobis distance within position group, primary sample. The two detectors are computed independently and rank the same players highly, and the flagged points are dominated by players carrying compound listed positions.

Table S8: The ten feature pairs whose Spearman correlation differs most across position groups. Every pair changes sign between at least two groups.

Feature A	Feature B	DF	MF	FW	Spread
Shots on target	Crosses	-0.01	0.43	-0.44	0.87
Crosses	Shot accuracy %	-0.22	0.26	-0.57	0.82
Crosses	Interceptions (adj)	0.21	-0.50	0.30	0.81
npxG	Crosses	-0.15	0.36	-0.43	0.79
Key passes	Tackles won (adj)	0.47	-0.32	0.26	0.78
Crosses	Tackles won (adj)	0.51	-0.25	0.48	0.76
xA	Shots on target	0.17	0.56	-0.19	0.75
Shots	Interceptions (adj)	0.22	-0.50	-0.04	0.72
xA	Tackles won (adj)	0.39	-0.30	0.15	0.69
xA	Interceptions (adj)	0.19	-0.50	0.16	0.69

S7 Statistics that change meaning by position

Correlations were computed within each position group and compared across groups, and the ten pairs whose correlation varies most between groups are reported in Table S8 and shown in Figure S12. The largest spread between position groups reaches 0.87, against a median spread across all pairs of 0.38.

Every one of the ten most divergent pairs changes sign between position groups. The relationship between shots on target and crosses is positive for midfielders and negative for forwards, which is to say that a midfielder who crosses often is generally a more active attacker while a forward who crosses often is generally not the player getting on the end of chances. The relationship between key passes and tackles won is positive for defenders and negative for midfielders, distinguishing the full back whose defensive work and creative output rise together from the midfield division of labour between a destroyer and a creator.

The practical implication is that a single pooled correlation matrix, of the kind that appears routinely in public football analysis, averages over these sign reversals and reports a number that describes no actual population of players. It also supplies the justification for the position stratified analysis used throughout this paper, since a global embedding fitted across all outfield players is fitted across relationships that point in opposite directions. The full correlation matrix the comparison is drawn from is Figure S13.

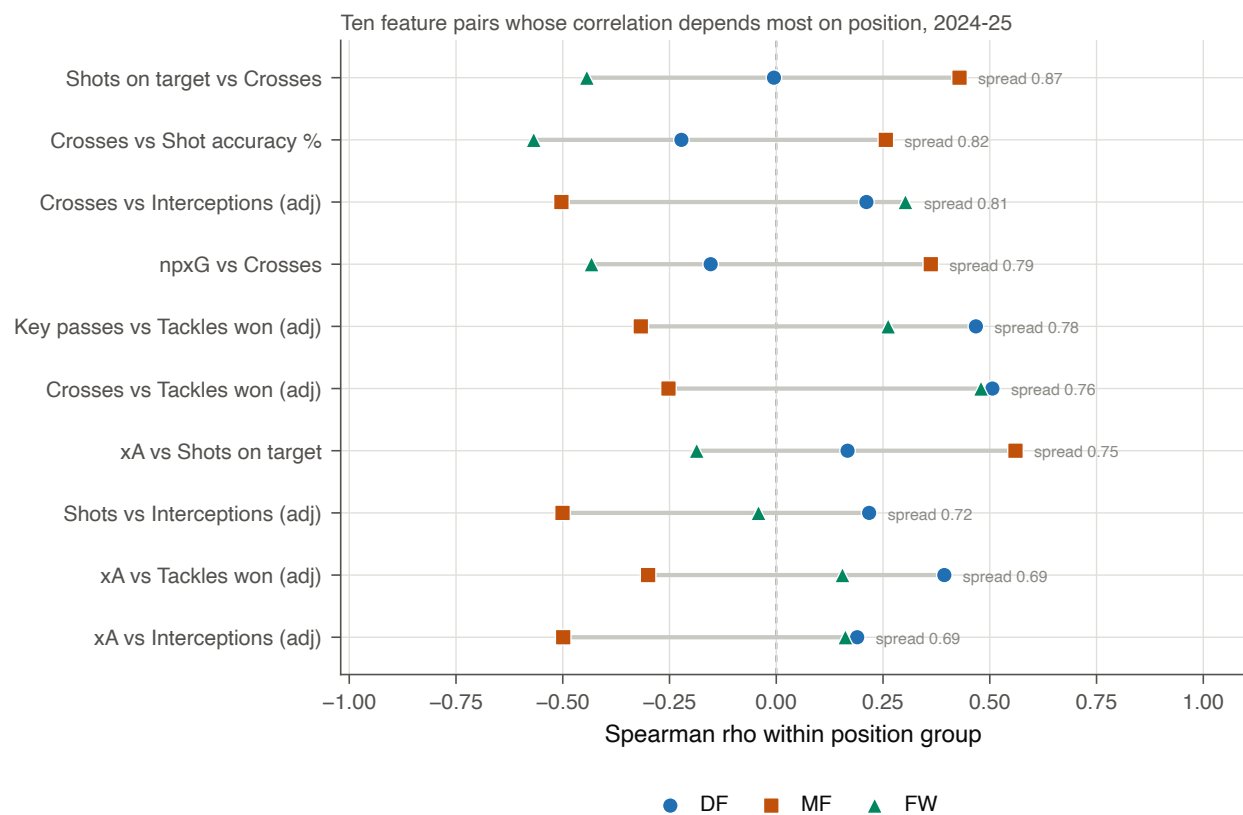


Figure S12: The ten feature pairs whose within-group correlation differs most across position groups, in the earlier of the two reduced-sample seasons. Every pair changes sign between at least two groups.

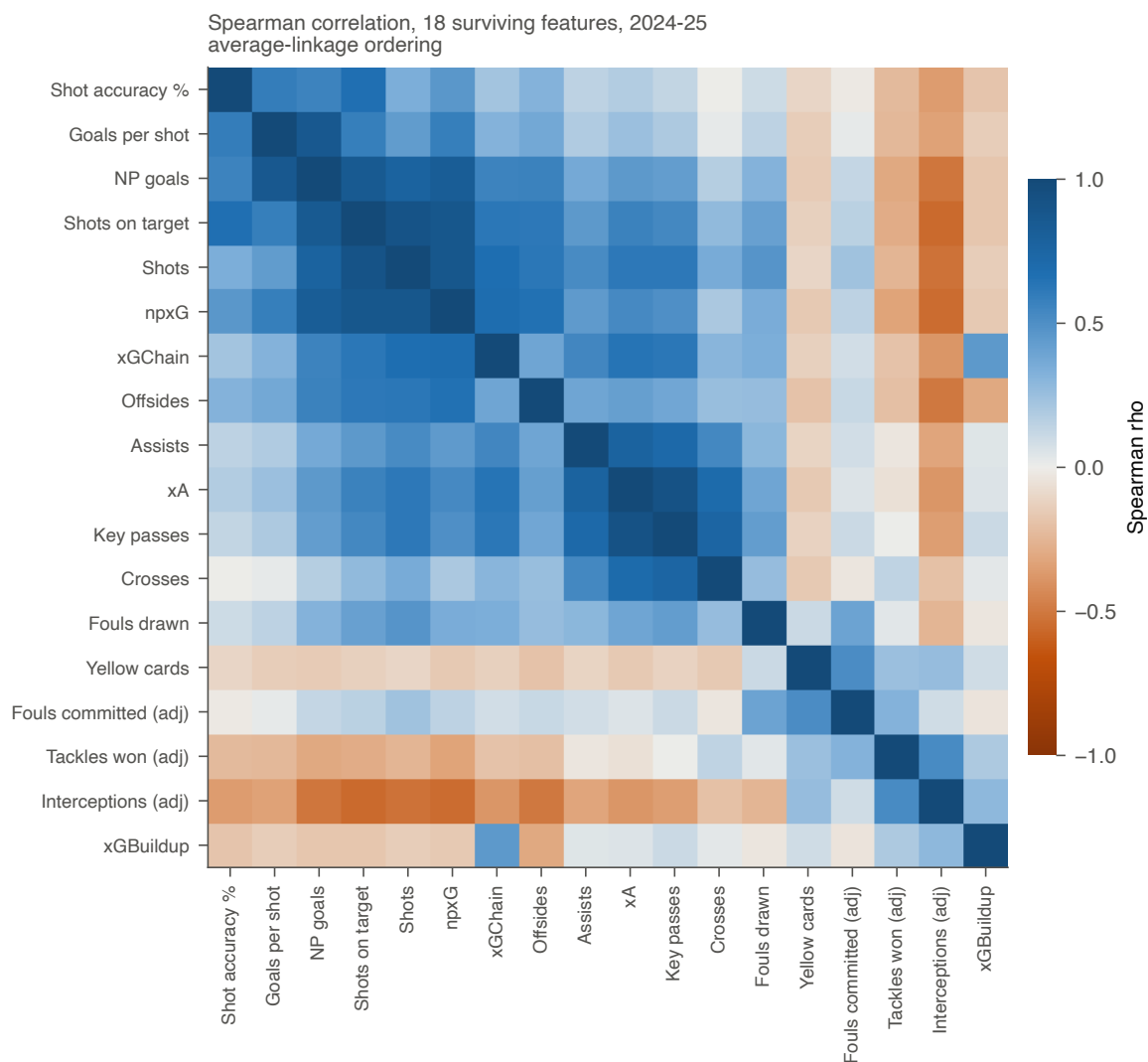


Figure S13: The full hierarchically clustered Spearman correlation matrix over the reduced outfield feature set, from which the comparison in Figure S12 is drawn.

S8 Analyses on the reduced sample

The analyses collected here run on the reduced sample described in Section 2.1: the seasons ending 2025 and 2026, pulled live after the withdrawal, with roughly half the features of the primary sample.

One point about scope needs stating plainly, because it applies to every figure in this section. Where a figure here shows Premier League players only, that is not a sampling choice made for convenience. It is the coverage of the live post-withdrawal pull that this robustness case is built on, and the reduced sample exists to test whether the main result survives a much poorer description of the same game rather than to characterise European football. The paper’s claims rest on the primary sample, which covers all 5 leagues and all 5 seasons, and every one of them is refitted independently in each league and each season in Section 3.4. Section S8.3 below additionally repeats the reduced-sample pipeline across all five leagues, so the reduced sample’s own generality is tested rather than assumed. The within-season drift analysis of Section S8.7 is narrower still, covering one club, and the reason is stated there.

S8.1 Cluster selection and stability

Applying the declared rule to the four selection criteria gives $k = 2$ globally and within each position group, matching the primary sample. Across 200 bootstrap resamples the mean adjusted Rand index against the reference partition is 0.914 globally and 0.815 for midfielders, both describing a partition that reappears almost unchanged under resampling. For defenders the mean falls to 0.607 with a standard deviation of 0.216, and for forwards to 0.602 with a standard deviation of 0.281. The minimum adjusted Rand index observed across forward resamples is -0.065, which is to say that in the least favourable resample the partition recovered is no better than chance. Forwards number only 49 players above the minutes threshold here, so this is partly a sample size problem, and it is a real limit on what the reduced sample can claim. The global partition scores an adjusted Rand index of 0.226 against the position group with a cluster purity of 0.579, so the clusters are not a rediscovery of the positions the source already records. Figure S19 gives the criterion curves, Figure S20 the co-assignment structure and Figure S21 the correspondence with position.

S8.2 Descriptive structure and embeddings

After filtering, 366 outfield players are eligible in the earlier of the two reduced-sample seasons, comprising 136 defenders, 181 midfielders and 49 forwards. Summary statistics by position group appear in Table S9. The feature set is strongly non-normal: all 18 of the 18 features reject the Shapiro-Wilk null at the five percent level, mean absolute skewness is 1.13, and Mardia’s test rejects multivariate normality decisively on both components. Principal component analysis requires no distributional assumption to be computed, so this does not invalidate the decomposition, but the components should be read as directions of variance rather than as latent factors, and it justifies the use of Spearman rather than Pearson correlation throughout.

Principal component analysis, UMAP and t-SNE were fitted to the same standardised features. The selected UMAP configuration used 10 neighbours at a minimum distance of 0. Trustworthiness is 0.855 for the linear projection, 0.932 for UMAP and 0.944 for t-SNE, while continuity is close to identical across the three. Nearest neighbour position purity at ten neighbours is 0.649 for the linear projection and 0.699 for UMAP, so roughly two thirds to three quarters of a player’s closest statistical neighbours share his listed position group. That figure is informative in both directions: it confirms the surviving features carry real positional signal, and it establishes that a substantial minority of every player’s nearest neighbours are listed at a different position, which is the space in which data-driven roles can differ from listed ones. All three embeddings display a continuous

Table S9: Headline per-90 statistics by position group in the earlier of the two reduced-sample seasons, reported as mean with standard deviation in parentheses.

Statistic	DF	MF	FW
np_xg_p90	0.05 (0.04)	0.16 (0.12)	0.41 (0.17)
xa_p90	0.06 (0.06)	0.15 (0.10)	0.16 (0.11)
key_passes_p90	0.51 (0.44)	1.26 (0.63)	1.05 (0.54)
xg_chain_p90	0.35 (0.16)	0.50 (0.20)	0.63 (0.20)
xg_buildup_p90	0.30 (0.14)	0.28 (0.13)	0.18 (0.09)
shots_p90	0.54 (0.33)	1.54 (0.85)	2.53 (0.69)
shot_accuracy_pct	29.22 (20.60)	31.91 (13.77)	40.30 (9.65)
crosses_p90	1.22 (1.62)	2.26 (1.75)	1.40 (1.62)
interceptions_padj_p90	0.96 (0.37)	0.74 (0.44)	0.24 (0.17)
tackles_won_padj_p90	1.07 (0.48)	1.16 (0.54)	0.56 (0.34)
fouls_committed_padj_p90	0.87 (0.39)	1.31 (0.55)	1.28 (0.61)
offsides_p90	0.06 (0.07)	0.16 (0.17)	0.53 (0.33)

gradient from defenders through midfielders to forwards rather than separated groups, and that gradient is the honest description of the geometry. Figure S14 shows the distributions and the three embeddings and Figure S18 the principal component structure, while Figures S15 to S17 give the archetypes this sample returns.

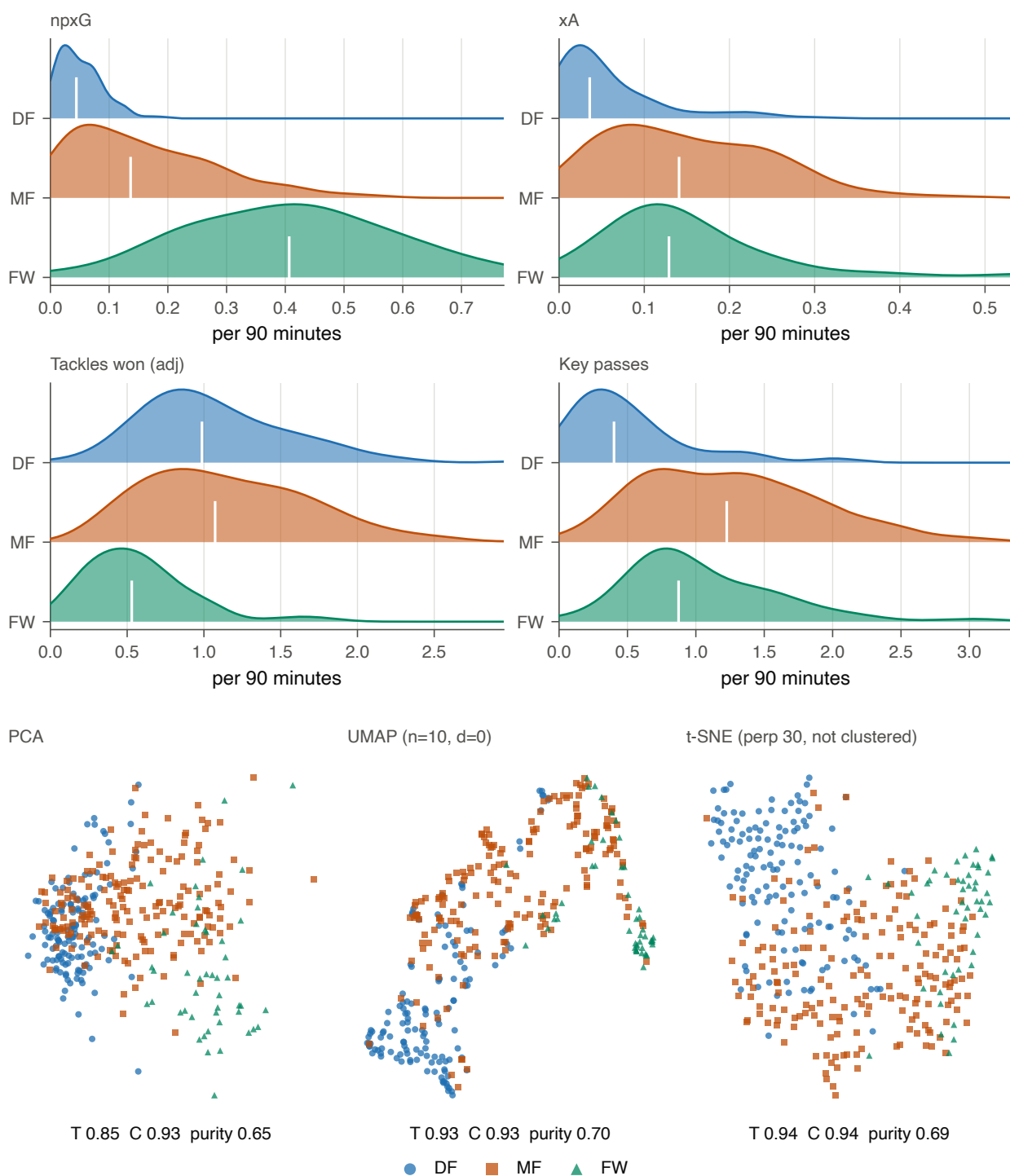


Figure S14: Reduced sample. Top, distributions of four representative per-90 statistics across position groups, where separation is clear for shot volume and defensive set actions and much weaker for creation. Bottom, two-dimensional embeddings of the same feature set coloured by listed position group; the defender to forward gradient is continuous in all three, and t-SNE is shown for visual contrast only and is never used for clustering.

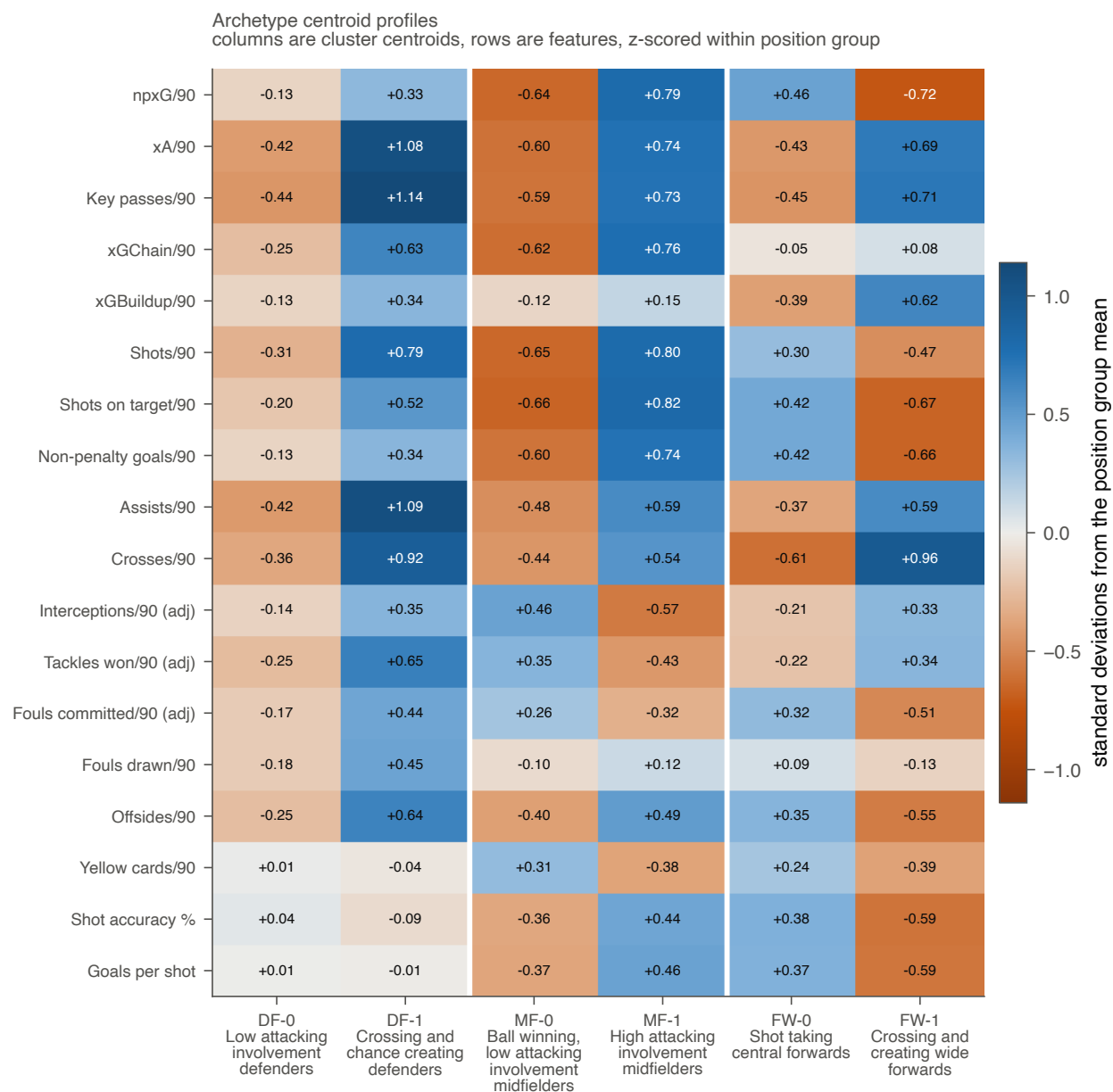


Figure S15: Centroid deviation of each archetype from its position group mean, reduced sample.



Figure S16: Archetype radars on the reduced sample. Top, defenders. Bottom, midfielders. The two sets of radars in this document are drawn on different feature counts, so their axes are not directly comparable and the useful comparison is of shape rather than extent.

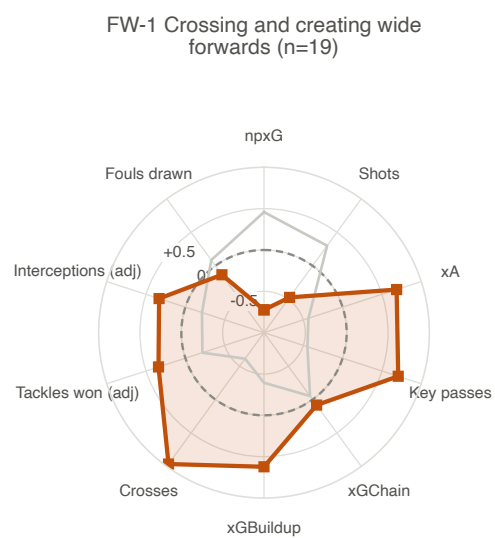
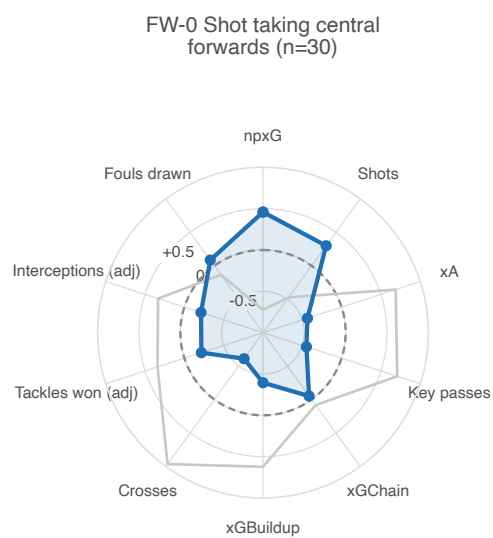


Figure S17: Archetype radars for forwards on the reduced sample, on the same axes as Figure S16.

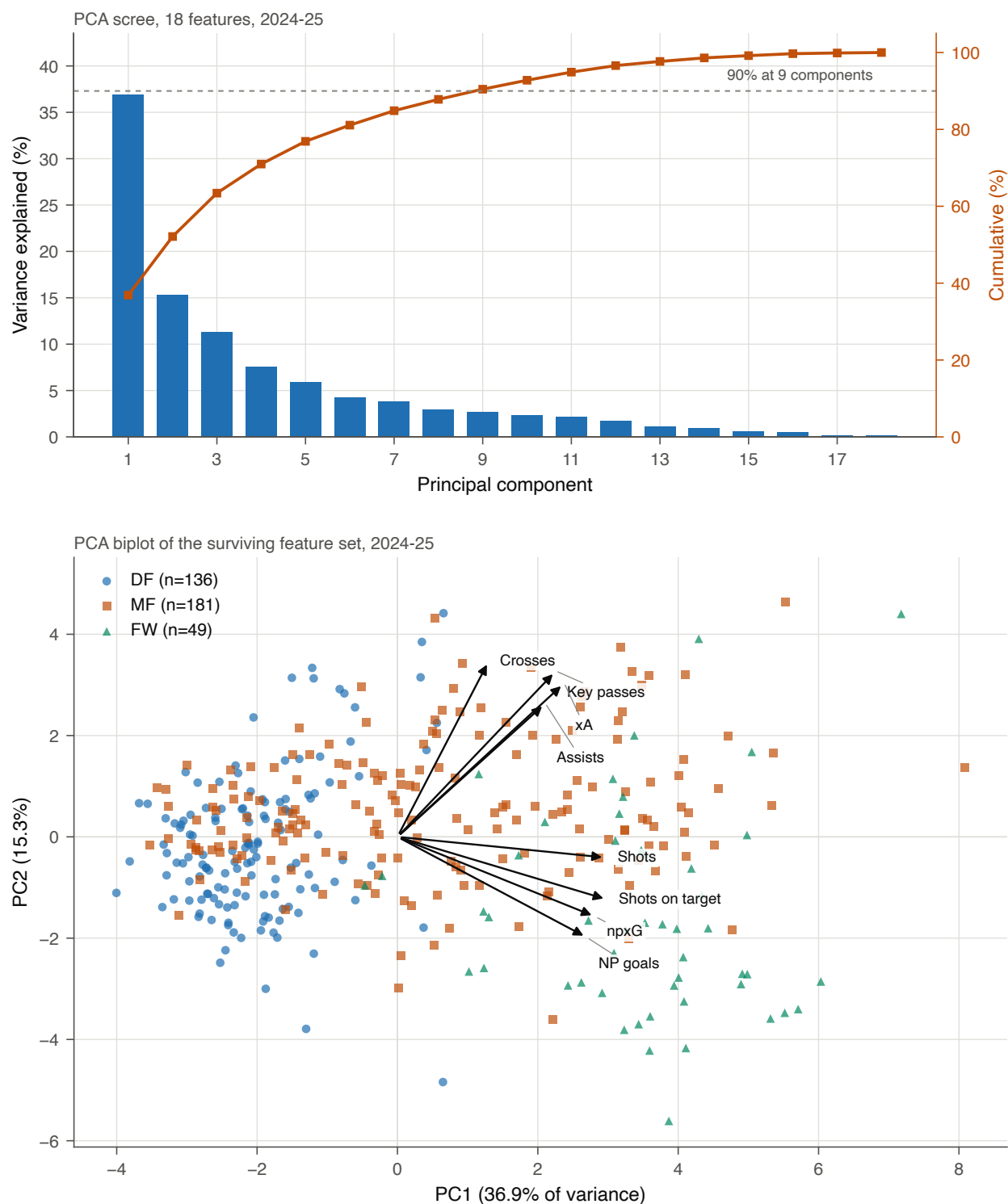


Figure S18: Principal component structure of the reduced outfield feature set. 9 components are required to reach ninety percent of the variance, so the feature set is not close to low rank, and the first three components together account for 0.634. Top, scree and cumulative variance. Bottom, the loadings biplot.

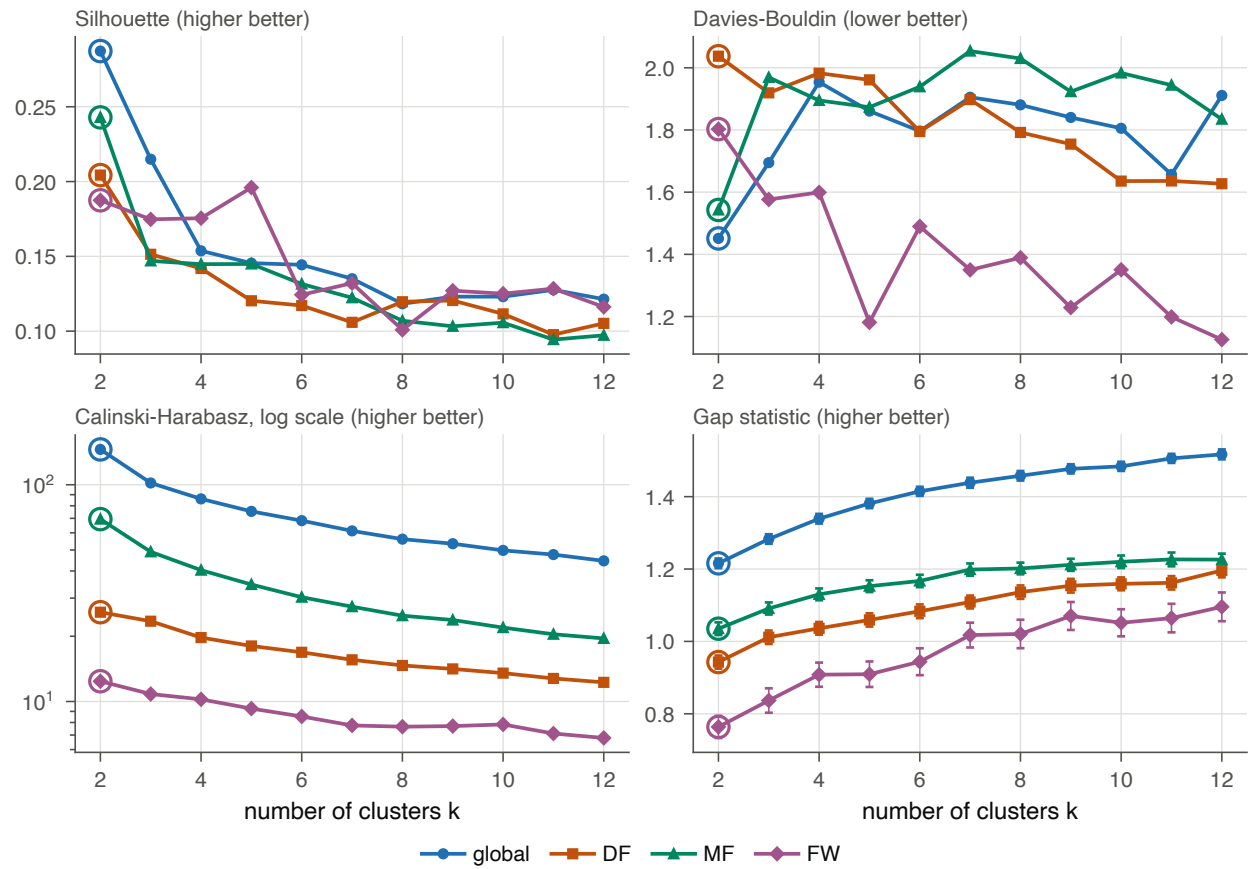


Figure S19: Cluster selection criteria across candidate cluster counts, reduced sample. The consensus rule was fixed before fitting and selects the same value in every scope.

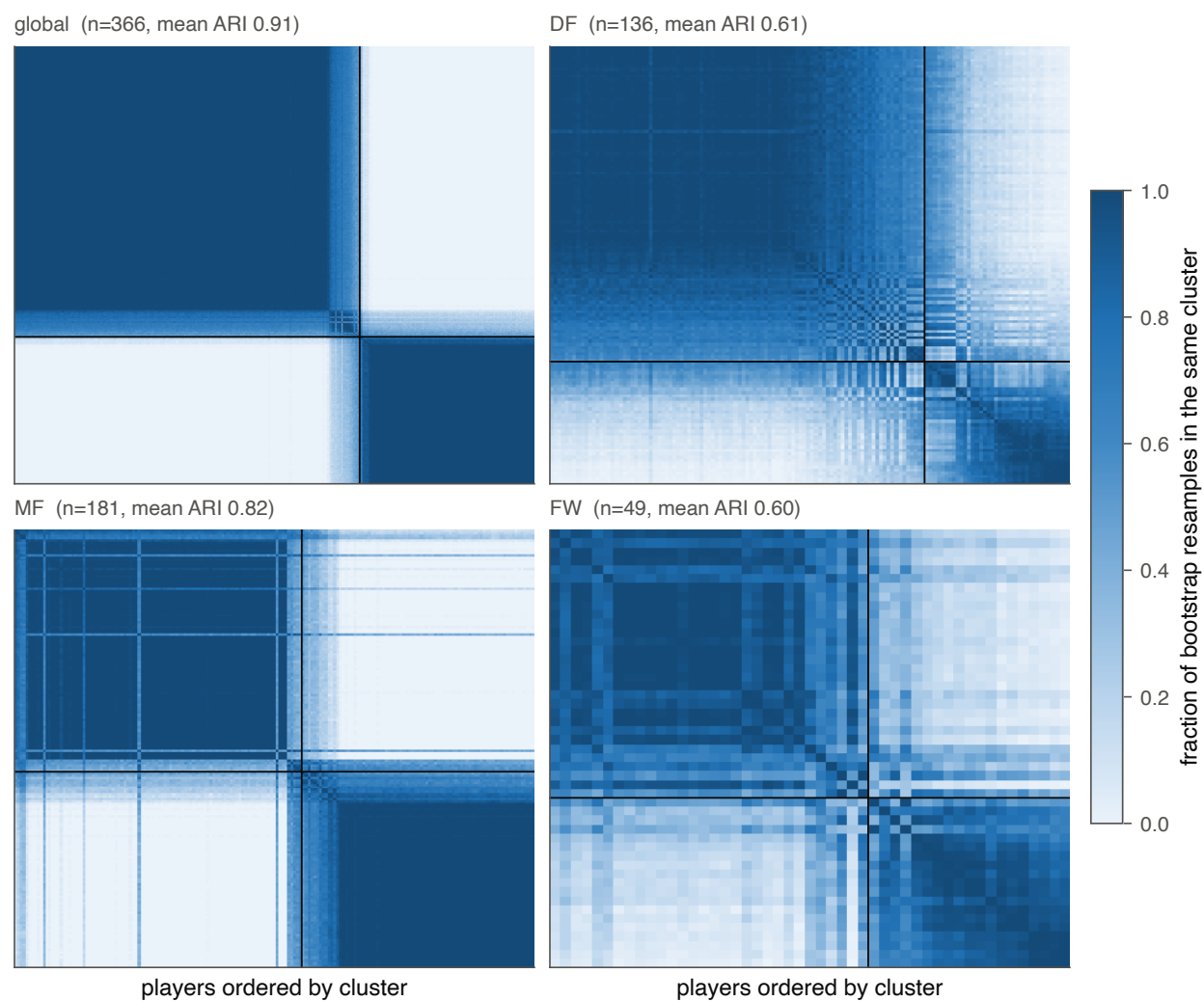


Figure S20: Bootstrap co-assignment structure on the reduced sample, one panel per scope, with players ordered by cluster.

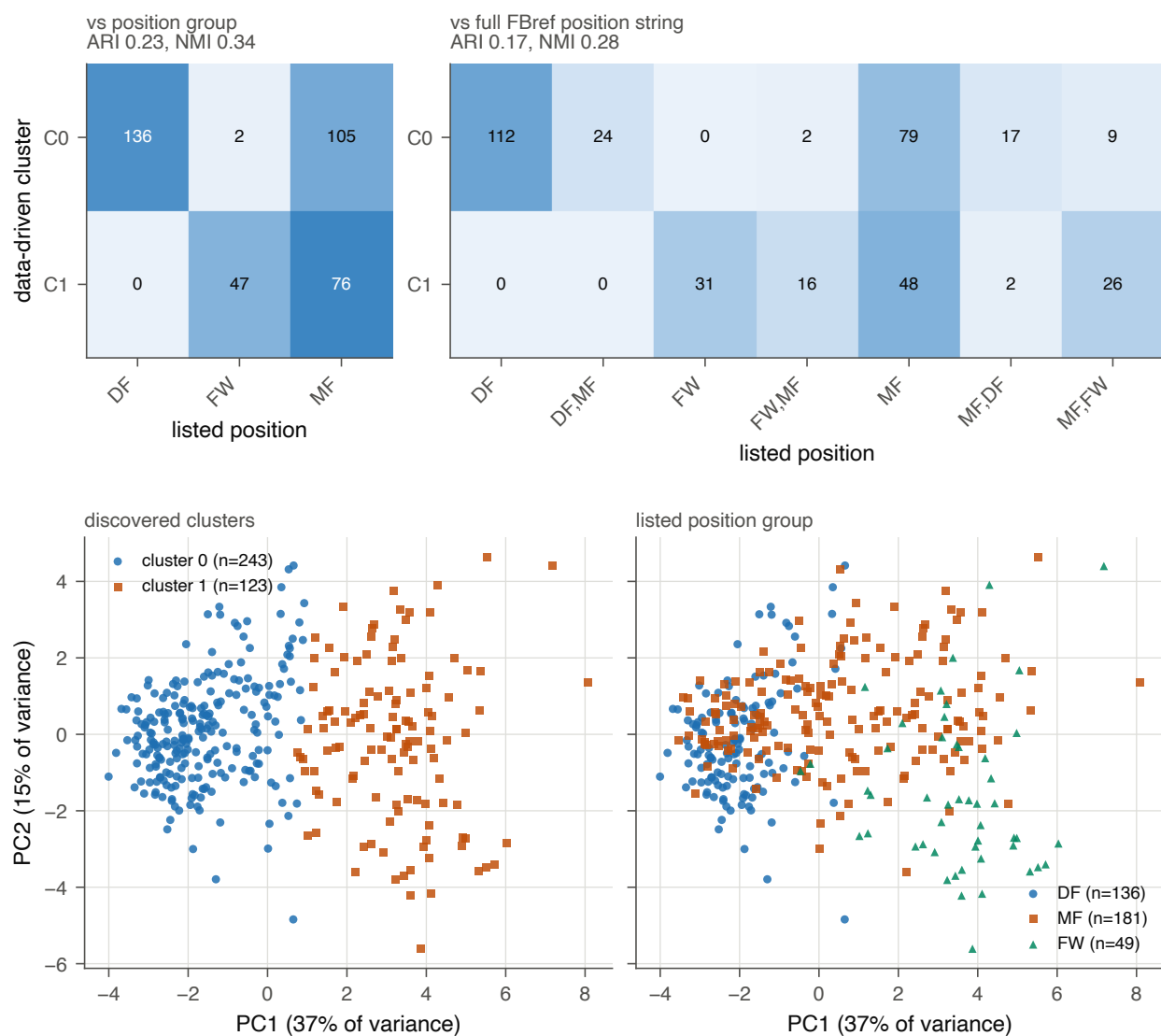


Figure S21: Reduced sample. Top, the selected partition against listed positions. Bottom, the same partition shown in the principal component space.

S8.3 The same pipeline across the big five

Repeating the reduced-sample pipeline unchanged on Spain, Germany, Italy and France for the earlier of the two seasons returns the same answer everywhere. The declared rule selects 2 clusters in all 5 leagues. The silhouette at that choice spans 0.255 to 0.286 across the five, bootstrap stability spans 0.907 to 0.946, and agreement with listed position groups spans 0.190 to 0.270. Each of those three ranges is narrow, and the third is uniformly weak: in no league does the partition recover the positions the source already records. The elasticity result generalises as well, since all 15 defensive elasticities fitted independently inside each league lie between 0.081 and 0.692, every one below the value of one that the conventional possession adjustment assumes. Figure S22 shows the comparison.

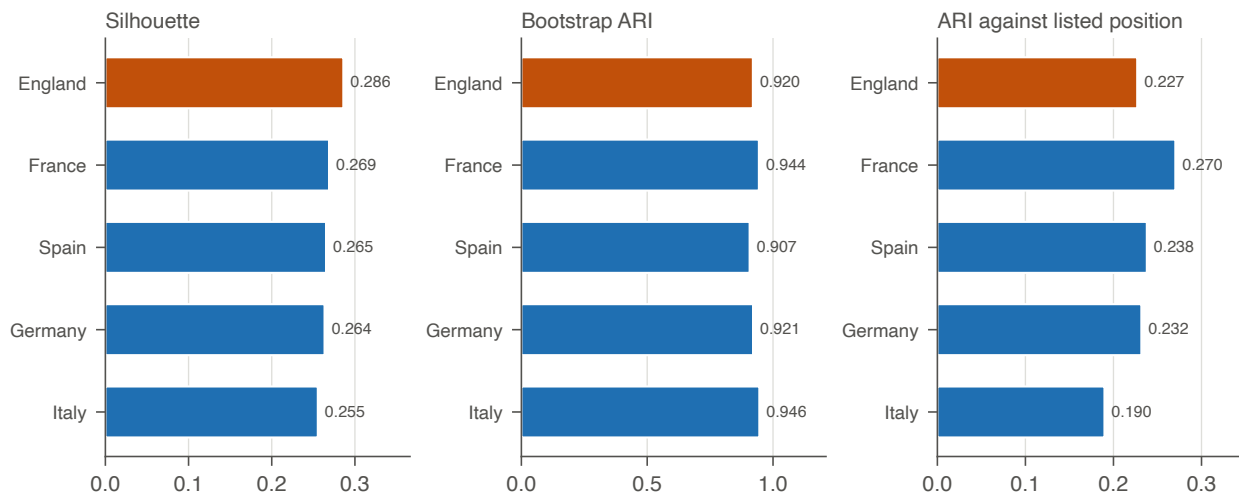


Figure S22: The reduced-sample pipeline applied unchanged to each of the big five leagues, with England, the competition the reduced sample is otherwise drawn from, in the accent colour. The silhouette at the selected number of clusters, bootstrap stability and agreement with listed positions are all close across competitions, and the declared rule selects the same number of clusters in every one.

S8.4 What a supervised model sees

Predicting the listed position group from the same 18 features, gradient boosting reaches an accuracy of 0.803 and a macro F1 of 0.769, and logistic regression reaches 0.811 and 0.786, against a majority class baseline of 0.494. The features clearly do carry position, so the weak agreement between clusters and positions reflects the clustering finding a different and equally real axis rather than an empty feature set.

The attribution carries a finding that follows directly from the withdrawal. With possession and passing volume gone, positions are identified largely by absence: defenders by low shooting and low key passes, forwards by low interceptions, and midfielders by an intermediate profile on both. A classifier of this kind trained before January 2026 would have identified a defender by what he did with the ball, and it now identifies him by what he does not do. Players the classifier places in the wrong group overlap with the players the clustering flagged as hybrids, at 0.556 for gradient boosting against an enrichment of 1.32 over independence, which is corroboration rather than a strong result because the hybrid flag is itself broad.

The finishing above role regression illustrates a hazard worth recording. Predicting non-penalty

expected goals per ninety from non-shooting behaviour returns a cross validated coefficient of determination of 0.531. An earlier specification that also included xGChain returned 0.892, which looks far better and is in fact circular, since Understat credits xGChain for possessions the player finished himself, so the difference between xGChain and xGBuildup contains the target and correlates with it at 0.904. Even at the honest value roughly half the variance is unexplained, so the residual leaderboard of Figure S35 is an illustration rather than a measurement of finishing ability. Figure S23 shows the confusion and the per-class importance, and Figure S24 the full attribution.

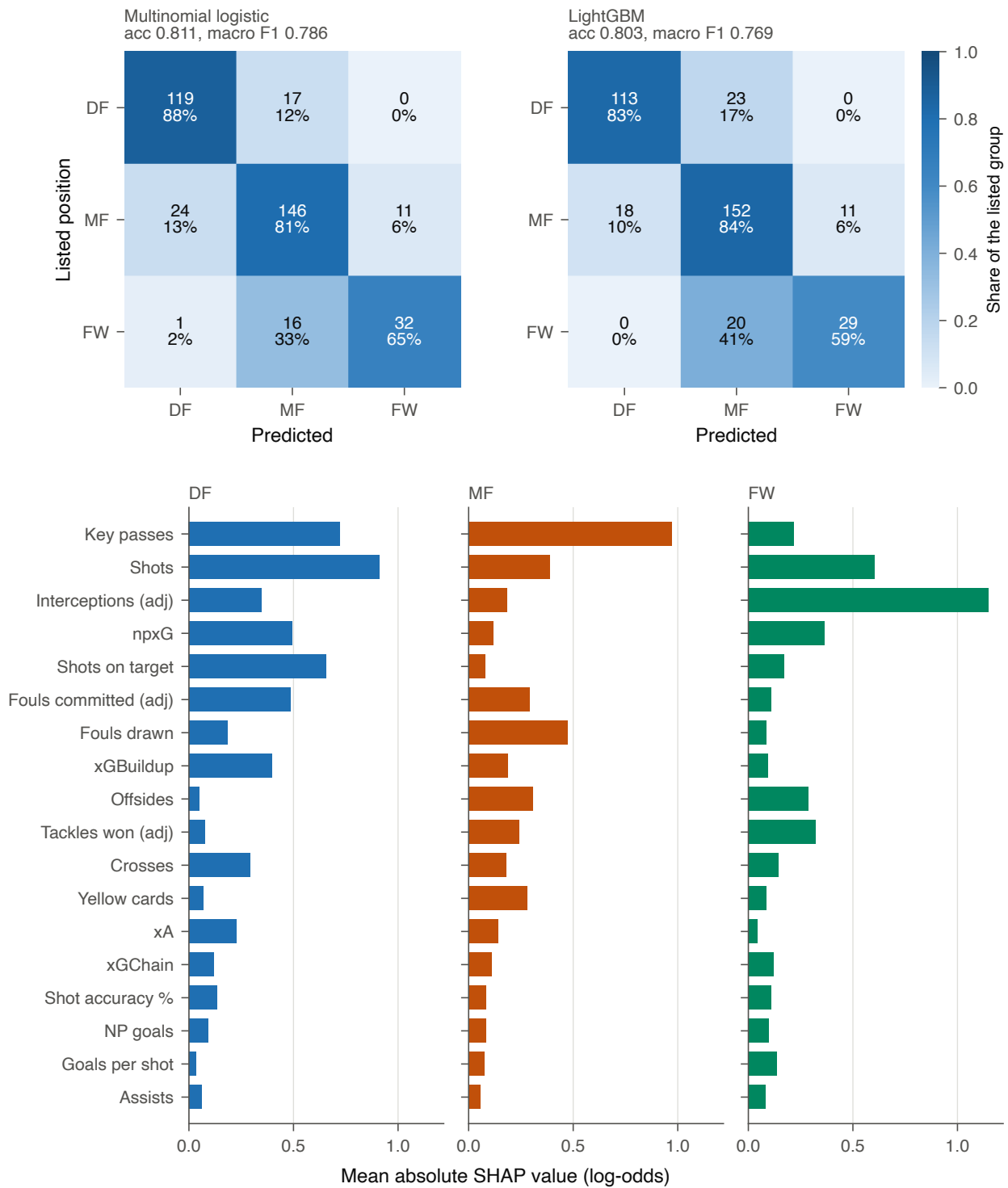


Figure S23: Supervised validation of the surviving feature set, reduced sample. Top, confusion between listed positions. Bottom, mean feature importance per position.

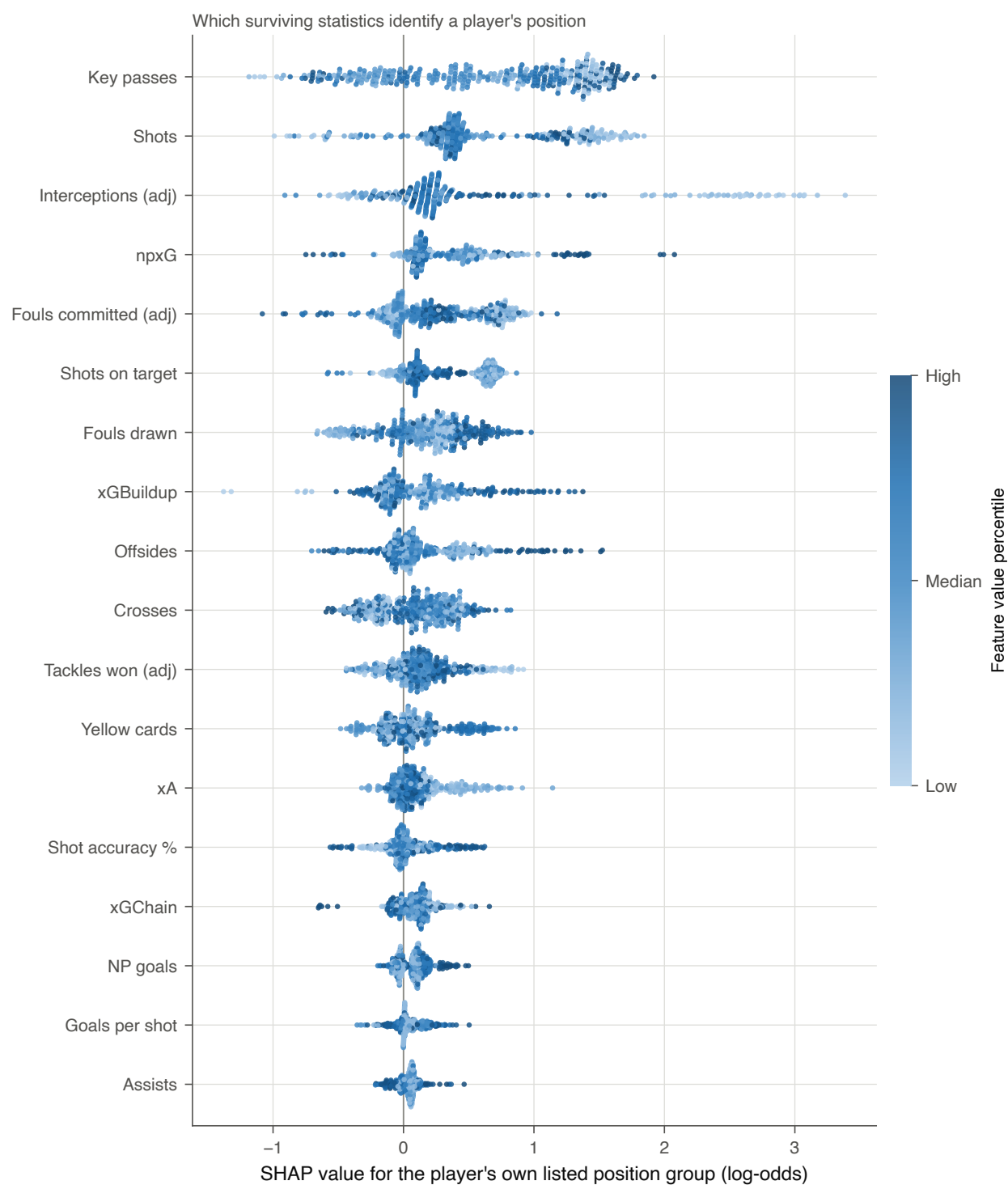


Figure S24: Global SHAP values for the gradient boosted position classifier on the reduced sample. Shooting volume and interceptions carry most of the signal, so a player's position is identified largely by the actions he does not perform.

S8.5 The third dimension and team signatures

Two components are a convenient way to draw the space and a poor way to describe it. On the reduced sample the first two carry shooting volume and wide creation and together account for 0.521 of the variance, while the third carries duels and discipline, loading on fouls committed, cards and tackles won, and is orthogonal to both of the others, so a planar projection does not compress it, it deletes it. Retaining it raises captured variance to 0.634. Adding the dimension does not change the headline geometry, since the defender to forward gradient remains continuous rather than resolving into islands.

The third dimension does change the club picture. Club centroids correlate with final league position at -0.861 on the first component and -0.695 on the second, but only 0.209 on the third. Shooting volume and wide creation both track league position; the physical dimension of duels, fouls and cards essentially does not. A club does not finish higher by fouling more or less, which needs stating because the duelling axis is a full component of player variation carrying almost no information about team success. The three-dimensional view also settles a structural question by measurement rather than assertion: drawing each archetype centroid as a direction from the origin, the two centroids within a position group have a cosine of -1 for defenders, -1 for midfielders and -1 for forwards. They are exactly anti-parallel, which is forced by standardising within group before a two-way split, so six archetypes occupy three lines through the origin.

Aggregating players into clubs produces the one result in the reduced sample that is both strong and stable. The minutes-weighted position of a club's outfield players on the first principal component correlates with final league position at a Spearman coefficient of -0.771 in the earlier season and -0.783 in the replication season, close to an exact reproduction. Since the first component is essentially attacking involvement, this says that clubs whose players collectively shoot and create more finish higher, which is unsurprising in itself; the near identical coefficient across two independent seasons shows the team level summary is measuring something stable even where the player level partitions are not. Hierarchical clustering on archetype composition groups the 20 clubs into 3 style families. Figures S25 and S26 show the three-dimensional views, Figure S27 the club signatures, and Figure S28 each club's players against the rest of the league in the shared embedding, on axes shared across all twenty panels.

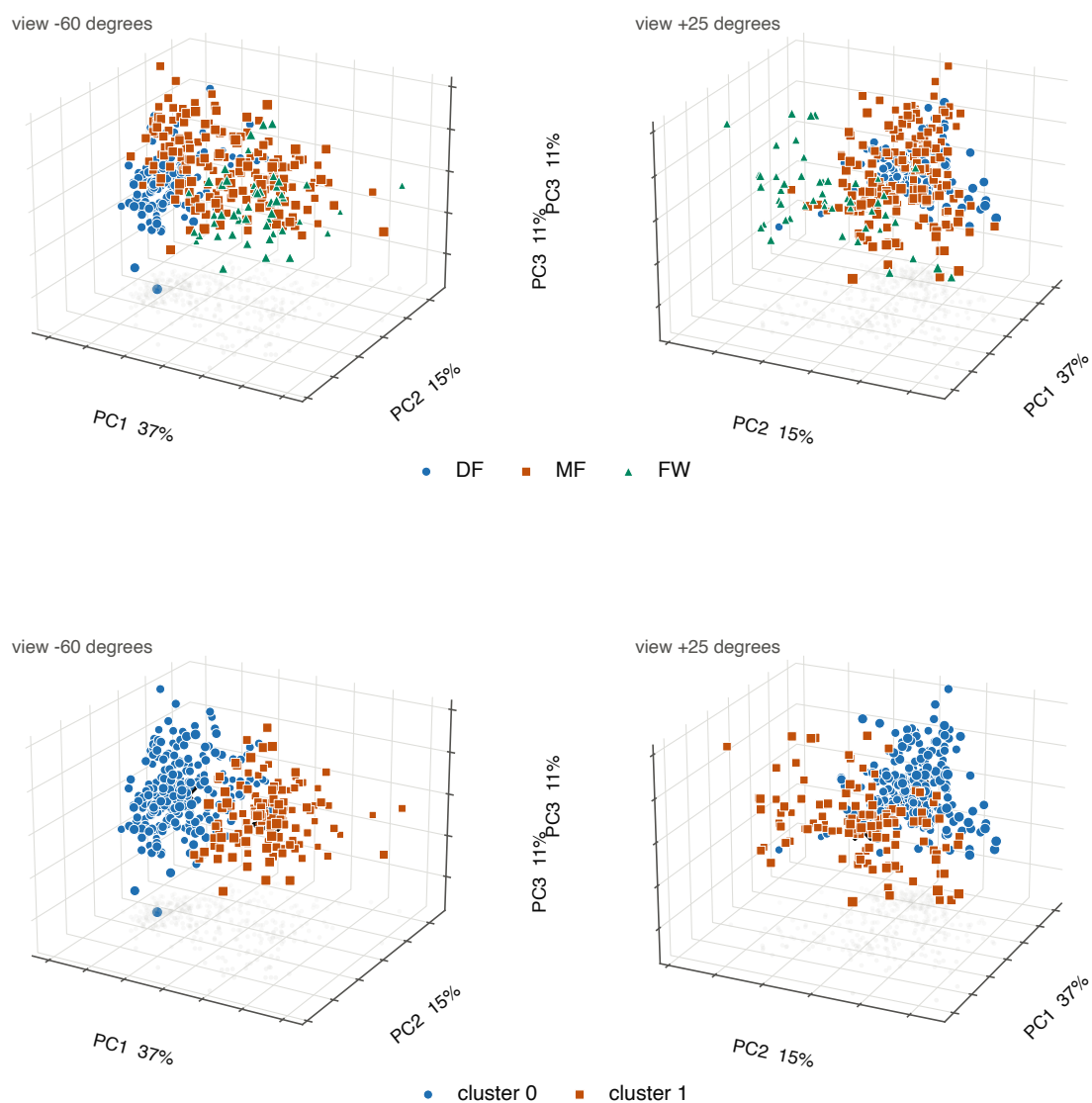
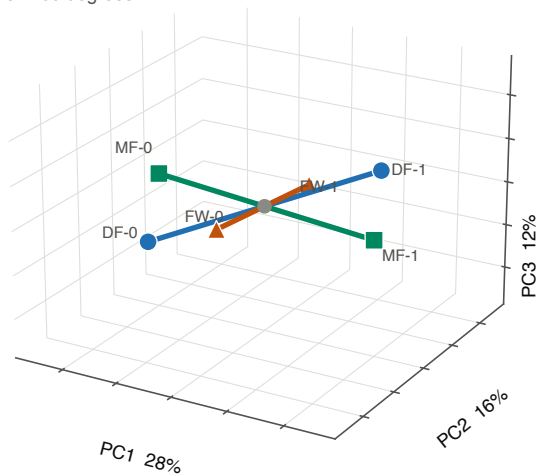
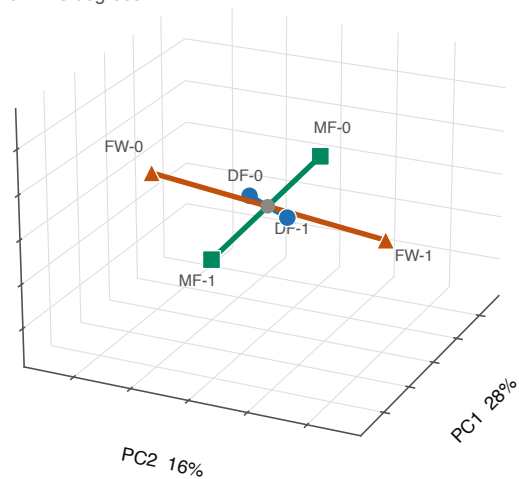


Figure S25: The reduced-sample outfield space in three principal components, each drawn from two viewpoints so that structure can be told apart from the accident of a single projection. Top, players by listed position group, with grey points on the floor supplying a depth cue. Bottom, the selected global partition with centroids marked by crosses.

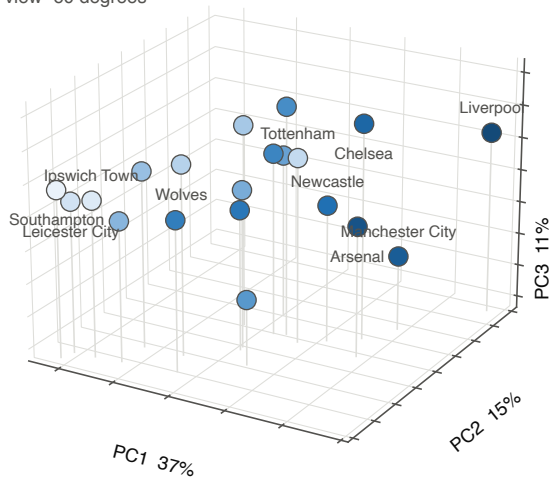
view -60 degrees



view +25 degrees



view -60 degrees



view +25 degrees

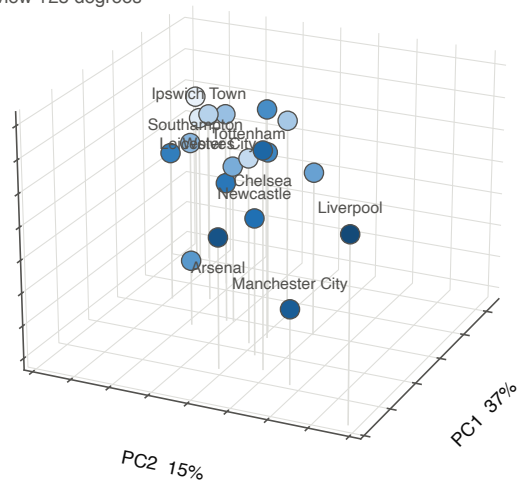


Figure S26: Top, archetype centroid directions drawn as unit vectors, so the figure shows orientation rather than magnitude; each position group forms a single line through the origin, so six archetypes describe three axes. Bottom, clubs as minutes-weighted centroids in three components, shaded by final league position with darker shading for a higher finish.

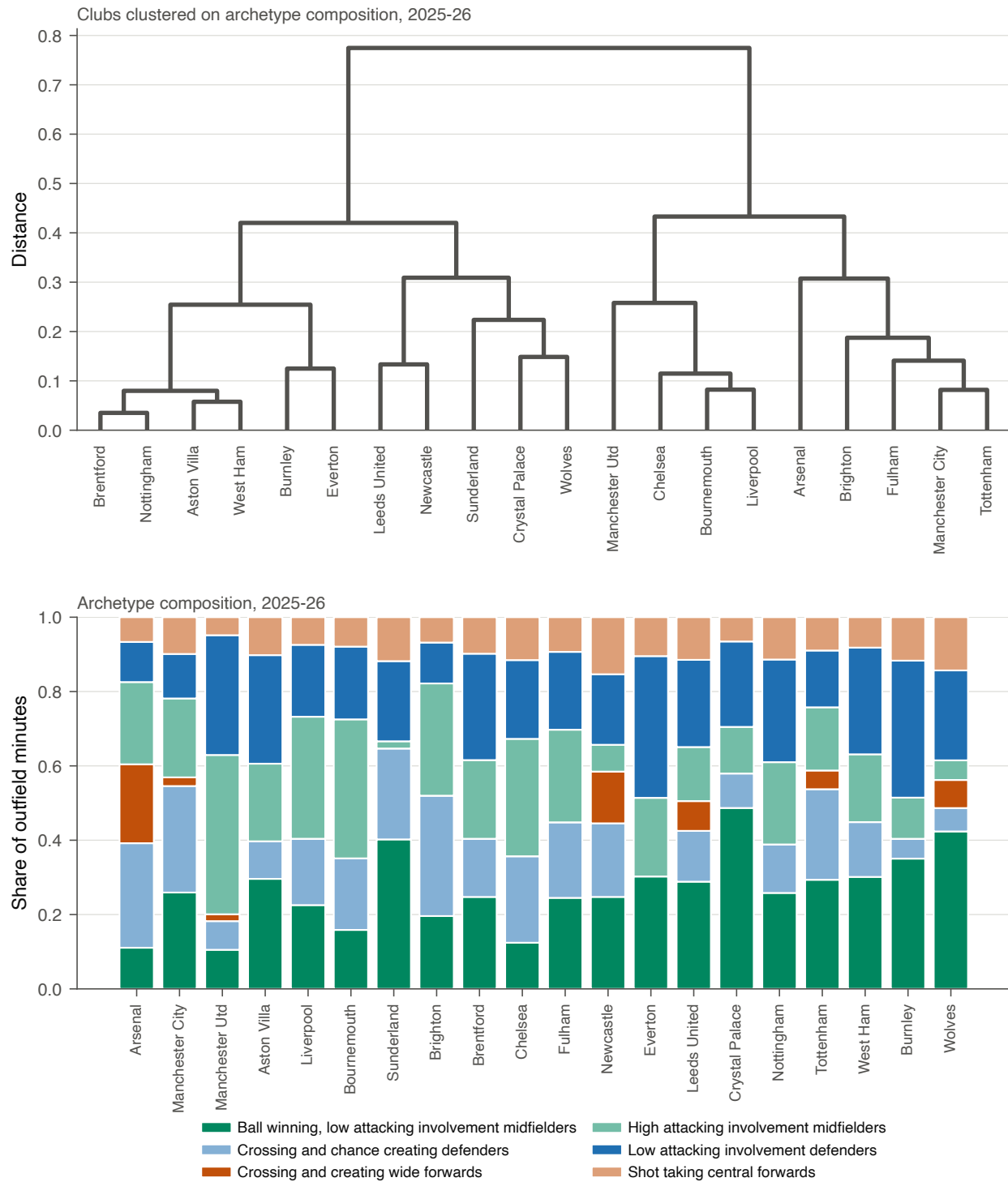


Figure S27: Club signatures on the reduced sample. Top, style families from hierarchical clustering on archetype composition. Bottom, archetype composition by club, ordered by final league position.

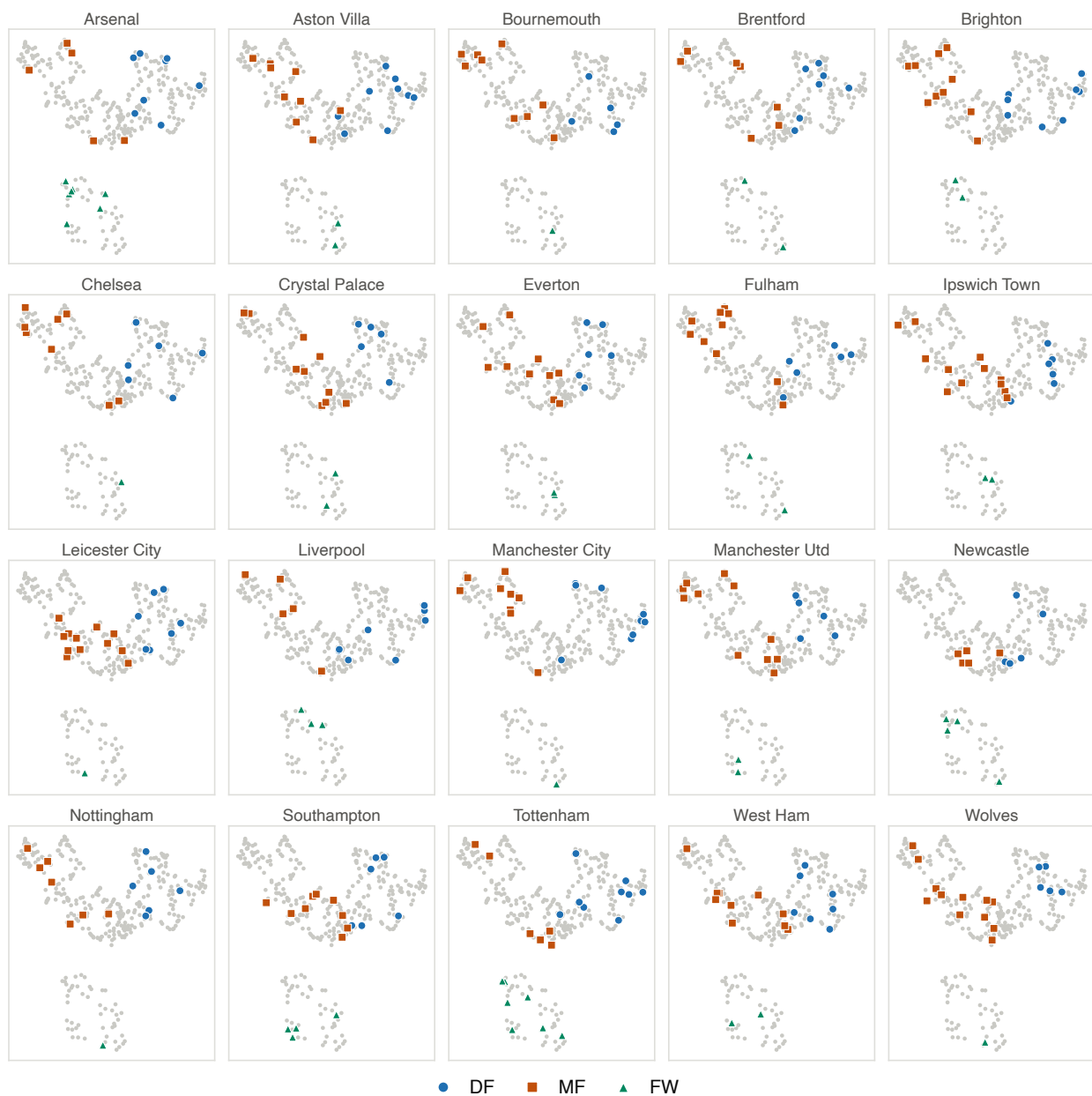


Figure S28: Each club's outfield players highlighted in the shared embedding, with the rest of the league in grey. Axes are shared across all twenty panels, so clubs can be compared directly, and they differ visibly in how far their defenders sit toward the midfield region.

S8.6 Similarity, replication and sampling noise

Nearest neighbour search within position group recovers football sense for players whose roles are defined by the surviving statistics, and fails informatively for those whose roles are not. Full backs return full backs and penalty box forwards return penalty box forwards. The instructive failures are the midfielders: a holding midfielder's nearest neighbours are creative attacking midfielders, because with no tackles by pitch third, no blocks, no clearances, no pressures and no passing volume, the only things left that distinguish him are his set piece deliveries and key passes. The metric finds the players who share that signature and misses the entire defensive half of his job, which Figure S29 makes visible.

242 players are eligible in both reduced-sample seasons. The declared rule independently selects the same number of clusters in every scope in both seasons, so the structure replicates, while membership replicates only partly: the adjusted Rand index between seasons is 0.598 globally, falling to 0.526 for defenders, 0.381 for midfielders and 0.453 for forwards, with the transitions in Figure S30.

Empirical Bayes shrinkage toward the position group mean, with the degree of shrinkage set by minutes played, gives the sharpest test of whether the partitions describe playing roles or sampling noise. The global partition survives: only 5 of 366 players change cluster, the adjusted Rand index between the original and shrunk partitions is 0.945, and the rate at which players move falls monotonically with minutes played, exactly as the noise account predicts. The within group partitions fare worse and not equally. For defenders the index falls to 0.453, a partition largely dissolved by removing sampling noise, while midfielders retain 0.770 and forwards 0.626. Figure S31 shows the change rate against minutes and Figure S32 the effect on every feature.

Read alongside the bootstrap and replication results, the three tests do not deliver a single uniform verdict on the reduced sample, and reporting them as though they did would overstate the case. Defenders fail all three there, forwards fail two of the three, and midfielders pass all three. The conclusion that survives every test on the reduced sample is narrower than a single summary sentence would suggest: the global division between attacking and defensive involvement is a real property of players, the midfield split is trustworthy, and the defender split should not be used to characterise an individual on that feature set. The primary sample, with twice the features and six times the players, supports all three splits, which is the clearest single illustration of what the richer description buys.

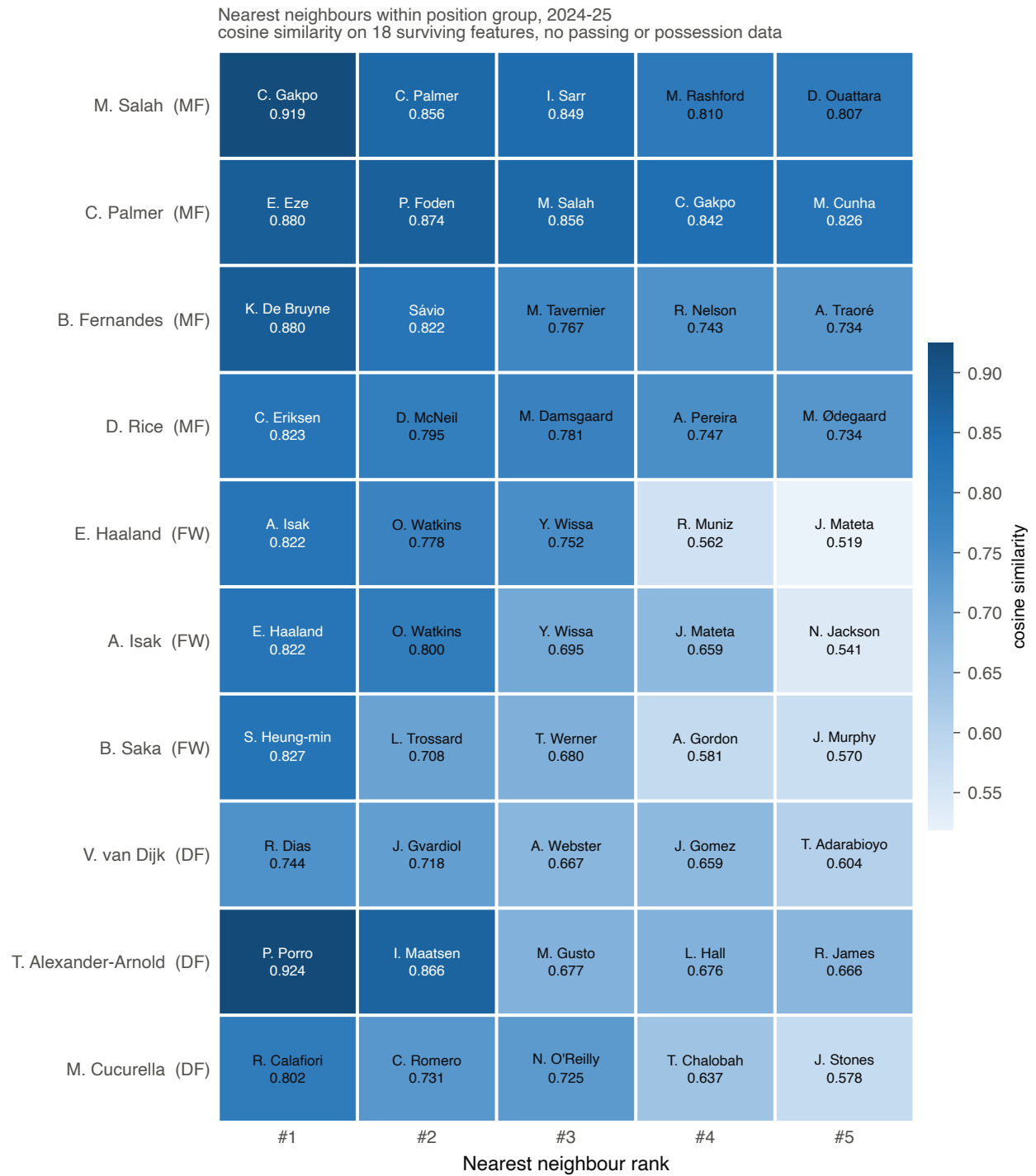


Figure S29: Nearest neighbour similarity structure within position group, reduced sample.

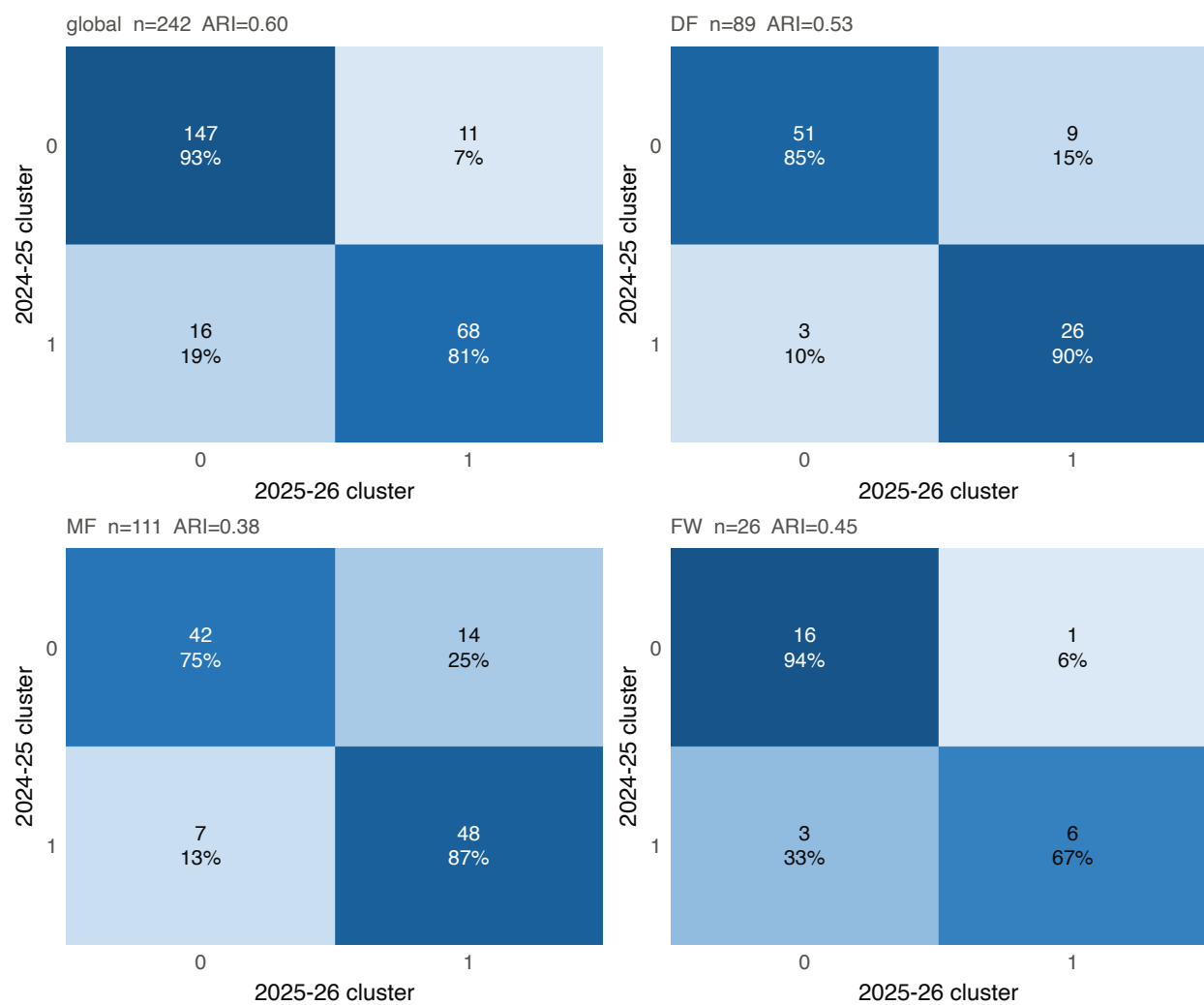


Figure S30: Between-season cluster transitions on the reduced sample, for the 242 players eligible in both seasons.

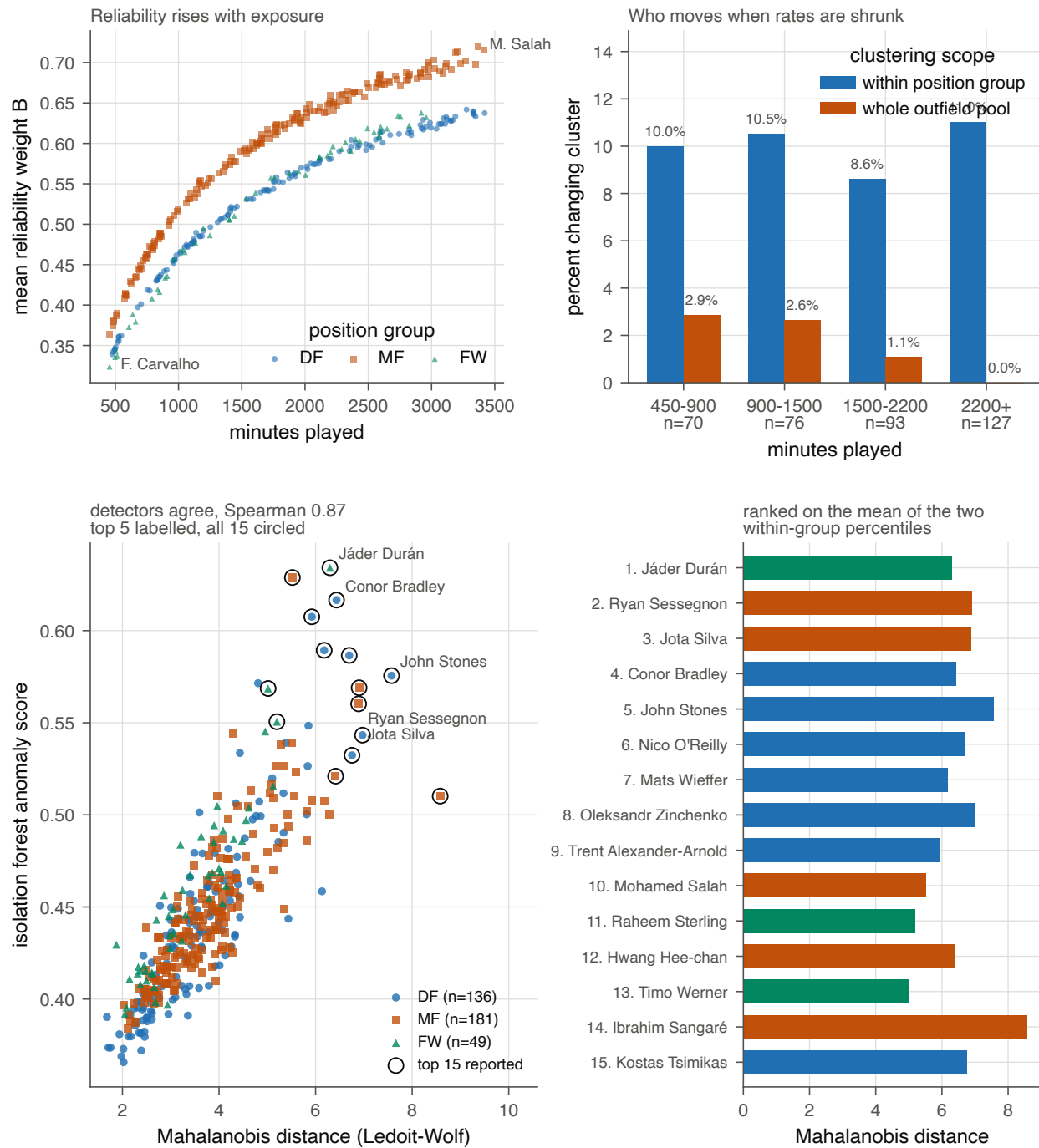


Figure S31: Reduced sample. Top, reliability and cluster change rate against minutes played, which falls monotonically as the noise account predicts. Bottom, the two outlier detectors against each other within position group.

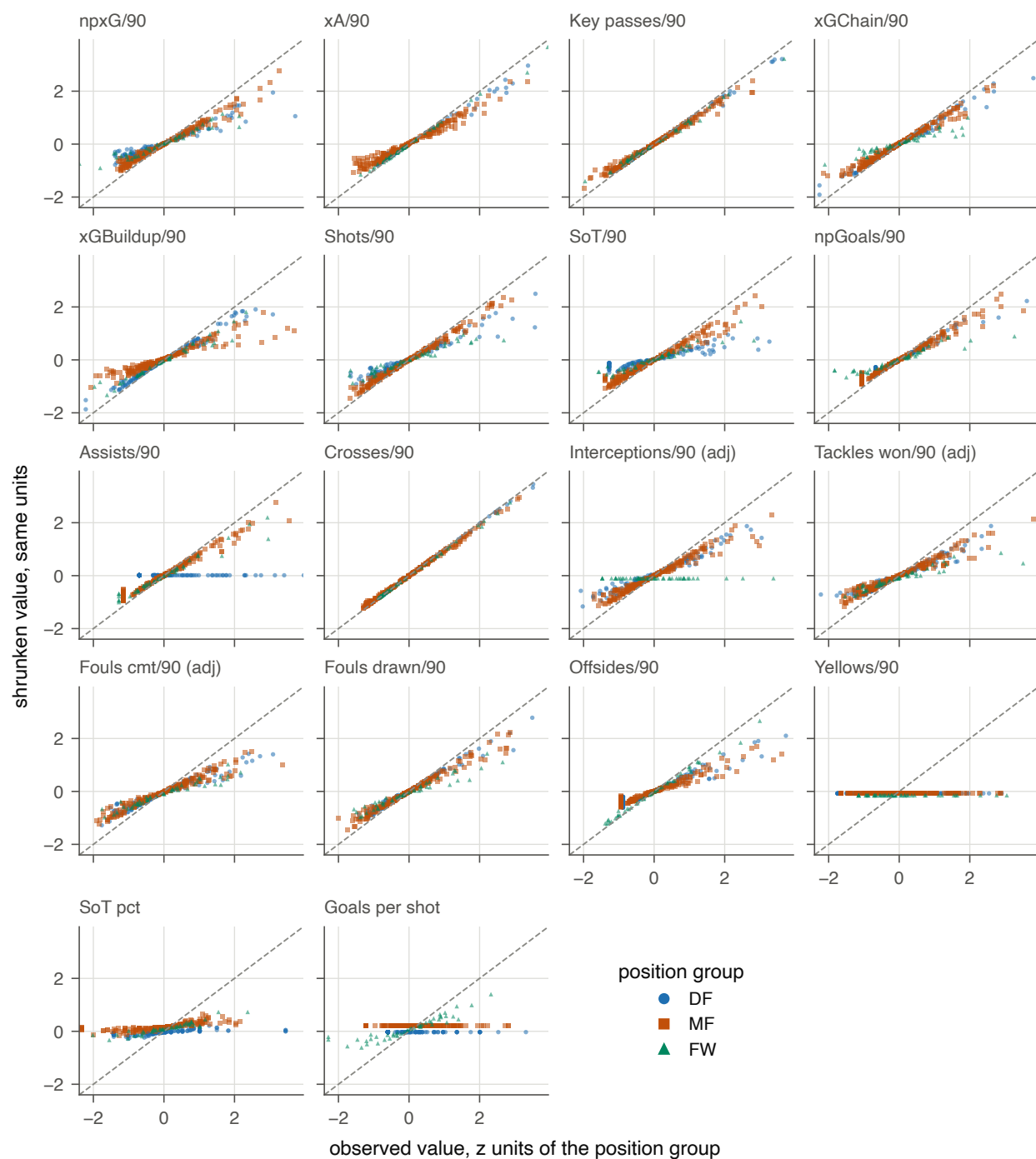


Figure S32: The effect of empirical Bayes shrinkage on every feature of the reduced sample, with the shrunk value against the observed one, one panel per feature.

S8.7 Role drift within a season

Every result elsewhere treats a season as one observation per player, which assumes a player's profile is homogeneous across it. Match reports make that assumption testable, and because the withdrawal removed the affected columns from the match schema rather than emptying them, what remains there is fully populated. A full season is 380 reports at seven seconds each, so the scrape is restricted to one club, Tottenham, chosen because its eligible players span all six archetypes and therefore give drift the widest range to move across. 38 matches were retrieved and 26 players accumulated at least one complete rolling window of 6 matches. This is an illustration of a limitation rather than an estimate of its size, and it should be read as one.

The match feature set has 10 of the eighteen features used elsewhere, since the Understat expected goals block has no match equivalent. Rather than assume the coarser space is a fair stand-in, we refit the season clustering using only those features and compare it to the partition reported throughout; the two agree at an adjusted Rand index of 0.903, so drift measured within the reduced space is interpretable.

Most players do not drift. 5 of 26, or 19 percent, change cluster at least once during the season, and the remainder occupy a single cluster throughout. The players who stay put are the ones whose roles are least ambiguous, holding midfielders on one side and the recognised attackers on the other, while the movers are predominantly full backs and wide players whose attacking involvement varies with the fixture. For four fifths of players a season aggregate describes a single role and the homogeneity assumption holds; for the remaining fifth it averages over two. Figure S33 shows the trajectories.

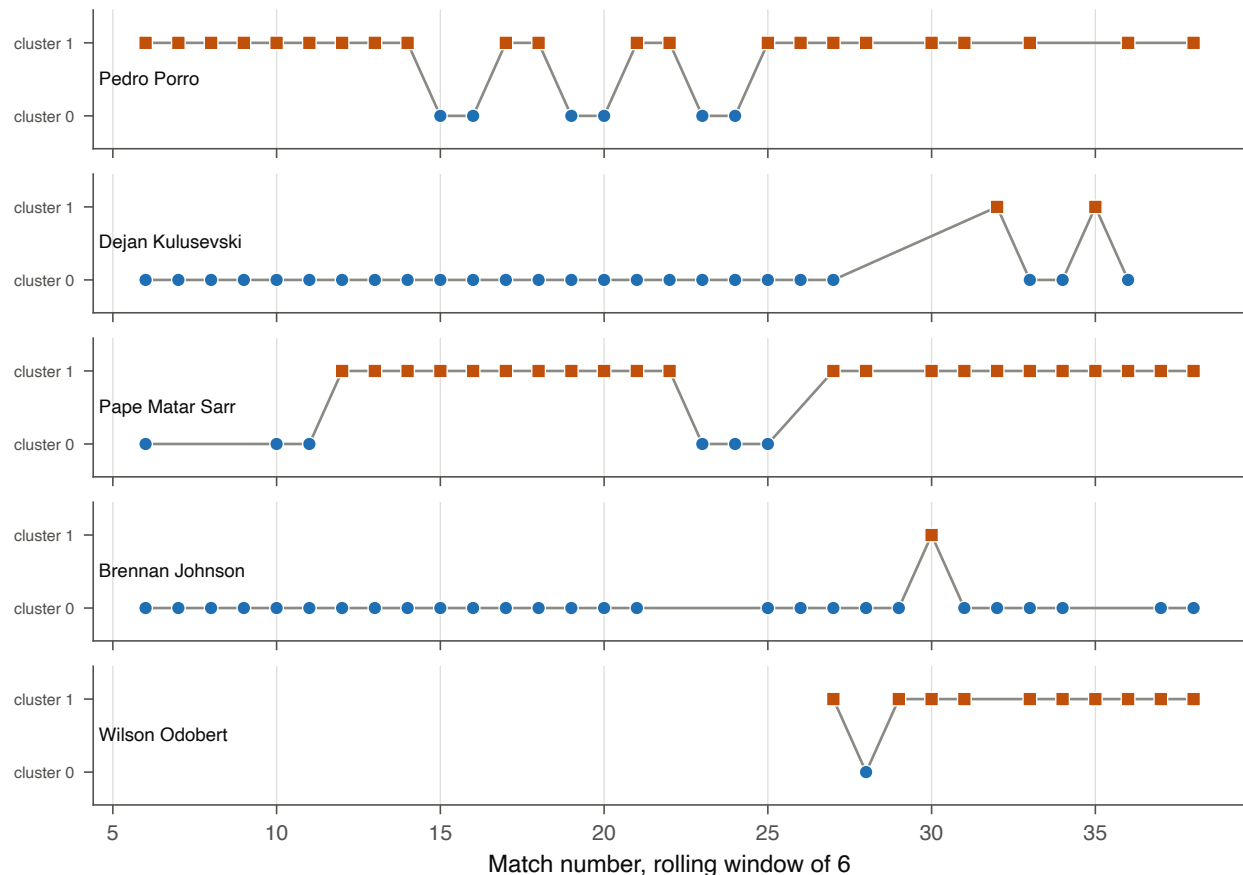


Figure S33: Rolling cluster assignment across the season for the Tottenham players who change cluster at least once, on a 6 match window. Players not shown occupied one cluster for the whole season.

S8.8 Age and archetype

The two archetypes within a position group differ mainly in attacking involvement, and involvement is plausibly something a career passes through rather than a fixed trait, so the association with age is a natural test. It is tested within position group, because archetypes are nested inside positions and a pooled table would largely recover the fact that forwards and defenders have different age profiles.

Only defenders show a relationship. The crossing and chance creating defenders average -1.65 years younger than the low involvement group, a difference the rank test places at 0.015, and the share of the crossing archetype falls monotonically from the youngest band to the oldest. Midfielders show nothing at all, with a gap of 0.12 years at 0.698, and the global partition is unrelated to age at 0.415. Forwards point the same way as defenders but cannot support the claim, since six of the ten cells in their table have expected counts below five.

The defender result also illustrates a point about test choice that is easy to get wrong. A chi-square on the banded table returns 0.194 and would ordinarily be read as no association. That test spends its degrees of freedom on any departure from independence, while the pattern here is monotonic, and a rank correlation against age recovers it at 0.209 with a p-value of 0.015. The banded test is not wrong, it is underpowered against the alternative that actually holds, and reporting only the chi-square would have converted a real finding into a null one. Figure S34 shows the shares.

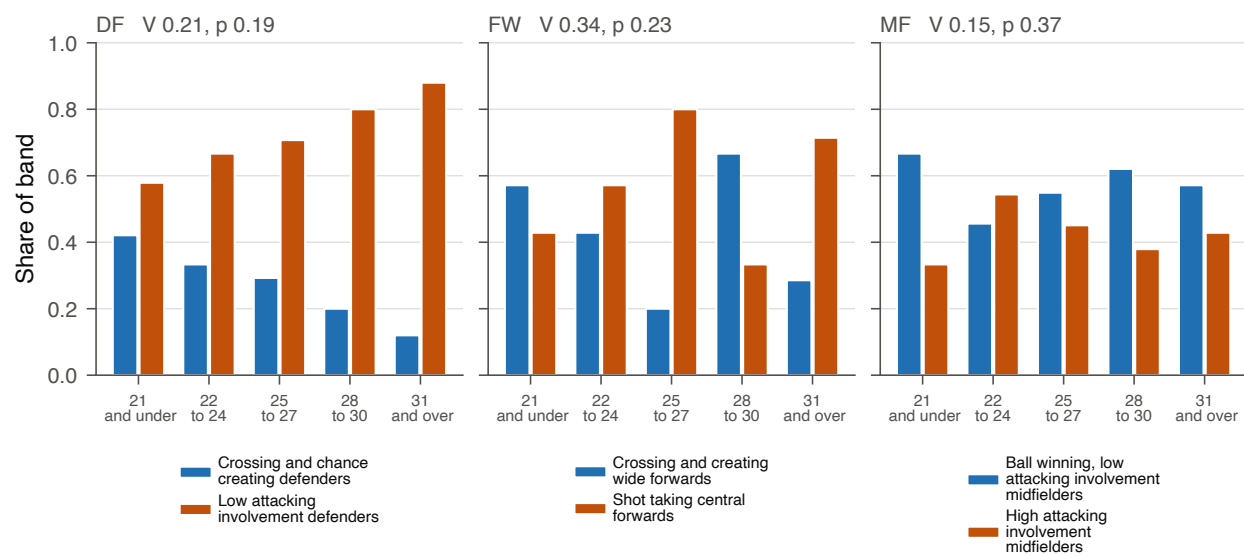


Figure S34: Share of each age band taken by each archetype within position group, reduced sample. The defender panel falls monotonically with age, which a chi-square on bands does not detect and a rank correlation does.

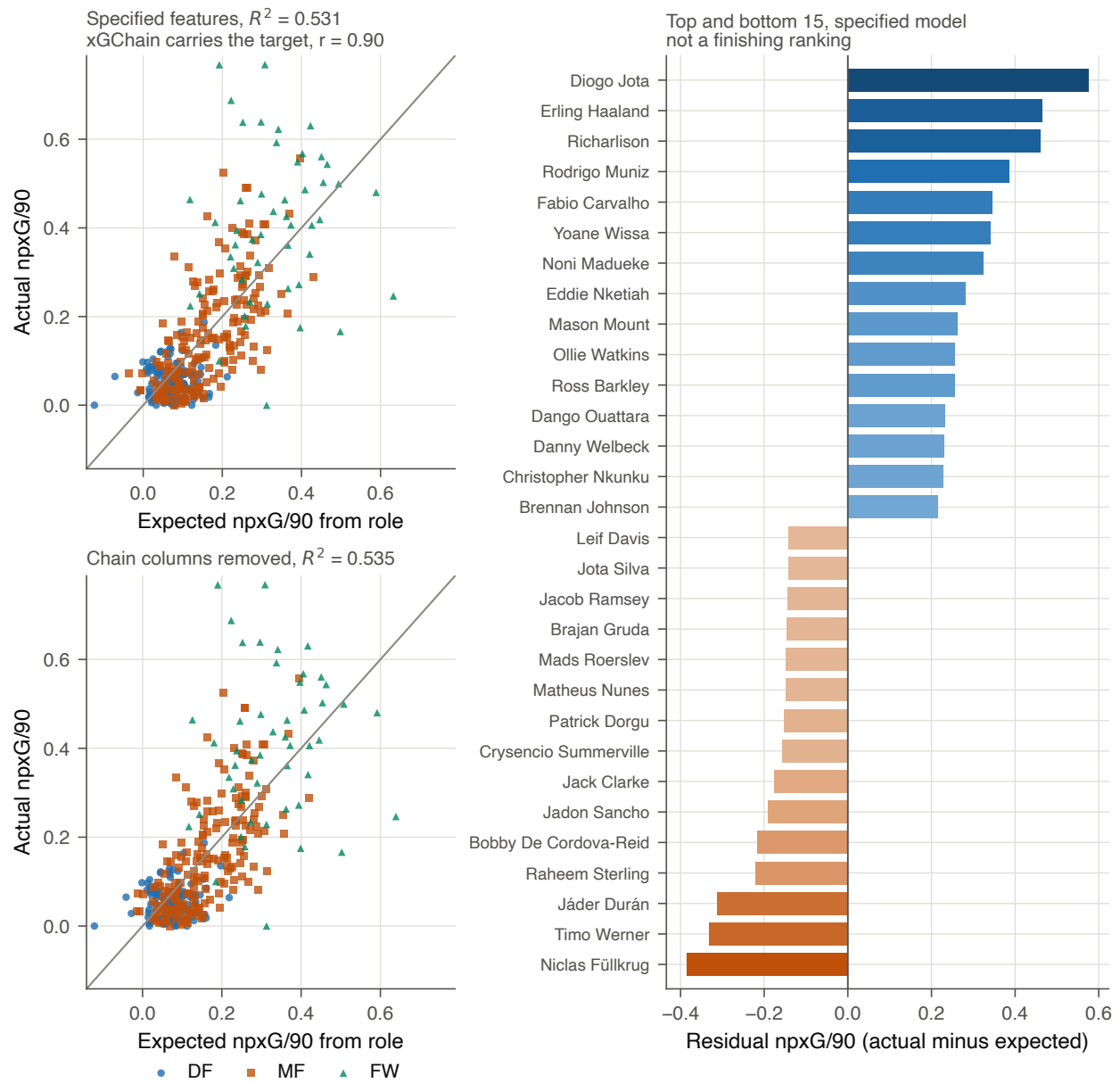


Figure S35: Residuals from the role model on the reduced sample, which should be read as an illustration rather than as a measurement of finishing ability, for the reasons given in the text.

S8.9 Goalkeepers on the reduced sample

Post-shot expected goals, sweeper actions, distribution and cross handling are all absent from the reduced sample, so only shot stopping and workload survive. That version builds two composites instead of five and substitutes the residual from a regression of saves on shots on target faced for post-shot expected goals. The substitute adjusts for the volume of work a goalkeeper receives but not for its difficulty, and it is reported as the weaker instrument it is. For the primary sample, Figure S36 gives the shot stopping ranking, Figure S37 the correlation structure of the goalkeeper block and Figure S38 the composites that block supports. For the reduced sample, Figure S39 gives the clusters and the two surviving composites, Figures S40 and S41 the shot stopping views that remain, and Figure S42 the correlation structure and composite distributions.

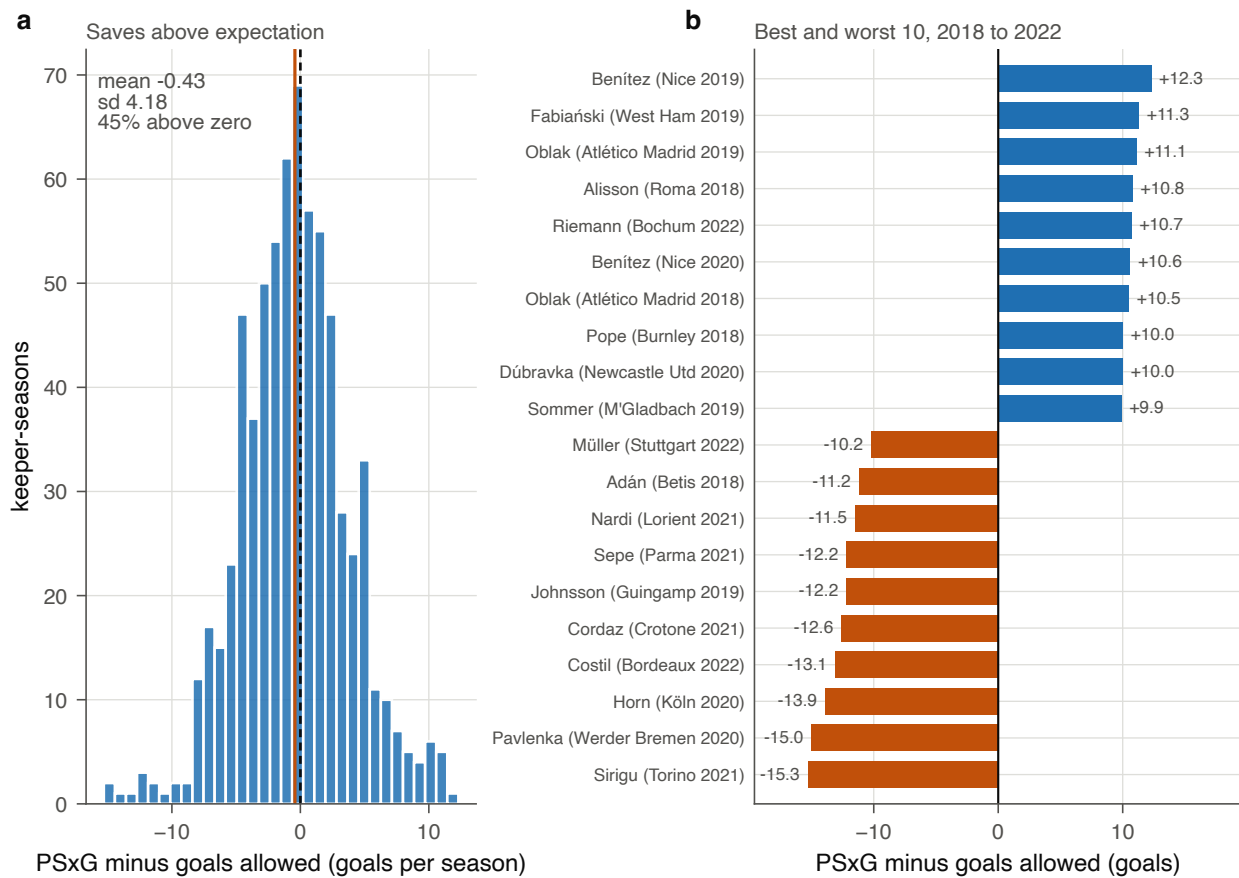


Figure S36: Saves above expectation on the primary sample, as post-shot expected goals minus goals allowed, which conditions on the difficulty of the shots faced rather than only their volume. **a**, the distribution over keeper-seasons. **b**, the best and worst keeper-seasons in the sample.

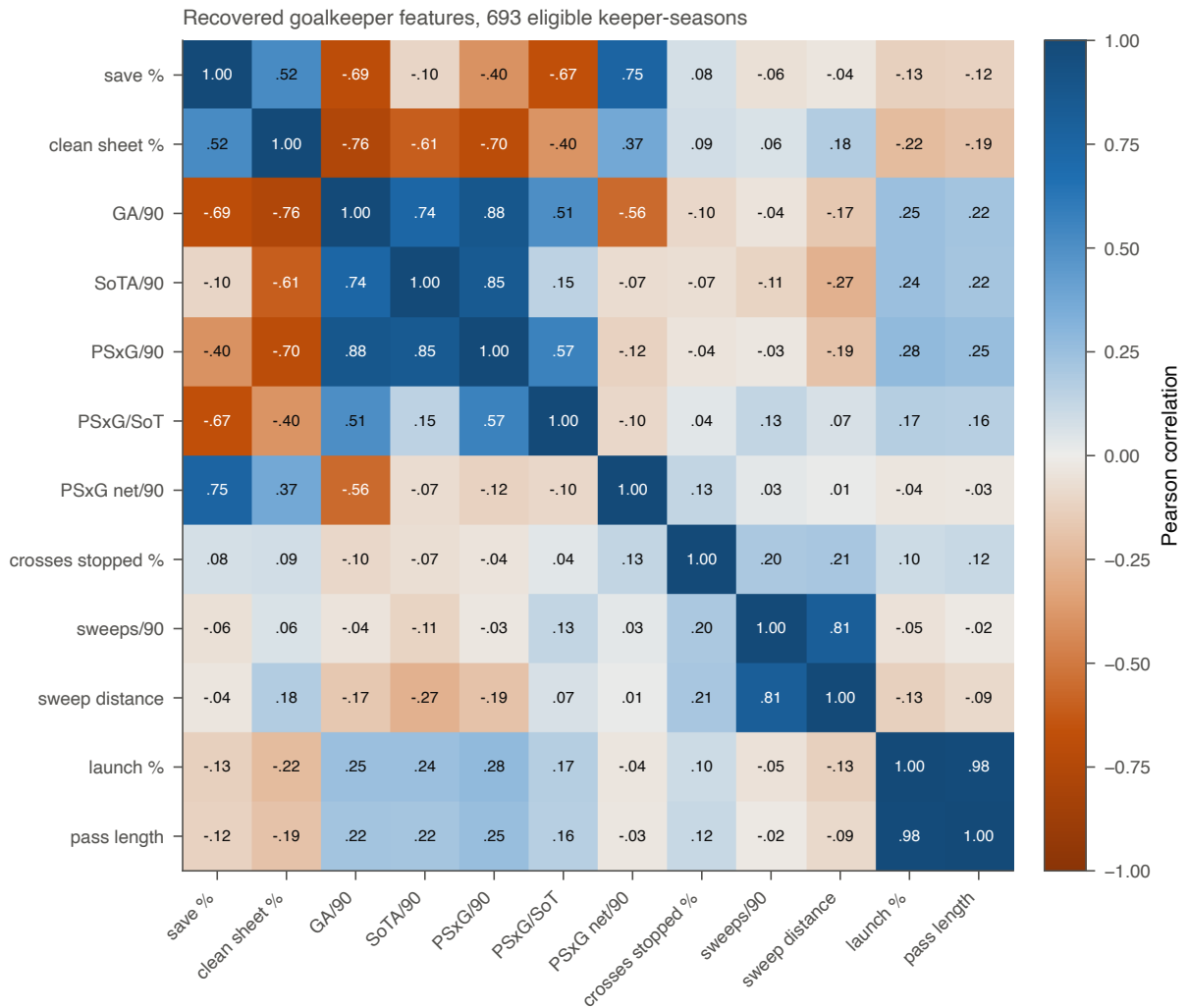
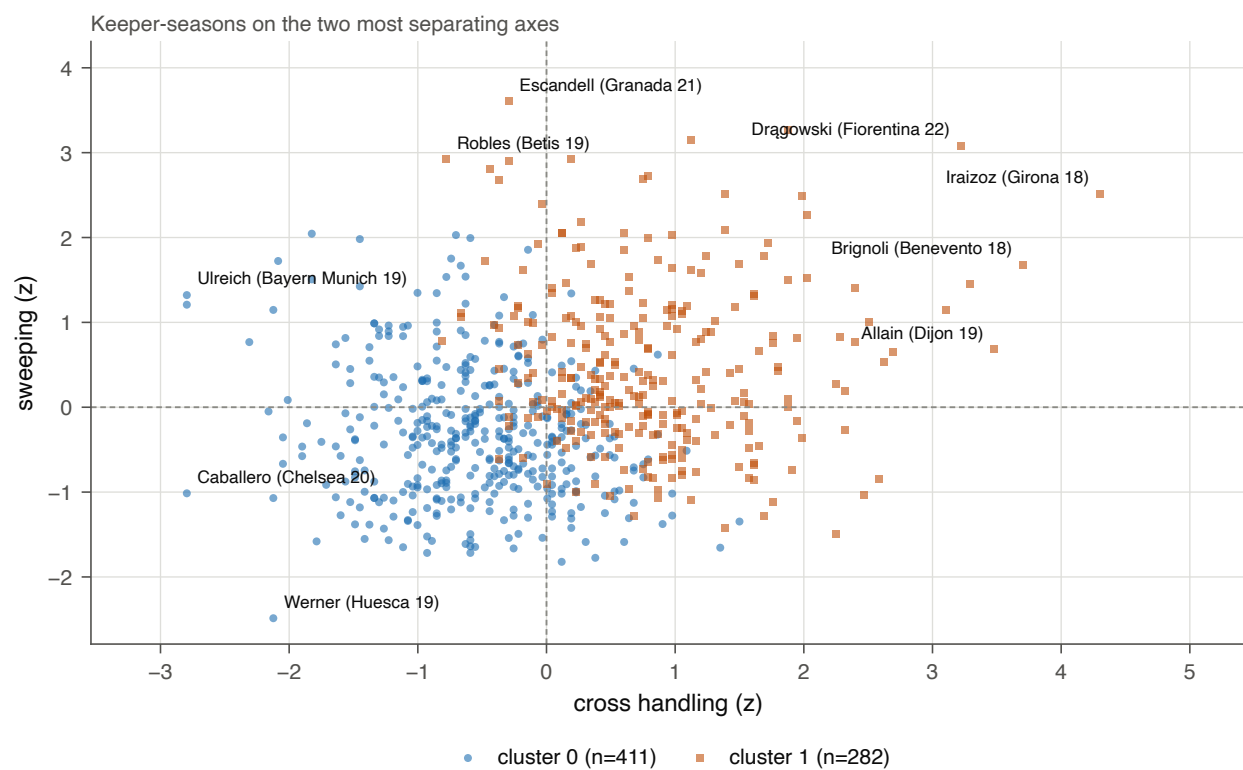


Figure S37: Correlation structure of the 13 goalkeeper features on the primary sample, which is what allows the composite axes the reduced sample cannot build.



eta2 against the partition: cross handling 0.46, sweeping 0.17, workload 0.03.
 Axes chosen for separation, not for football expectation. The most extreme keeper-seasons are named.

Figure S38: Distribution of the goalkeeper composite axes on the primary sample.

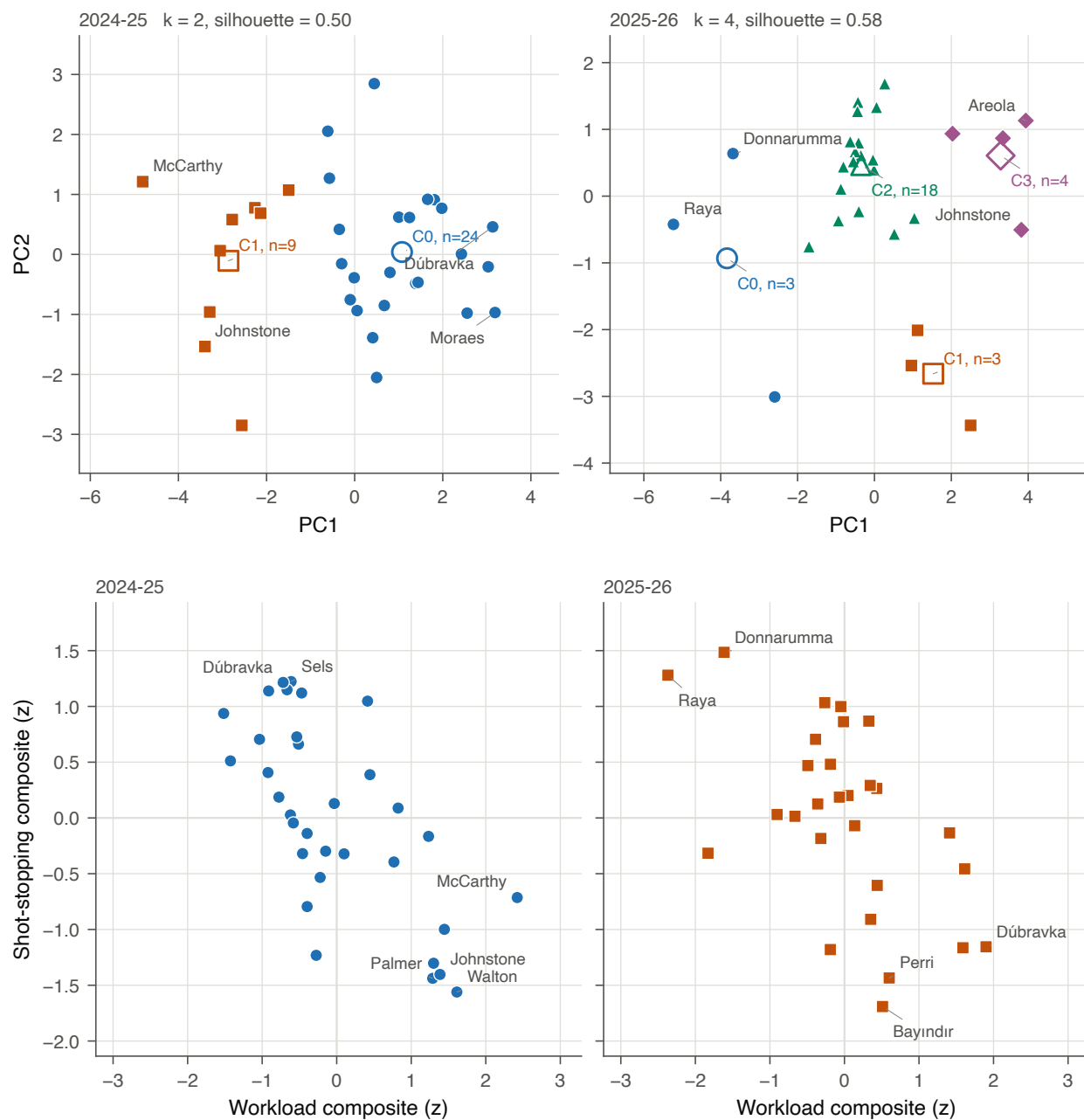


Figure S39: Goalkeepers on the reduced sample. Top, clusters in principal component space. Bottom, the two surviving composite axes.

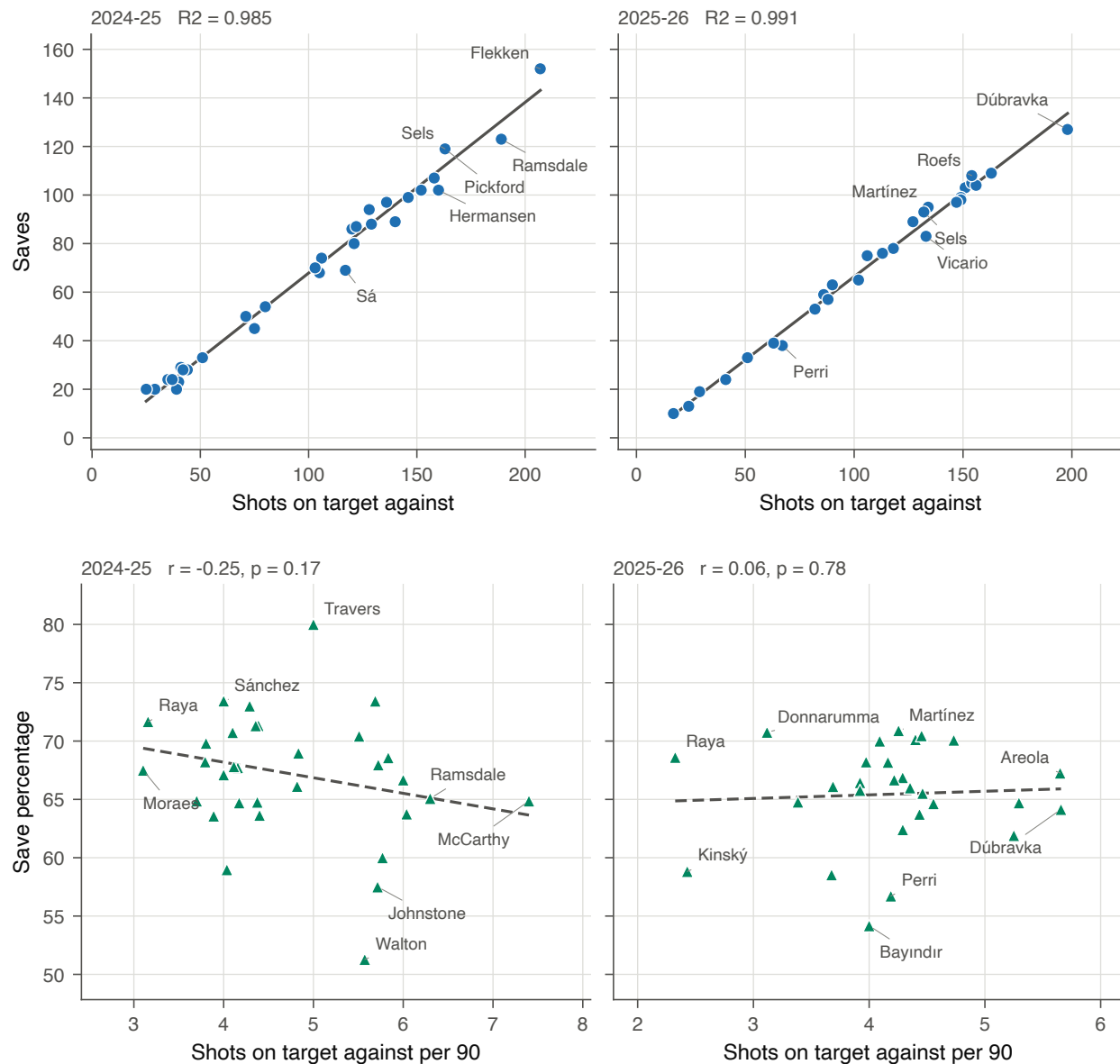


Figure S40: Shot stopping on the reduced sample. Top, saves against shots on target faced. Bottom, save percentage against workload. With expected goals on target withdrawn these are the only shot stopping views the reduced sample supports, and neither separates difficulty from quantity.

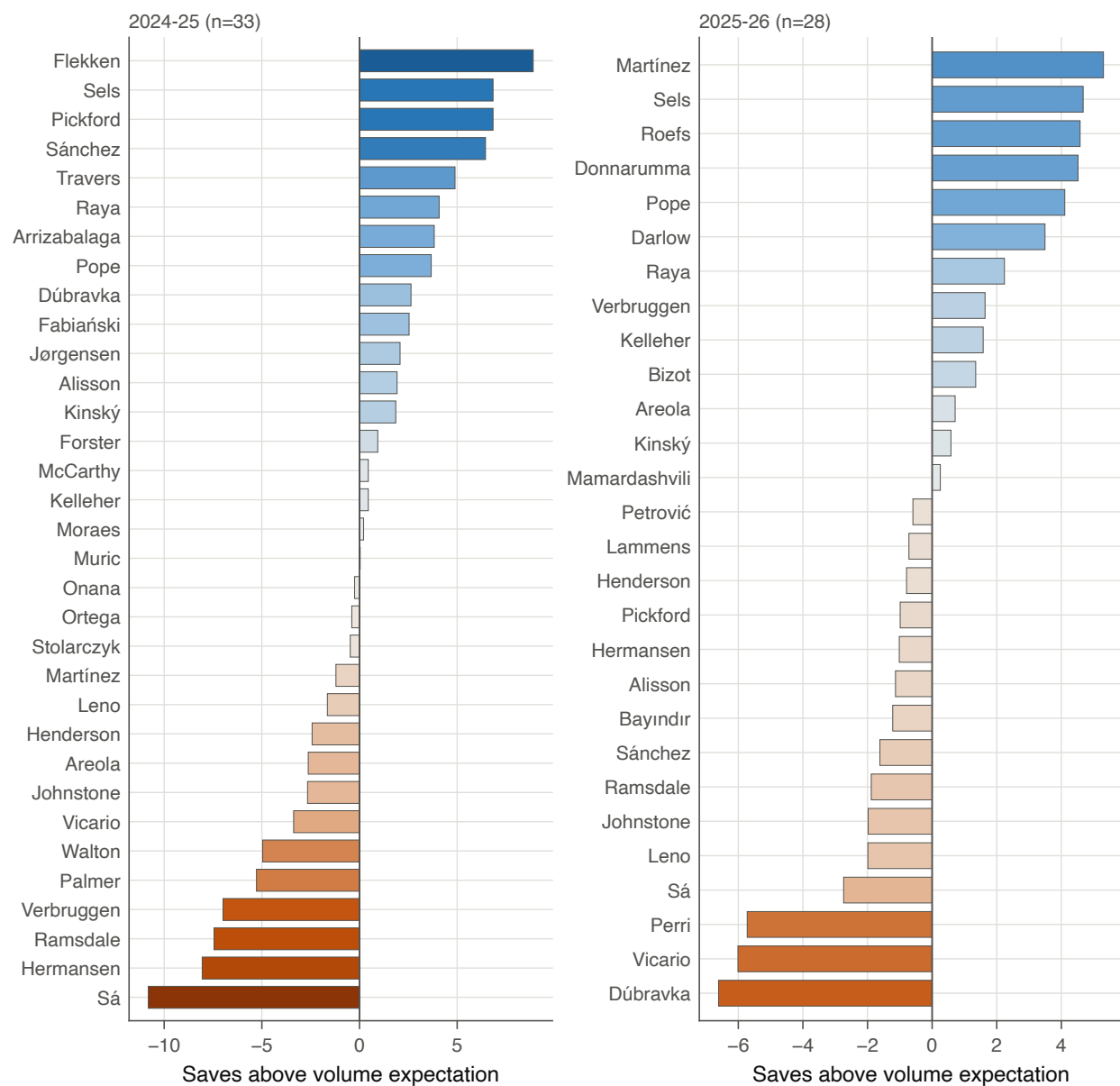


Figure S41: The shot stopping ranking that the two surviving reduced-sample composites support.

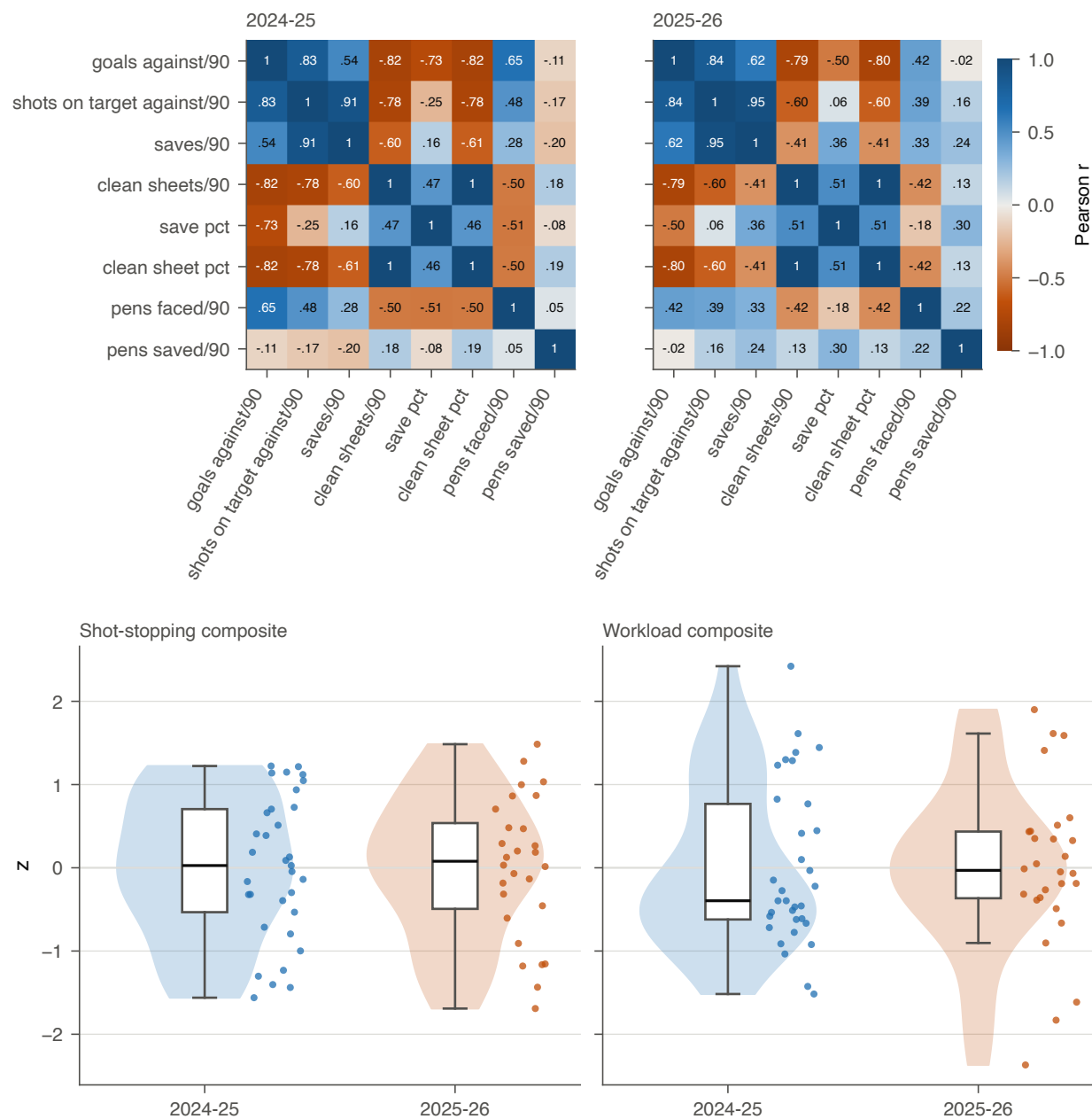


Figure S42: Reduced-sample goalkeeper features. Top, the correlation structure of what survived the withdrawal, which is what constrains the number of composites that can be built. Bottom, the distribution of the two composites.

S9 Cluster membership

Every player's archetype assignment is published as CSV in the accompanying repository rather than reproduced here, for both samples. The file `results/membership_primary.csv` carries the 9,263 eligible player-seasons of the primary sample with club, league, season, listed position, minutes and assigned archetype, and `results/membership_reduced.csv` carries the reduced sample with the same fields plus each player's distance from his cluster centroid. Both are regenerated by the pipeline, so they cannot drift from the numbers reported above.